

ELECTRONICS



ΠΑΝΕΠΙΣΤΗΜΙΟ
ΔΥΤΙΚΗΣ ΑΤΤΙΚΗΣ
UNIVERSITY OF WEST ATTICA

DEPARTMENT OF INFORMATION AND COMPUTER
ENGINEERING

PROJECT 2
CRYSTAL DIODE (SILICON) - ZENER DIODE

STUDENT DETAILS

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STUDENT STATUS : UNDERGRADUATE

PROGRAMME OF STUDY : UNIWA

LABORATORY SECTION : ELECT 17 FRIDAY 13:00 - 15:00

LABORATORY TEACHERS : TSALERA ELENA - DIAMANTOPOULOS MICHAEL

DELIVERY DATE : 18/11/2022

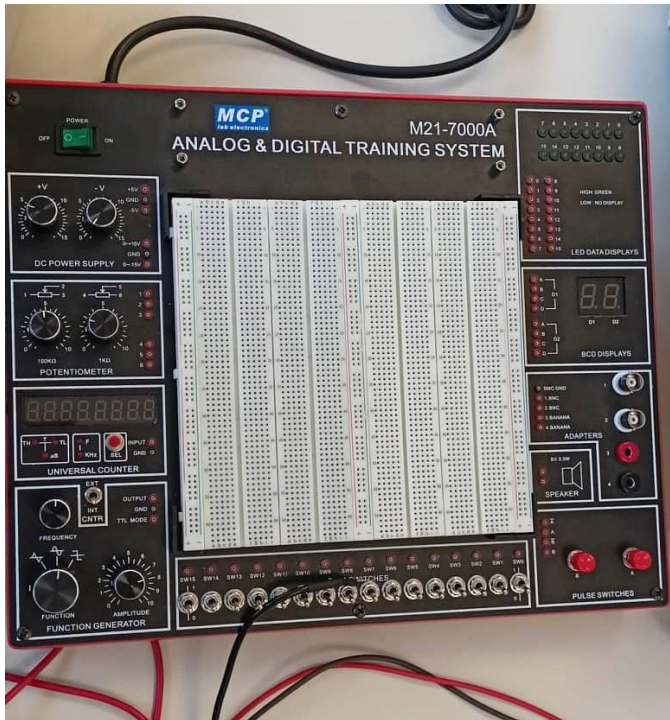
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STUDENT PHOTO :

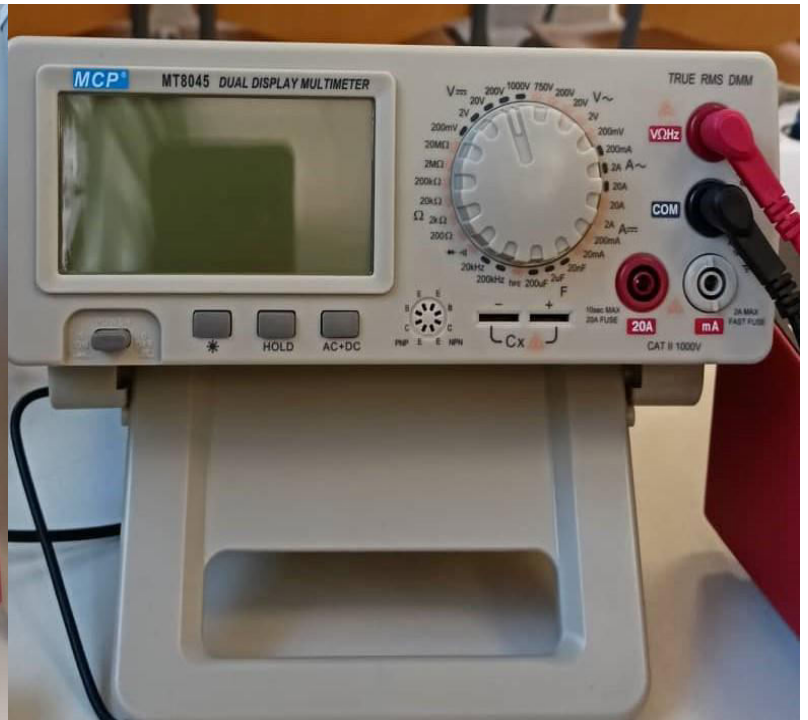


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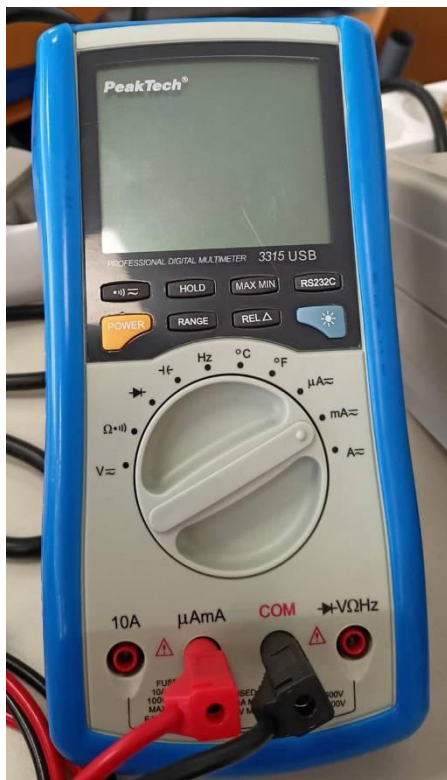
PHOTOGRAPHS OF EQUIPMENT USED IN THE LABORATORY



Breadboard M21-7000A



*Digital
Multimeter Bench MT8045*



Digital Multimeter



Cable

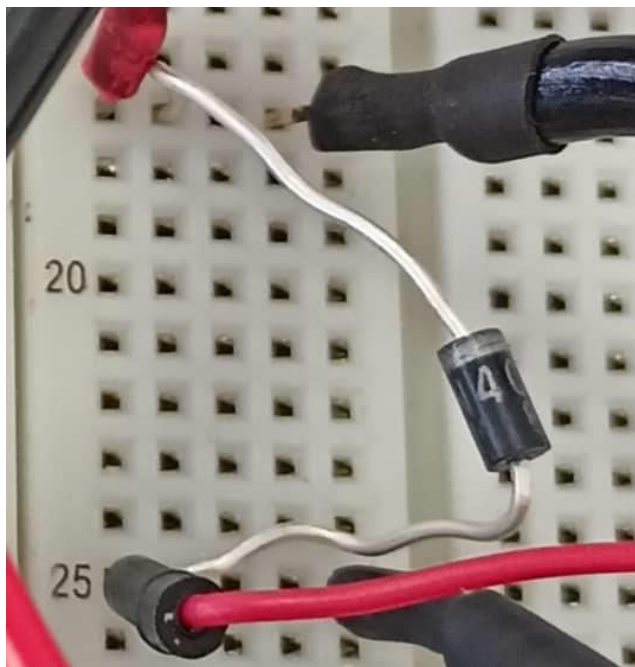
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Resistor of 4.7 KOhm



150 Ohm resistor



Silicon Diode



Zener Diode

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1. CRYSTALLODIODE (SILICON DIOXIDE)

1.1 FORWARD BIAS

General

The chapters "Simulation Solution" and "Experimental Solution" present in detail the components used, the recorded measurements and the observations made during the experiment with snapshots from the simulation software "Multisim" and photographs of the experiment from the laboratory environment respectively, while the chapter "Theoretical Solution" analyses the mathematical proof of the experiment.

Theoretical Solving

Figure 1.1.1 shows the schematic representation of the right polarization of a silicon diode from the design software "MS Visio".

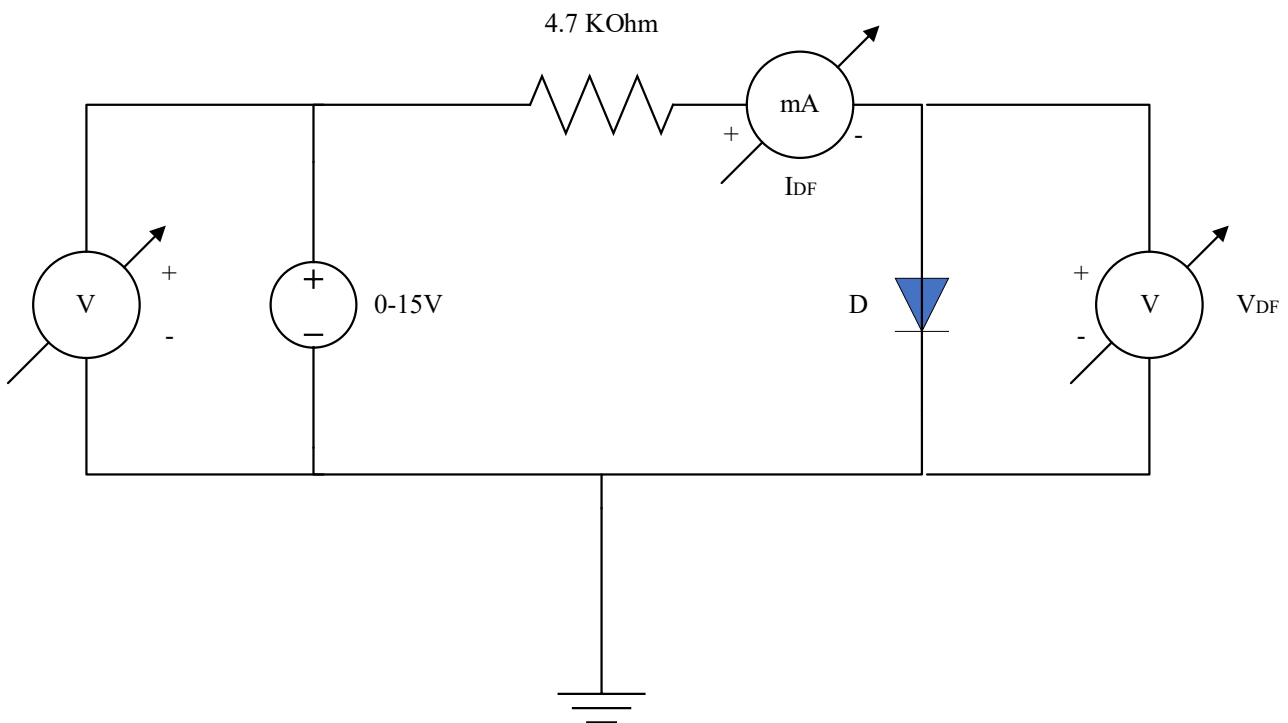


Figure 1.1.1.1 Schematic representation of a silicon diode polarization in "MS Visio"

Table 1 shows the measurements made in the silicon diode right polarization experiment, in terms of the current " I_{DF} " flowing through the diode and the potential difference " V_{DF} " between the ends of the diode. The measurements were made for different source supply values and a resistor with a resistance of 4,7 KOhm is connected to the circuit.

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Table 1.

	Spring (V)	0	0.1	0.3	0.6	0.9	1.5	2	4	7	9	12	15
Si	V _{DF} (V)	0	0.07	0.2	0.26	0.3	0.4	0.42	0.5	0.6	0.6	0.6	0.6
	I _{DF} (mA)	0	0.006	0.03	0.07	0.1	0.2	0.3	0.7	1.4	1.8	2.4	3.0

The observations recorded according to the data in Table 1 are as follows:

- The current "I_{DF}" flowing through the silicon diode increases each time the voltage at the ends of the source increases.
- The potential difference "V_{DF}" at the ends of the silicon diode increases each time the voltage at the ends of the DC source increases, where it stabilizes at about 7 Volt and beyond.

According to Kirchhoff's 2^o rule (law of voltages), the sum of the potential differences in the individual branches of a loop is equal to zero. By Ohm's law, it is true that the potential difference at the ends of a branch is proportional to the intensity of the current flowing through the branch. From the above laws, the measurements recorded in Table 1 are verified. For example, at 7 volts source we have:

$$V_{\pi\eta\gamma\eta} + V_R + V_{DF} = 0 \rightarrow 7 \text{ Volt} - (I_{DF} \times R) + 0.6 \text{ Volt} = 0 \rightarrow$$

$$7 \text{ Volt} - (0.0014 \text{ Ampere} \times 4700 \text{ Ohm}) + 0.6 \text{ Volt} = 0 \rightarrow$$

$$7 \text{ Volt} - 6.58 \text{ Volt} + 0.6 \text{ Volt} = 0 \rightarrow 0 \cong 0$$

The voltage at the ends of the silicon diode is stabilized at about 0.6 Volt when the voltage source is ≥ 7 Volt, i.e., approximately the barrier potential difference of the diode. The internal resistance is small and current flows through it which increases for higher source voltages. The diode behaves as a conductor. Figure 1.1.2 shows the characteristic of a silicon diode during normal polarization, varying the supply voltage according to Table 1.

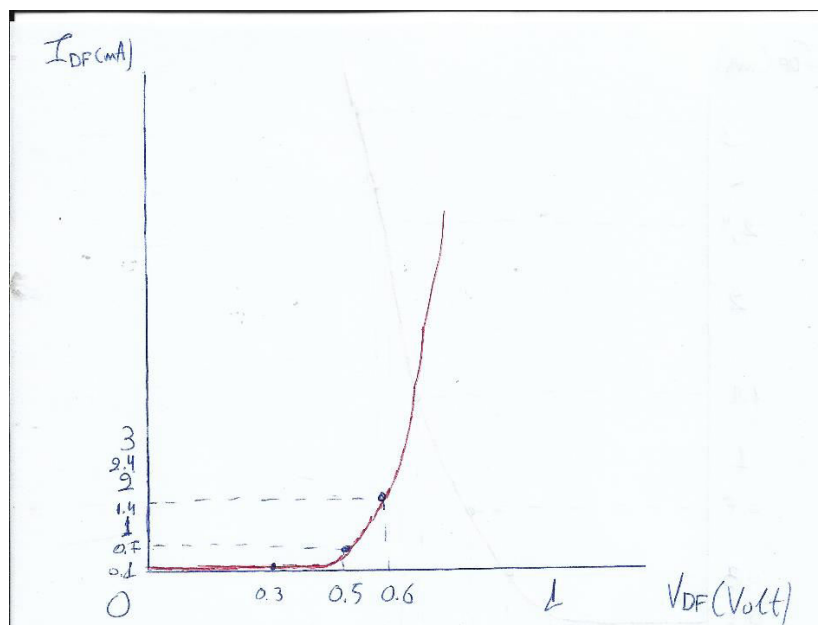


Figure 1.1.2 Silicon diode polarization characteristic I_{DF} - V_{DF}

Simulative Resolution

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Figure 1.1.3 shows the simulated illustration of the silicon diode polarization in the simulation software "Multisim".

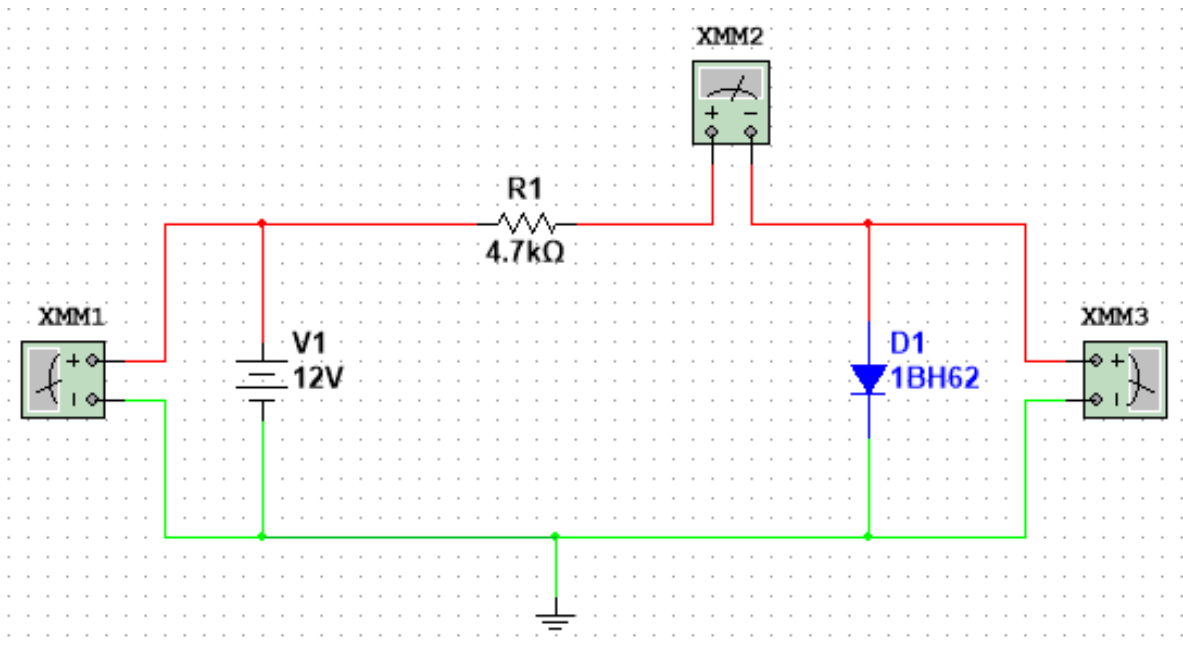


Figure 1.1.1.3 Simulation of a silicon diode polarization in "Multisim"

In Figures 1.1.4, 1.1.5, 1.1.6, 1.1.7, 1.1.8, 1.1.9, 1.1.10, 1.1.11, 1.1.12, 1.1.13, 1.1.14 and 1.1.15, the recorded measurements of Table 1 are simulatively verified in the simulation software "Multisim".

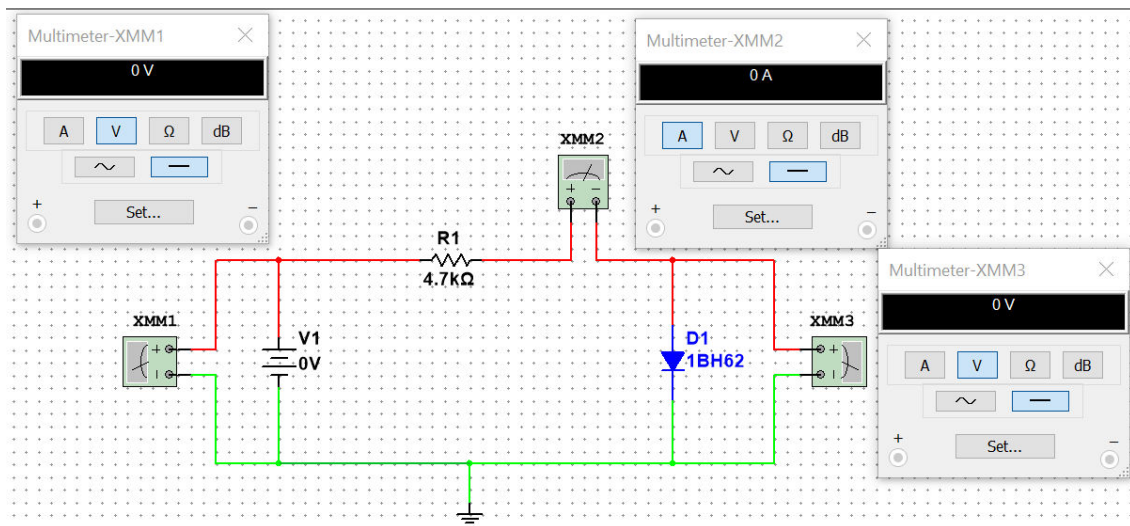


Figure 1.1.1.4 Correct polarization of a silicon diode with 0 Volt supply voltage in "Multisim"

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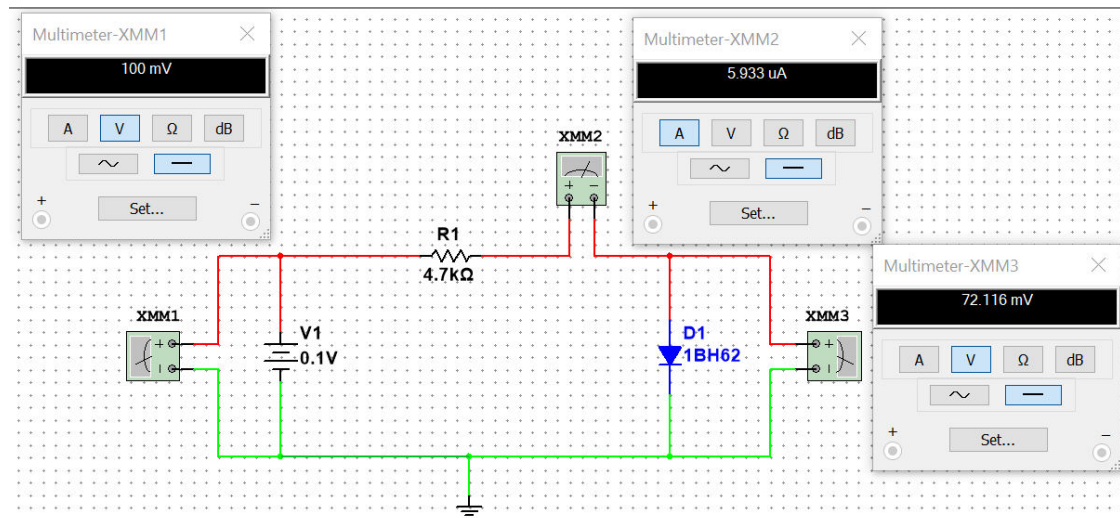


Figure 1.1.1.5 Correct polarization of a silicon diode with 0.1 Volt "Multisim" supply voltage

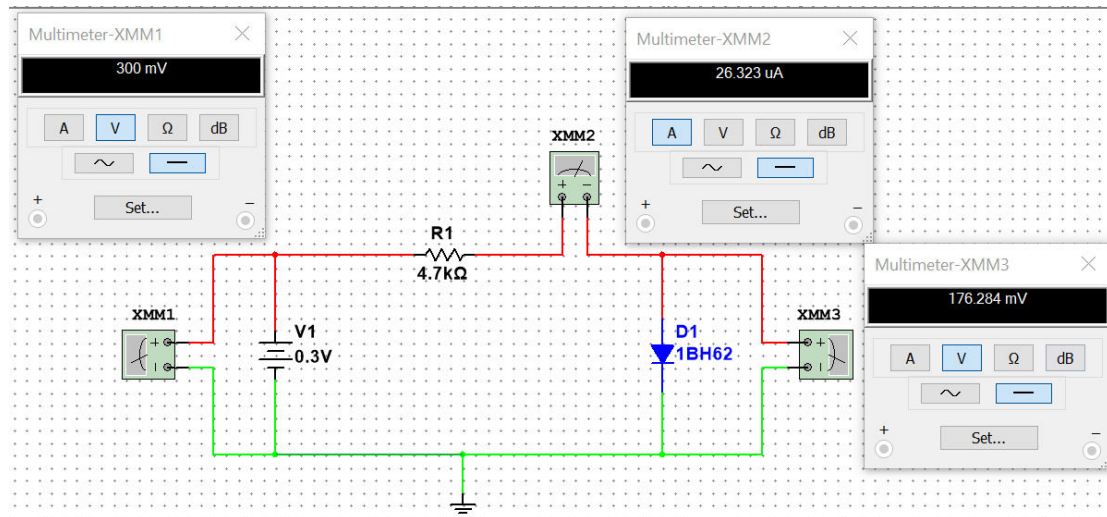


Figure 1.1.1.6 Correct polarization of a silicon diode with 0.3 Volt "Multisim" supply voltage

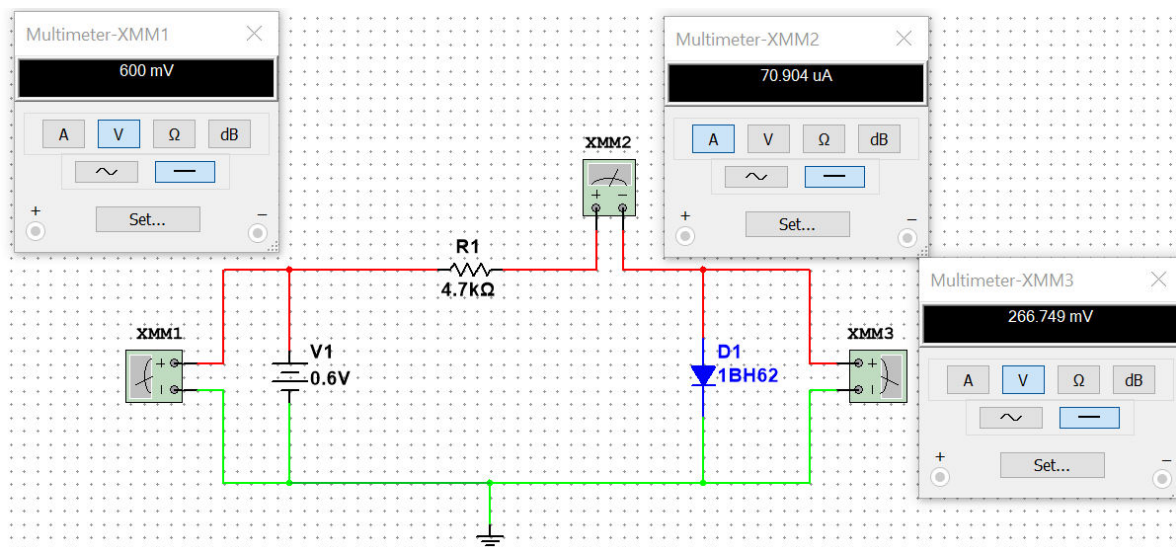


Figure 1.1.1.7 Correct polarization of a silicon diode with a supply voltage of 0.6 Volt in "Multisim"

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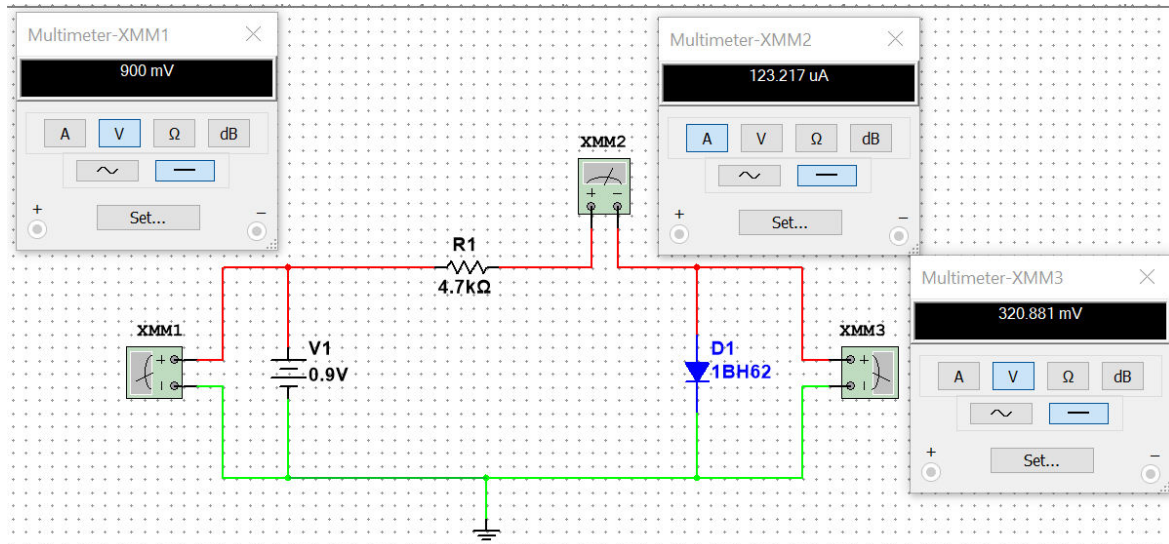


Figure 1.1.8 Correct polarization of a silicon diode with 0.9 Volt supply voltage in "Multisim"

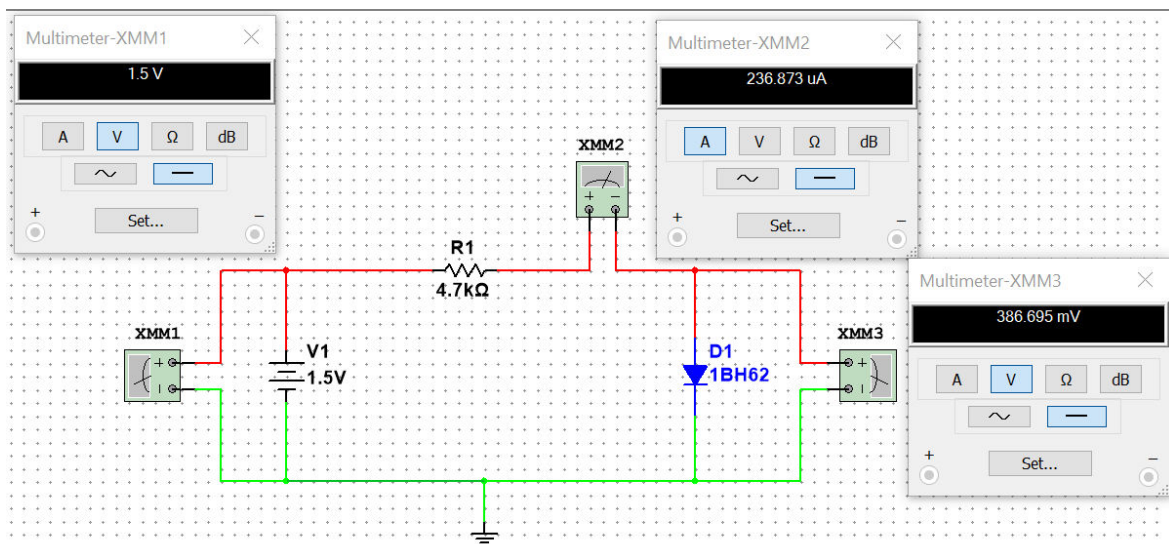


Figure 1.1.9 Correct polarization of a silicon diode with a supply voltage of 1.5 Volt in "Multisim"

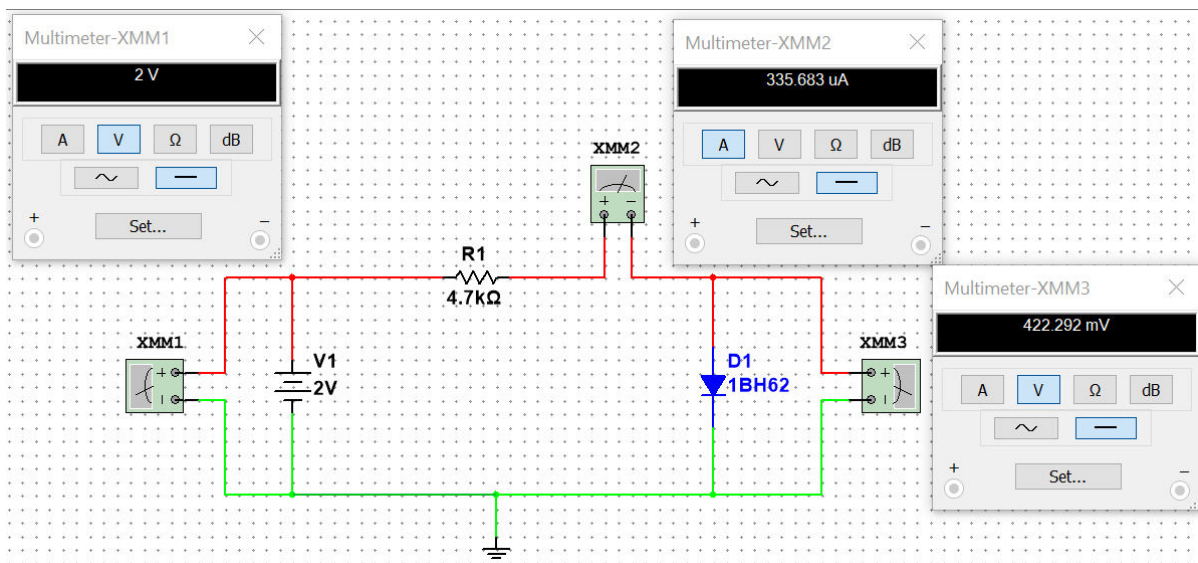


Figure 1.1.10 Correct polarization of a silicon diode with 2 Volt supply voltage in "Multisim"

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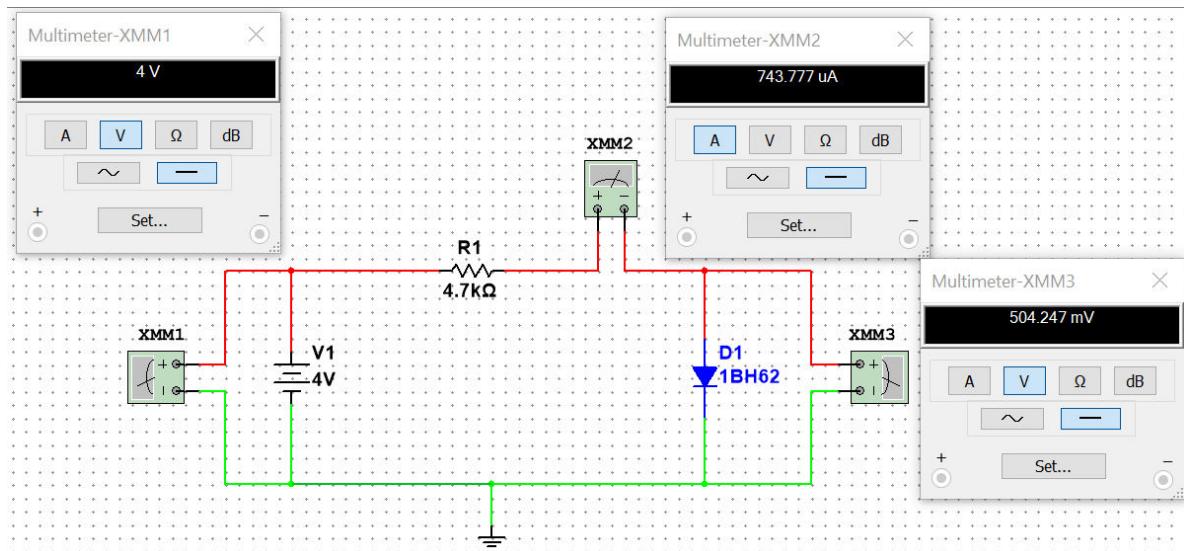


Figure 1.1.11 Correct polarization of a silicon diode with 4 Volt supply voltage in "Multisim"

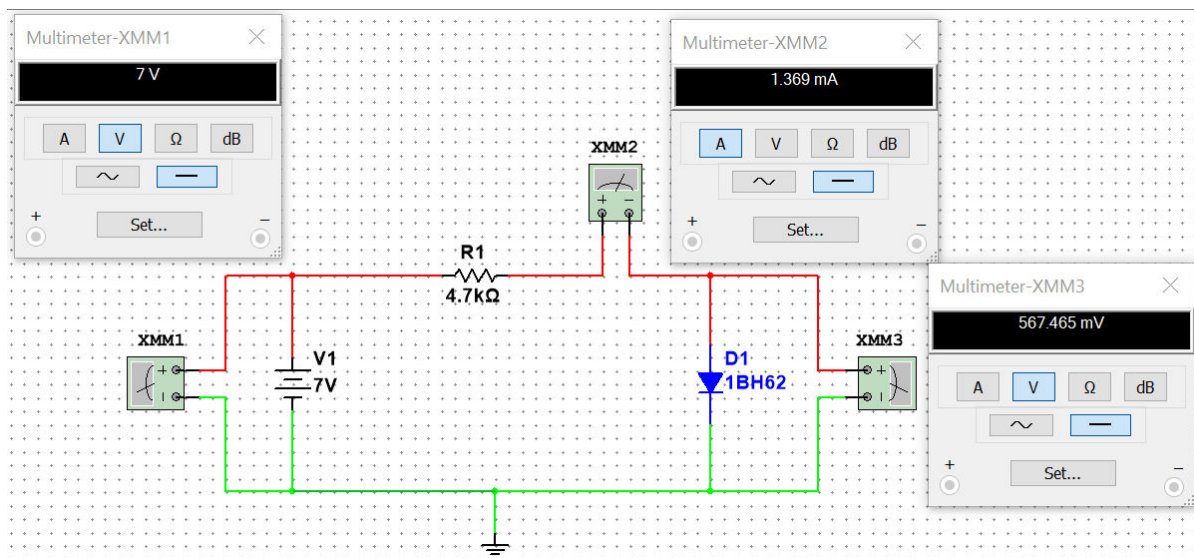


Figure 1.1.12 Correct polarization of a silicon diode with 7 Volt supply voltage in "Multisim"

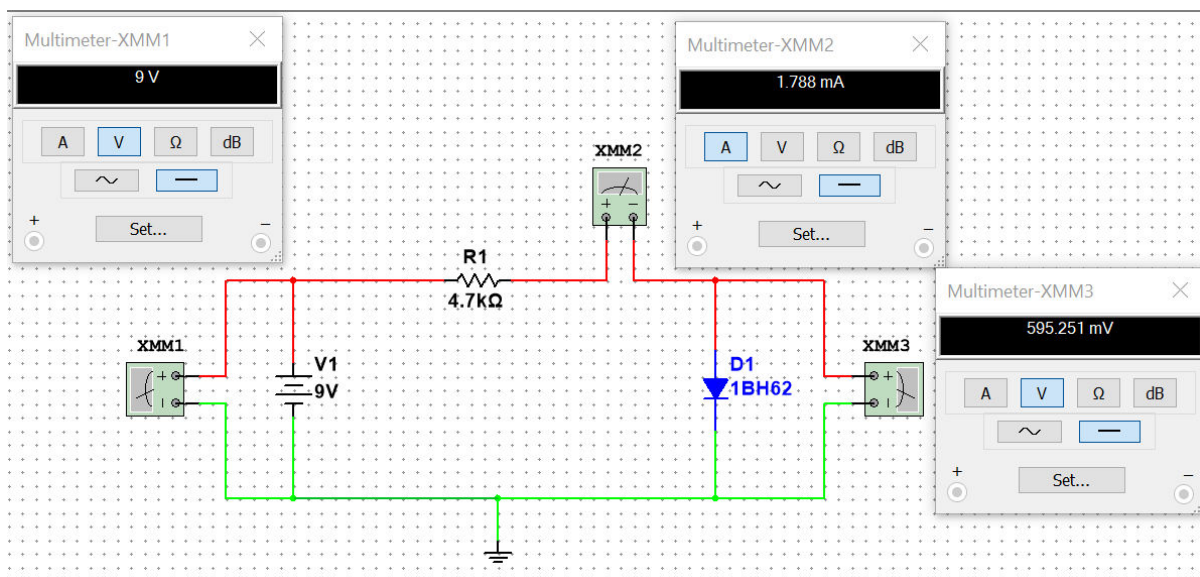


Figure 1.1.13 Correct polarization of a silicon diode with 9 Volt supply voltage in "Multisim"

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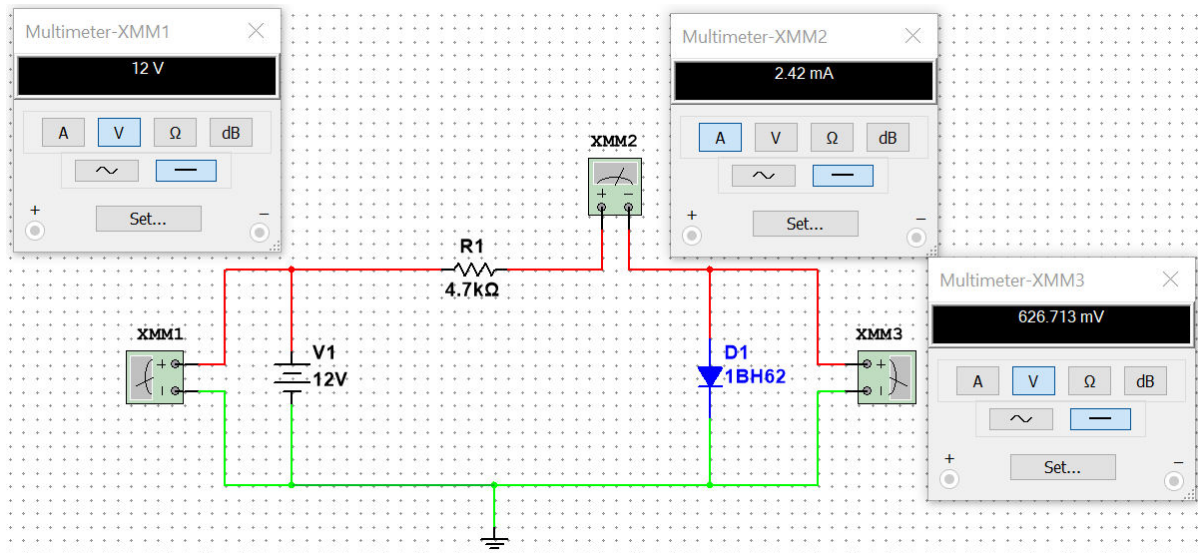


Figure 1.1.14 Correct polarization of a silicon diode with 12 Volt supply voltage in "Multisim"

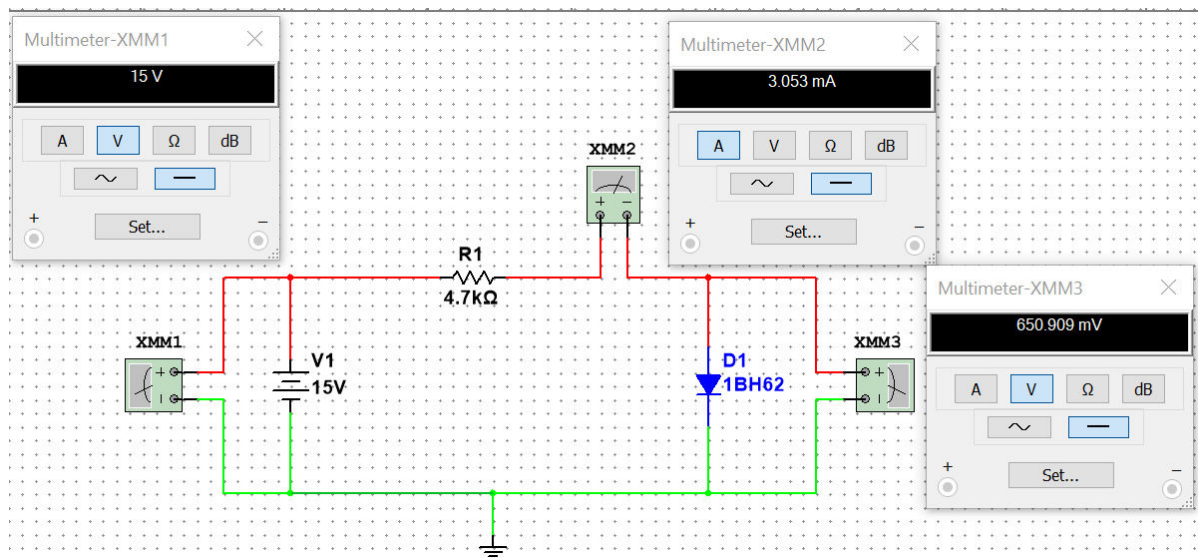


Figure 1.1.15 Correct polarization of a silicon diode with 15 Volt supply voltage in "Multisim"

Experimental Solution

Figure 1.1.16 shows an illustration of the silicon diode polarization on the "breadboard", where it was carried out in the laboratory.

First, we connect the voltage supply (0-15 Volt) and a digital bench top multimeter set to calculate voltage in Volts, in parallel to the ends of the supply in order to calculate the source voltage. Next, we connect the 4.7 KOhm resistor in series with the source, as well as connect the diode in series with the resistor. The diode will be correctly polarized, which means that contact N (grey line) will be biased towards ground, so that the anode voltage (contact P) is greater than the cathode voltage (contact N). We connect the simple digital multimeter set to calculate current intensity in mA, in series with the resistor and before the diode, to calculate the intensity of the current flowing through the diode and thus the whole circuit. Finally, we connect a digital bench multimeter set to calculate voltage in volts, in parallel to the ends of the diode, in order to calculate the potential difference at the ends of the diode. We also use a wire to connect the earth ground.

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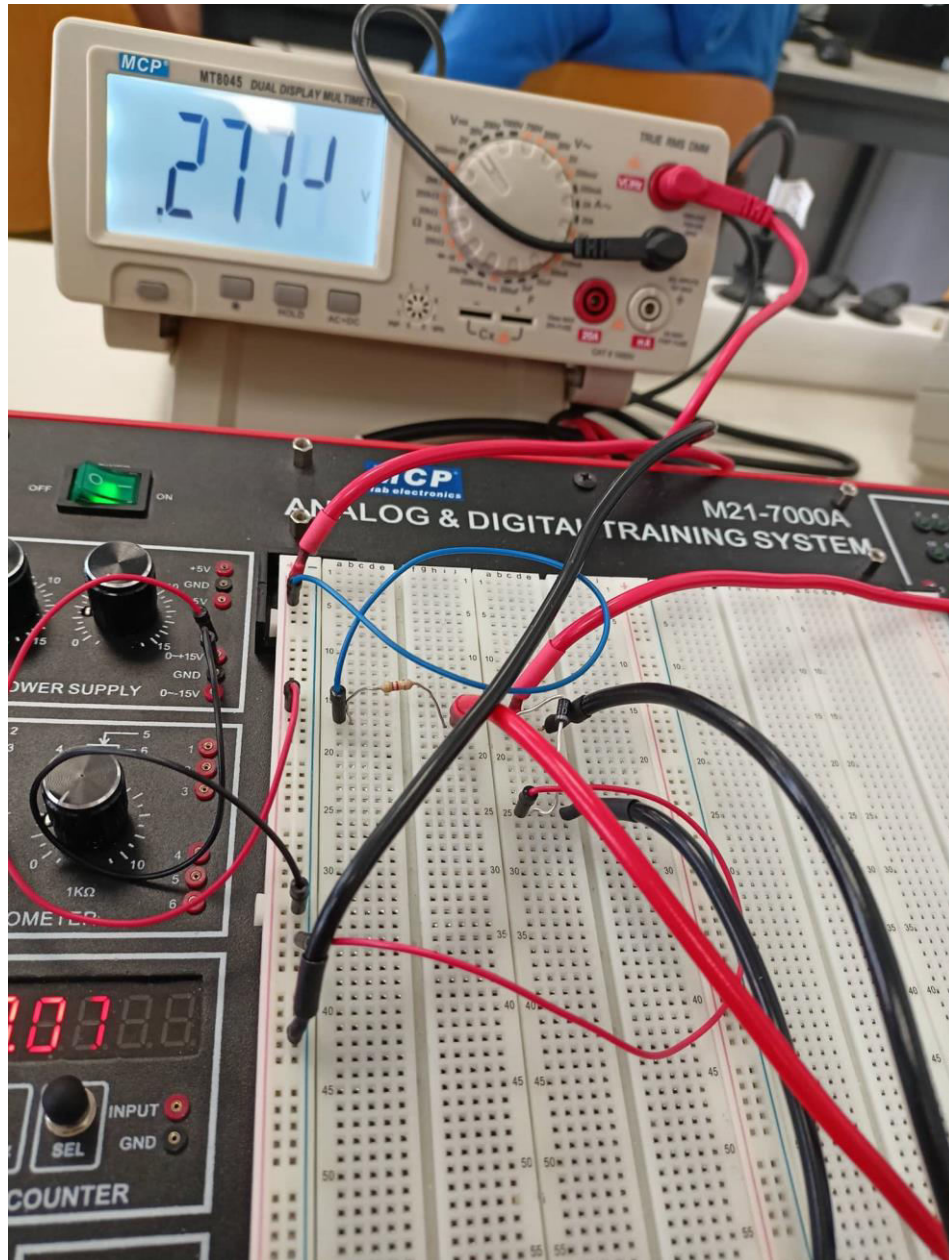


Figure 1.1.16 Experimental illustration of a silicon diode polarization in the "Breadboard"

In Figures 1.1.17, 1.1.18, 1.1.19, 1.1.20, 1.1.21, 1.1.22, 1.1.23, 1.1.24, 1.1.25, 1.1.26 and 1.1.27 the recorded measurements of Table 1 are experimentally verified on the "Breadboard" interface board from the laboratory environment. Absent is the right silicon diode bias with 0 Volt supply voltage on the "Breadboard".

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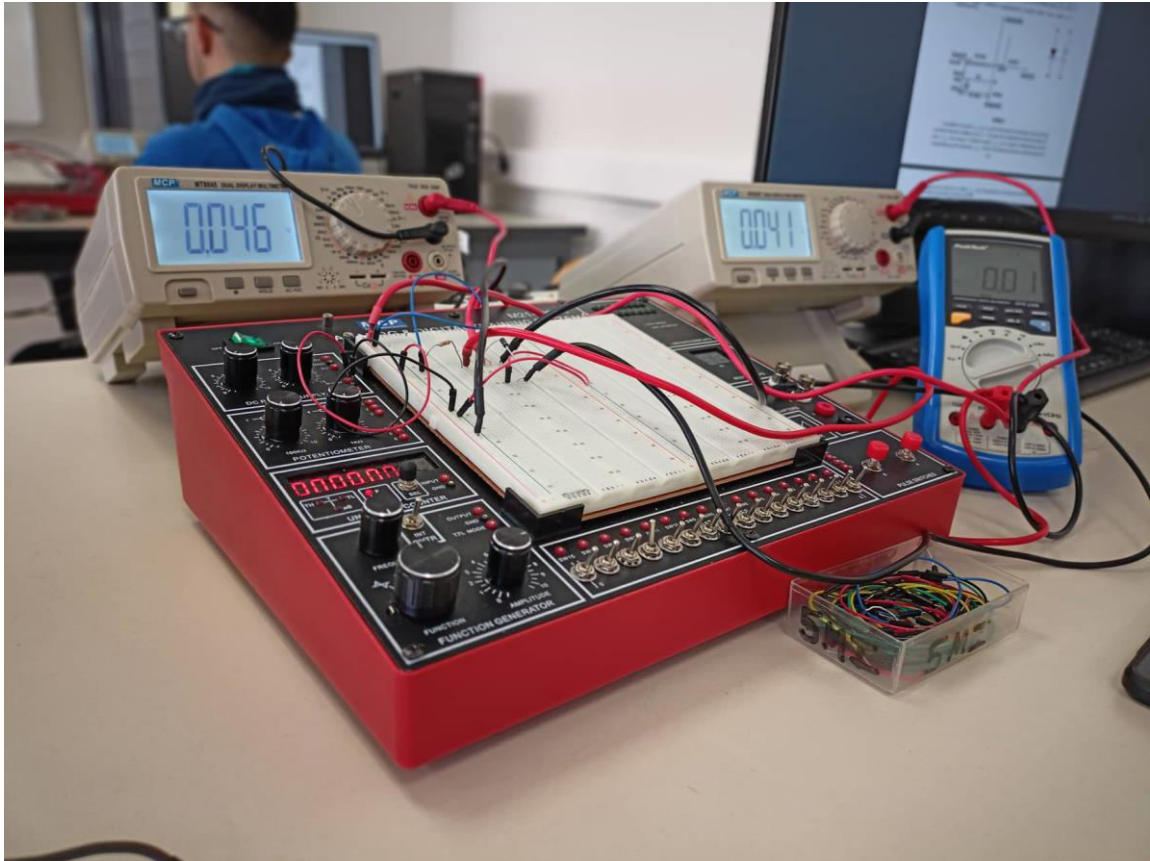


Figure 1.1.17 Correct polarization of a silicon diode with 0.1 Volt supply voltage on the "Breadboard"

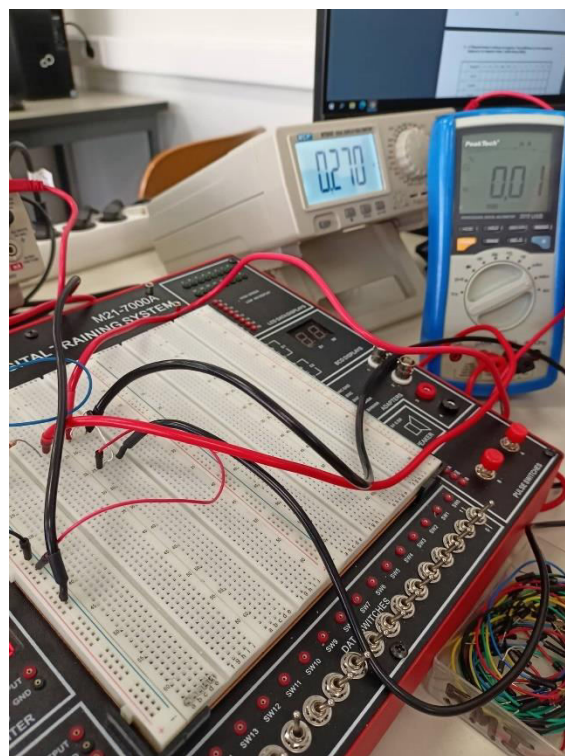


Figure 1.1.18 Correct polarization of a silicon diode with 0.3 Volt supply voltage on the "Breadboard"

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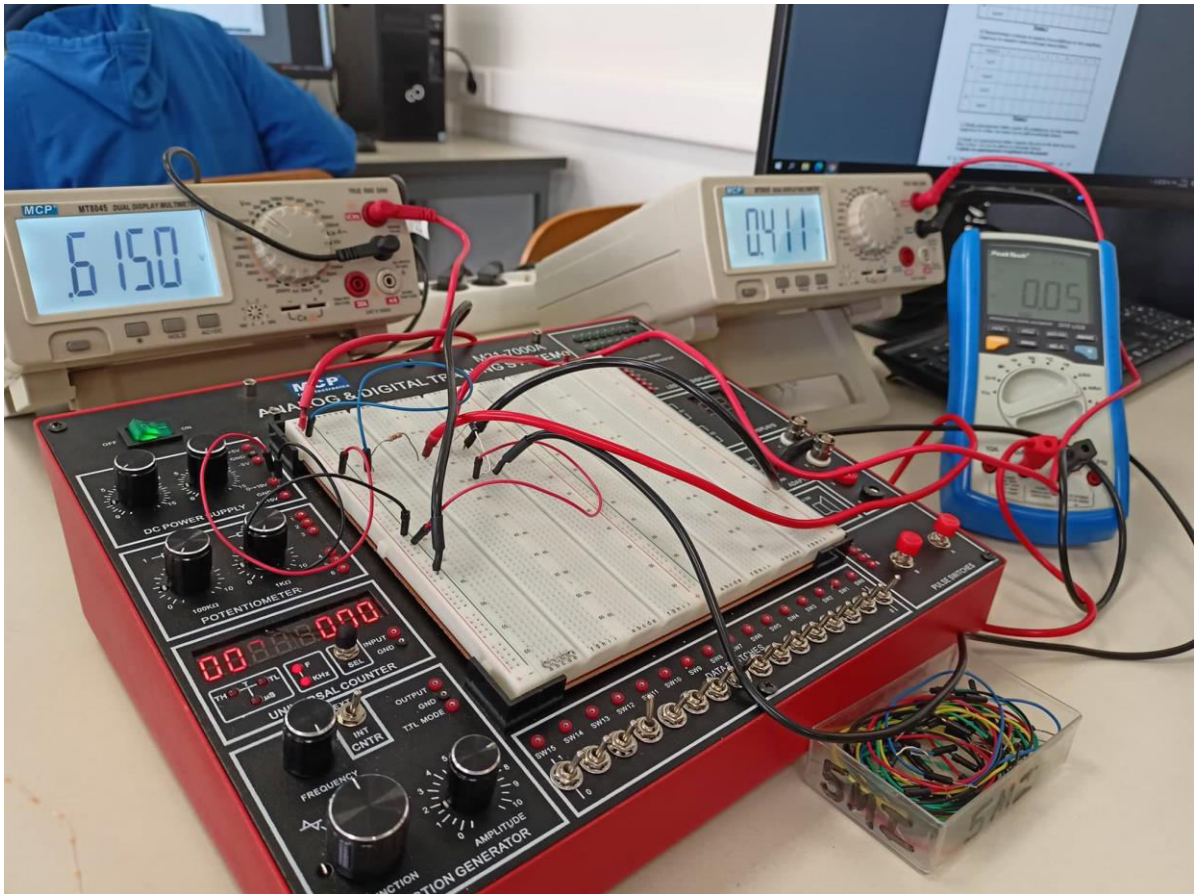
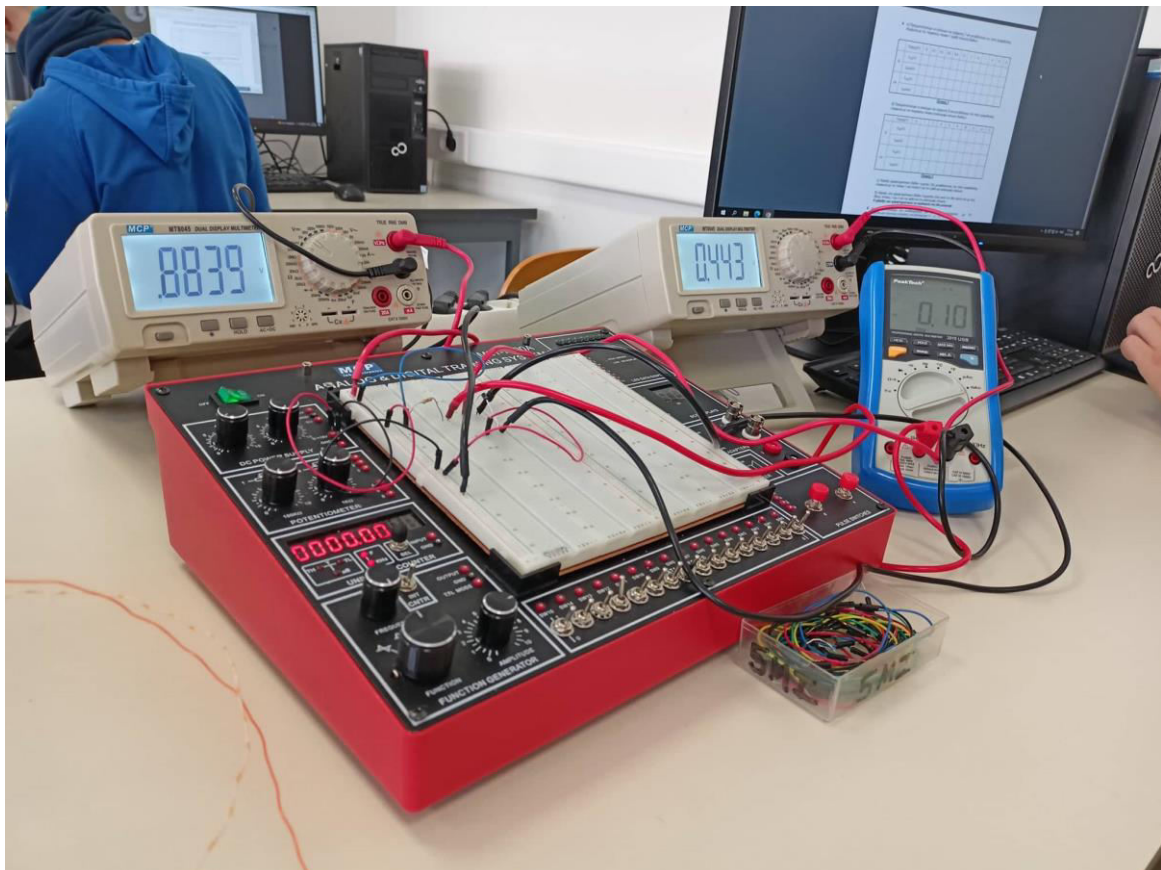


Figure 1.1.19 Correct polarization of silicon diode with 0.6 Volt supply voltage on "Breadboard"



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Figure 1.1.20 Correct polarization of a silicon diode with a supply voltage of 0.9 Volt on the "Breadboard"

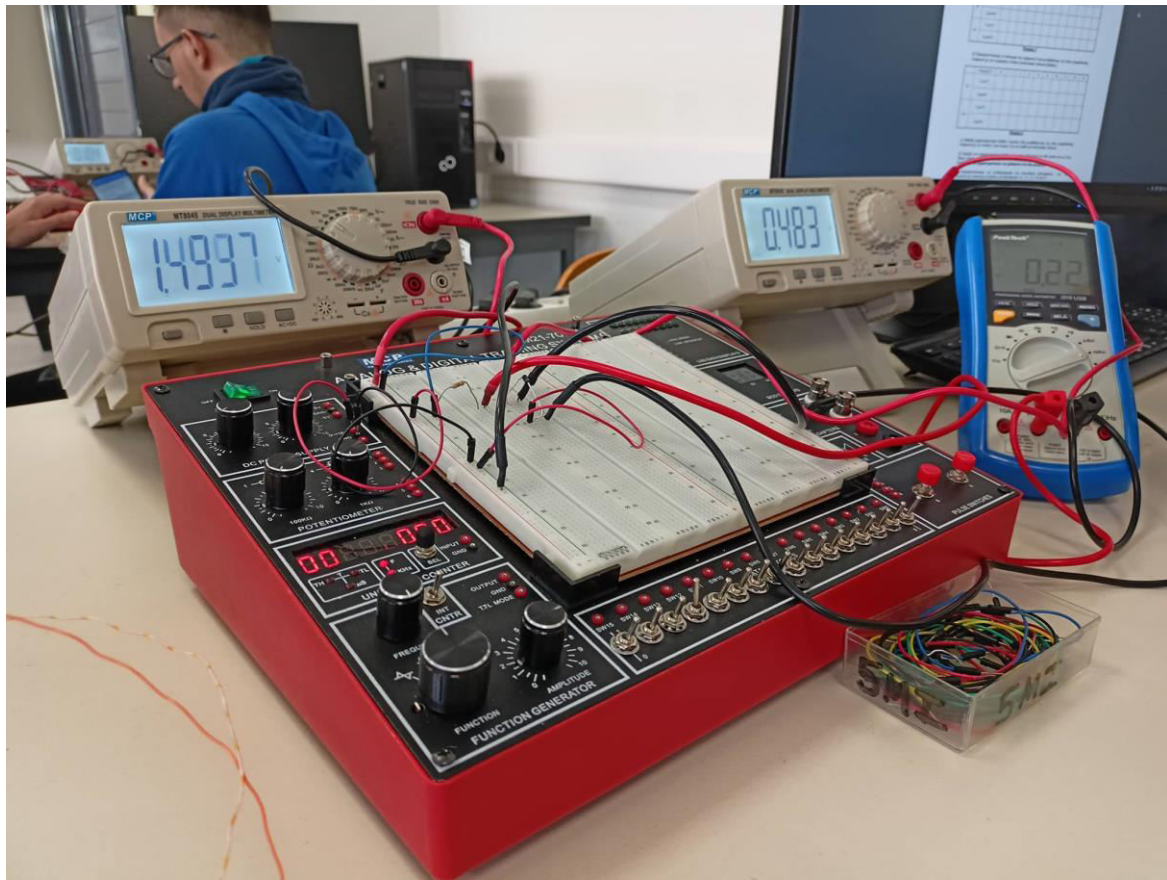


Figure 1.1.1.21 Correct polarization of a silicon diode with a supply voltage of 1.5 Volt on the "Breadboard"

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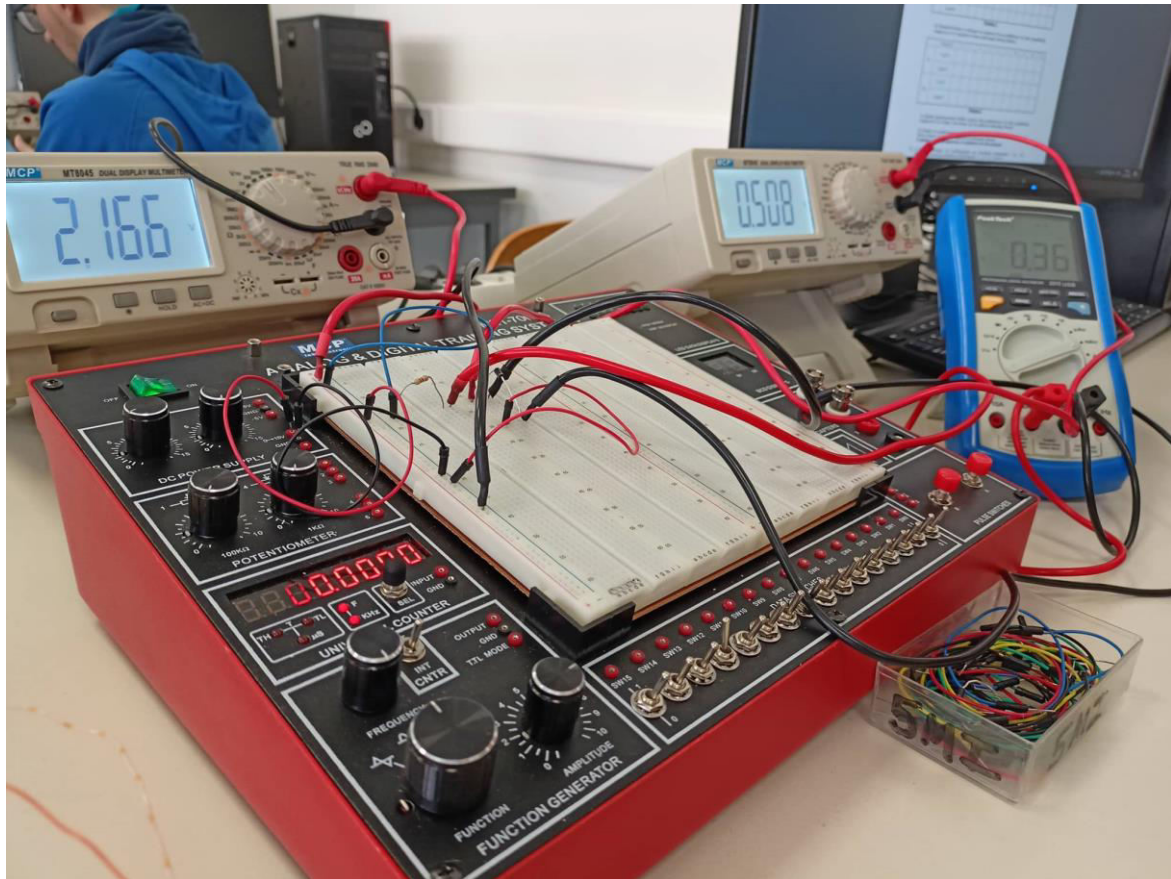


Figure 1.1.22 Correct polarization of a silicon diode with 2 Volt supply voltage on the "Breadboard"

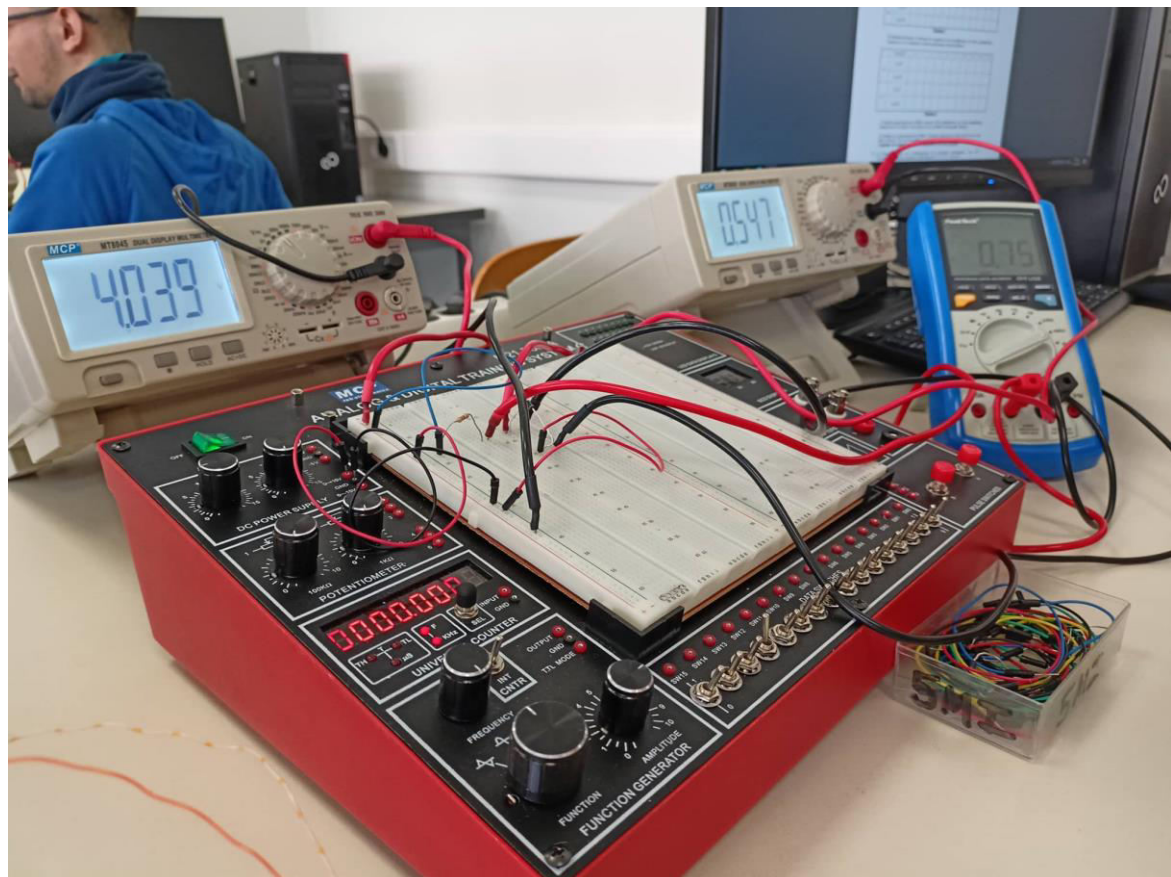


Figure 1.1.23 Correct polarization of a silicon diode with 4 Volt supply voltage on the "Breadboard"

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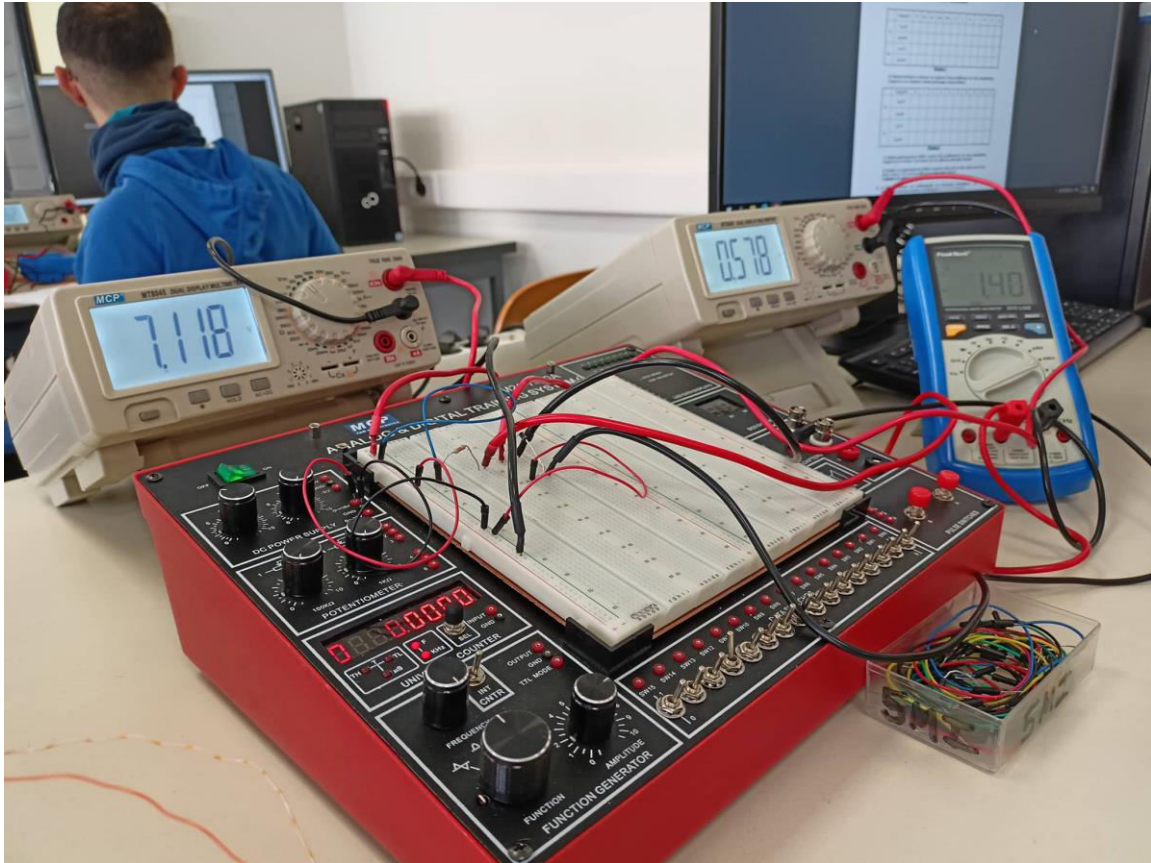


Figure 1.1.24 Correct polarization of a silicon diode with 7 Volt supply voltage on the "Breadboard"

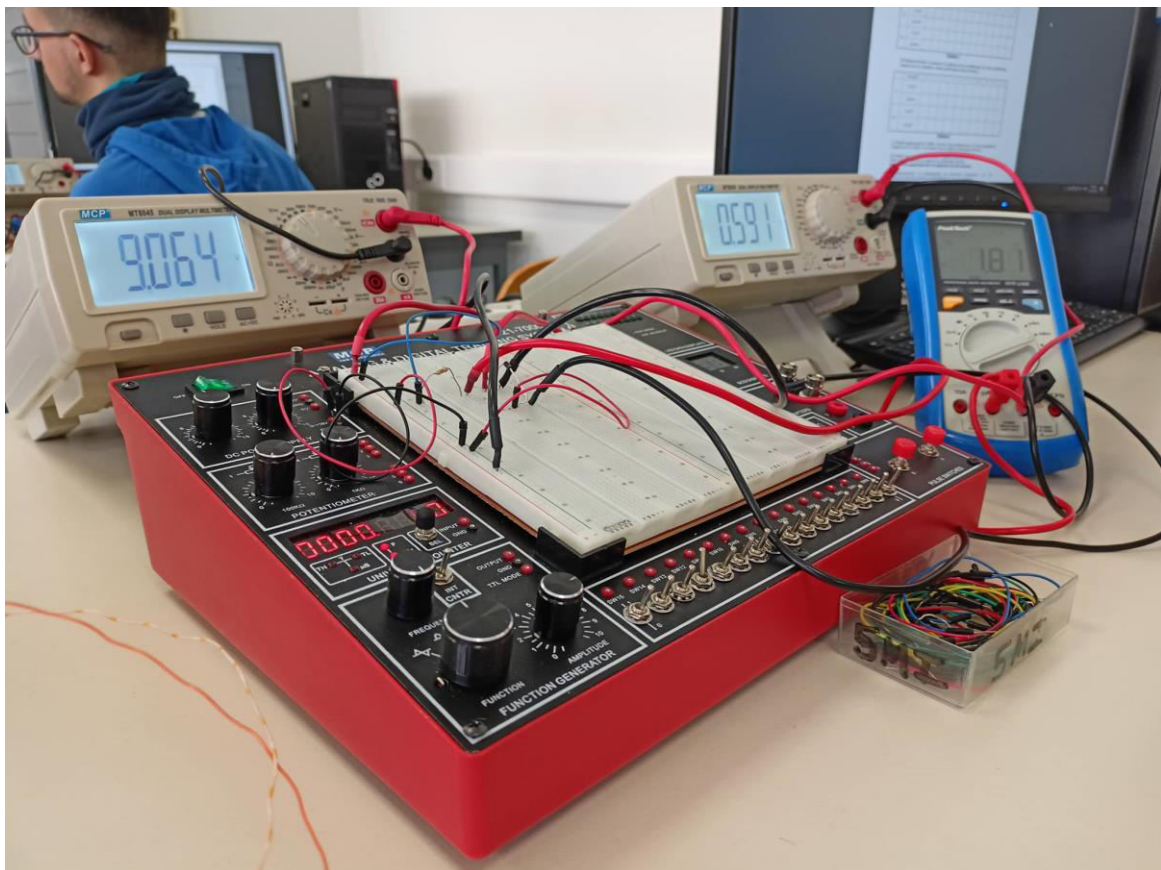


Figure 1.1.25 Correct polarization of a silicon diode with 9 Volt supply voltage on the "Breadboard"

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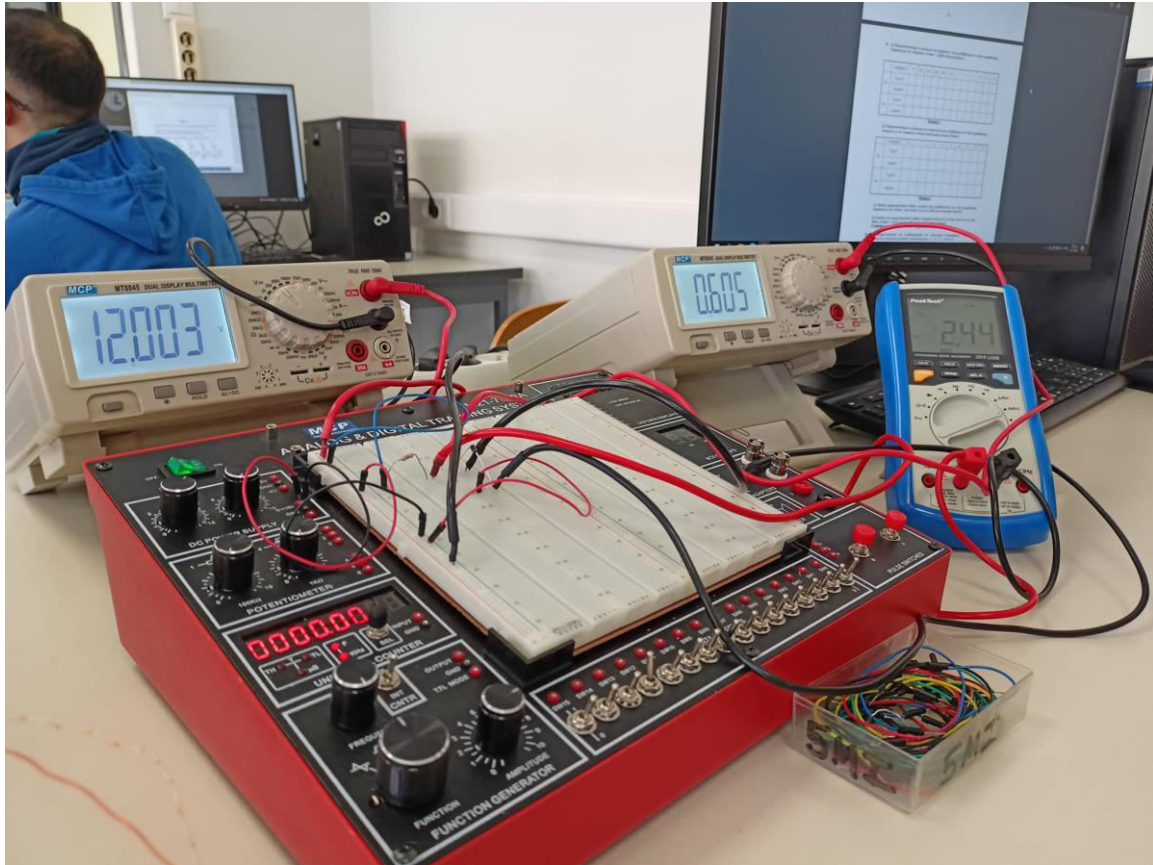
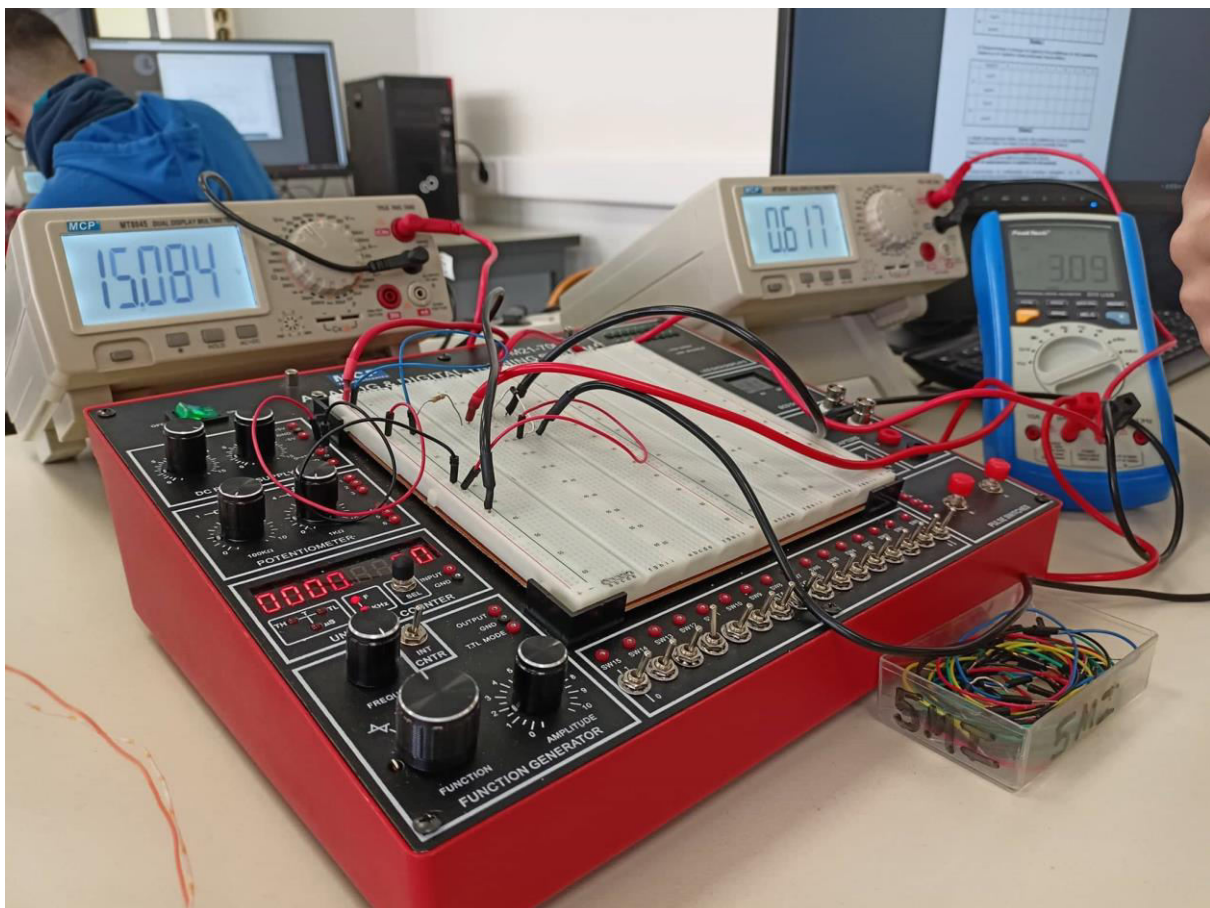


Figure 1.1.26 Correct polarization of a silicon diode with 12 Volt supply voltage on the "Breadboard"



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Figure 1.1.27 Correct polarization of a silicon diode with 15 Volt supply voltage on the "Breadboard"

1.2 INVERSE BIAS

General

The chapters "Simulation Solution" and "Experimental Solution" present in detail the components used, the recorded measurements and the observations made during the experiment with snapshots from the simulation software "Multisim" and photographs of the experiment from the laboratory environment respectively, while the chapter "Theoretical Solution" analyses the mathematical proof of the experiment.

Theoretical Solving

Figure 1.2.1 shows the schematic illustration of the inverted polarization silicon diode from the design software "MS Visio".

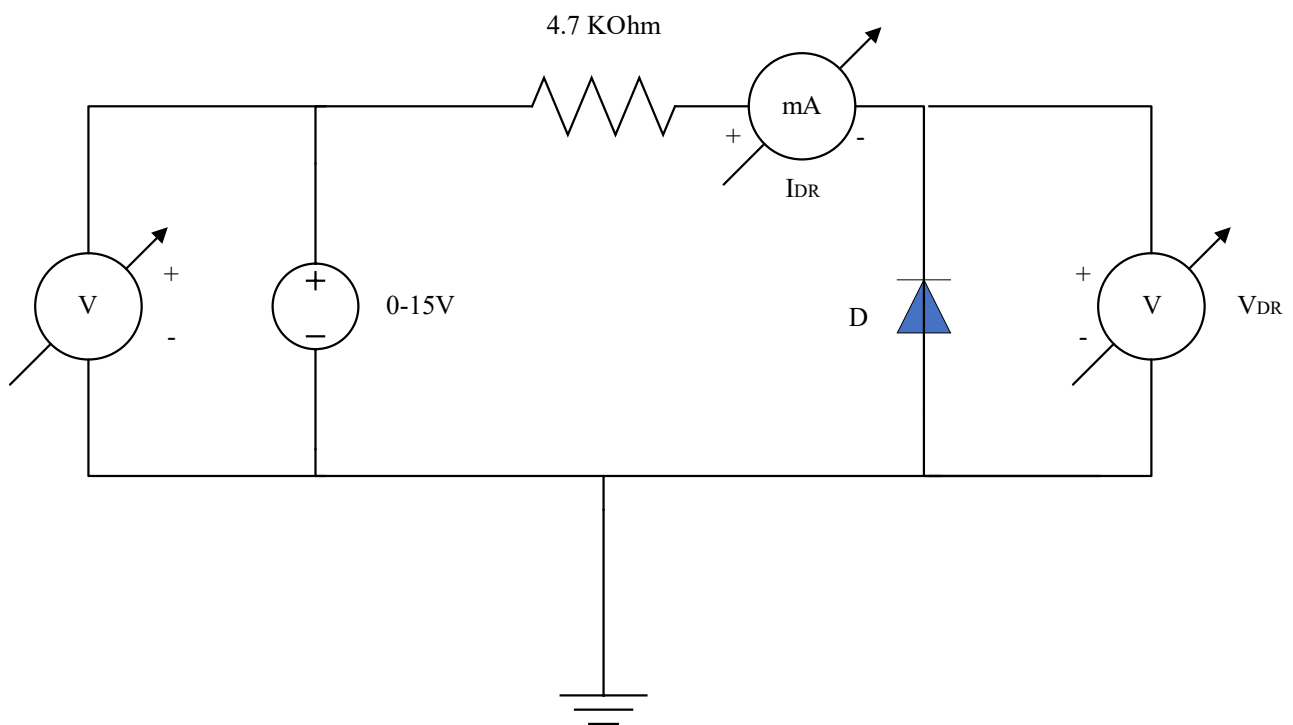


Figure 1.2.1 Schematic illustration of silicon diode reverse polarization in "MS Visio"

Table 2 shows the measurements made in the silicon diode reverse polarization experiment, in terms of the current " I_{DR} " flowing through the diode and the potential difference " V_{DR} " between the ends of the diode.

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The measurements were made for different source supply values and a resistor with a resistance of 4,7 KOhm is connected to the circuit.

Table 2.

	Spring (V)	0	1	2	4	6	8	10	12	14	15
Si	V _{DR} (V)	0	0.97	1.97	3.97	5.97	7.97	9.97	11.97	13.97	14.97
	I _{DR} (μA)	0	6	6	6	6	6	6	6	6	6

The observations recorded according to the data in Table 2 are as follows:

- The current "I_{DR}" flowing through the silicon diode remains approximately constant whenever the voltage at the ends of the source increases.
- The potential difference "V_{DR}" at the ends of the silicon diode remains approximately the same (due to the voltage drop from the resistor) as the supply voltage whenever the source voltage increases.

According to Kirchhoff's 2^o rule (law of voltages), the sum of the potential differences in the individual branches of a loop is equal to zero. By Ohm's law, it is true that the potential difference at the ends of a branch is proportional to the intensity of the current flowing through the branch. The measurements recorded in Table 2 are verified from the above laws. For example at 8 Volt source we have:

$$V_{\text{source}} + V_R + V_{DR} = 0 \rightarrow 8 \text{ Volt} + (I_{DF} \times R) - 8 \text{ Volt} = 0 \rightarrow$$

$$8 \text{ Volt} + (0.000006 \text{ Ampere} \times 150 \text{ Ohm}) - 8 \text{ Volt} = 0 \rightarrow$$

$$8 \text{ Volt} + 0.0009 \text{ Volt} - 8 \text{ Volt} = 0 \rightarrow 0 \cong 0$$

The voltage at the ends of the silicon diode remains approximately the same and the current flowing through it remains approximately constant. This is due to the fact that the diode is inversely polarized and therefore does not generate any current. Therefore, the internal resistance is very high which accounts for the very low current flowing through the diode (Ohm's law). The diode behaves as an insulator. Figure 1.2.2 shows the silicon diode characteristic during reverse biasing, varying the supply voltage according to Table 2.

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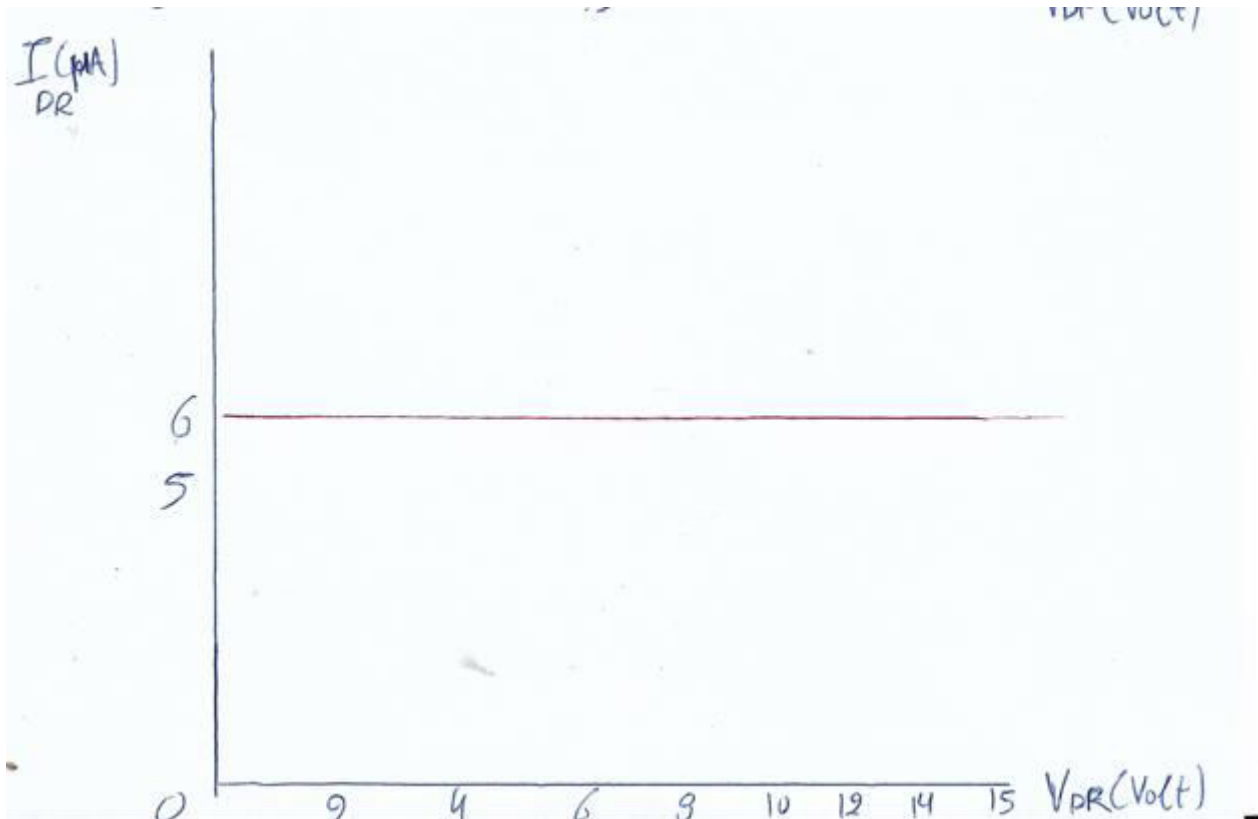


Figure 1.2.2 Reverse bias characteristic of a silicon diode $I_{DR} - V_{DR}$

Simulative Resolution

Figure 1.2.3 shows the simulation of the inverted polarization of a silicon diode in the simulation software "Multisim".

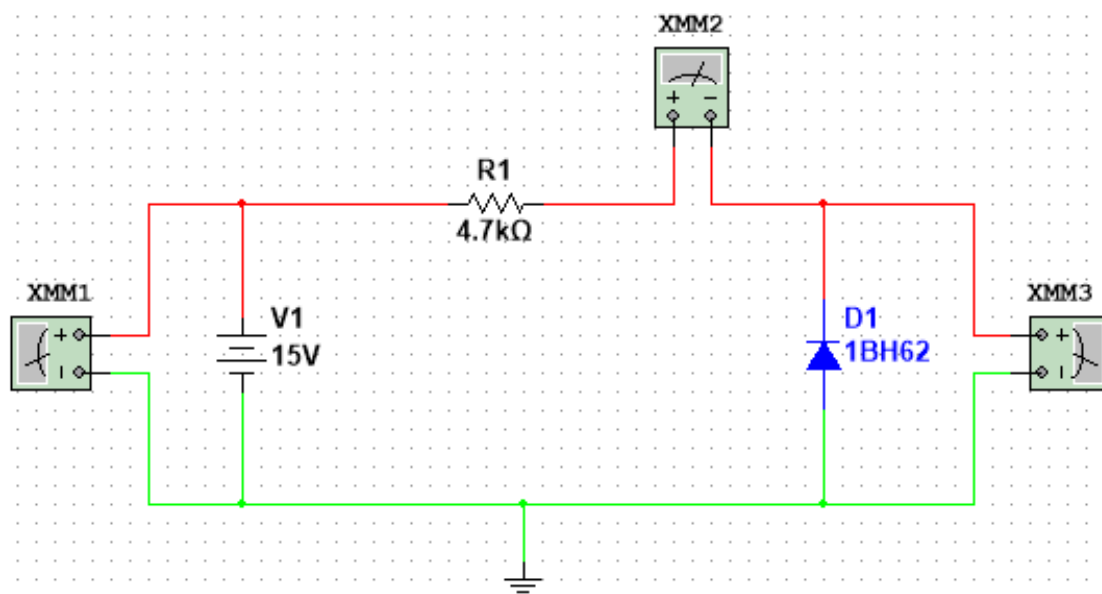


Figure 1.2.3 Simulation of silicon diode reverse bias in "Multisim"

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In Figures 1.2.4, 1.2.5, 1.2.6, 1.2.7, 1.2.8, 1.2.9, 1.2.10, 1.2.11, 1.2.12 and 1.2.13 the recorded measurements of Table 2 are simulated in the simulation software "Multisim".

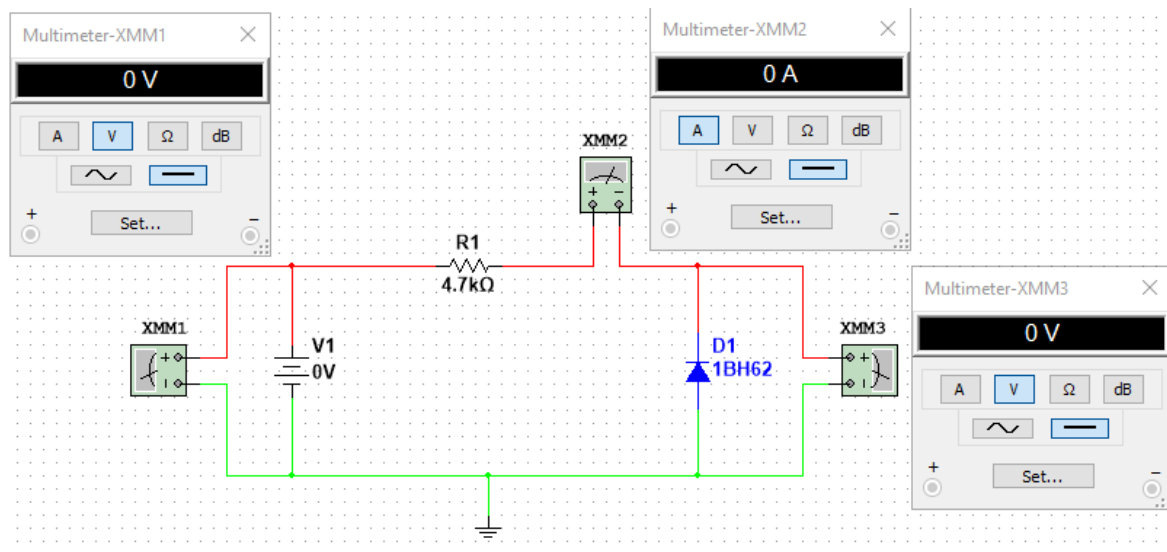


Figure 1.2.4 Inverted polarization of a silicon diode with 0 Volt supply voltage in "Multisim"

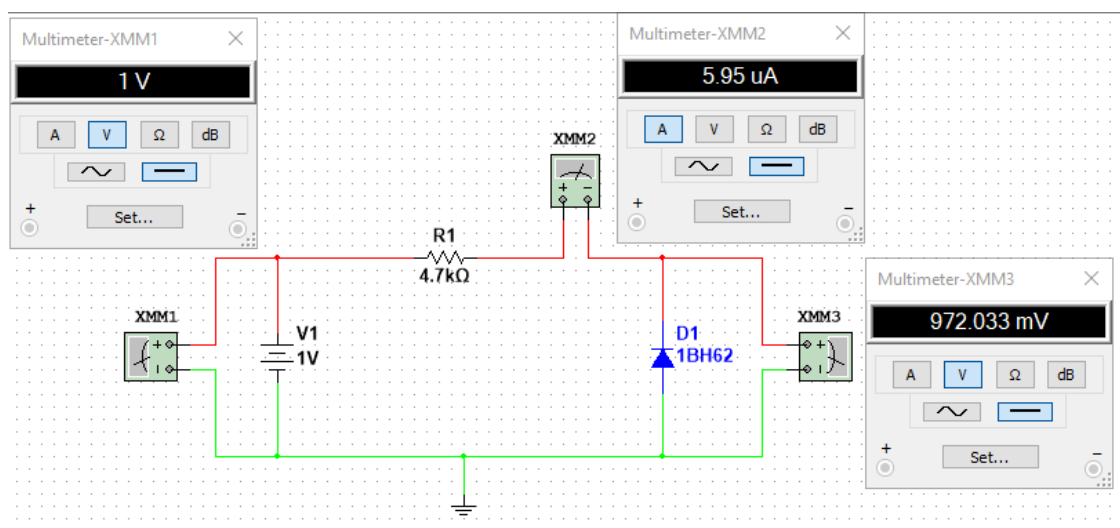


Figure 1.2.5 Inverted polarization of a silicon diode with 1 Volt supply voltage in "Multisim"

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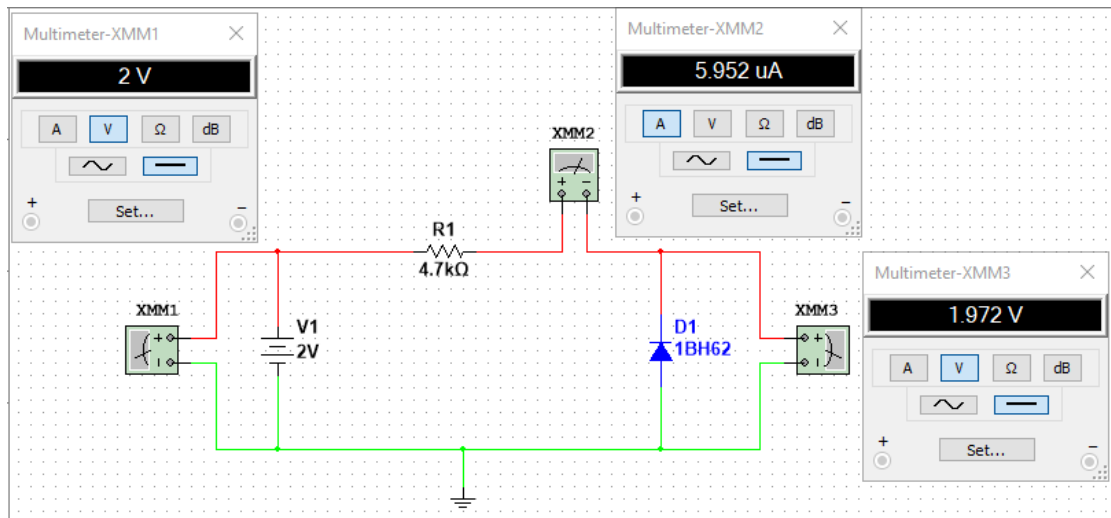


Figure 1.2.6 Inverted polarization of a silicon diode with 2 Volt supply voltage in "Multisim"

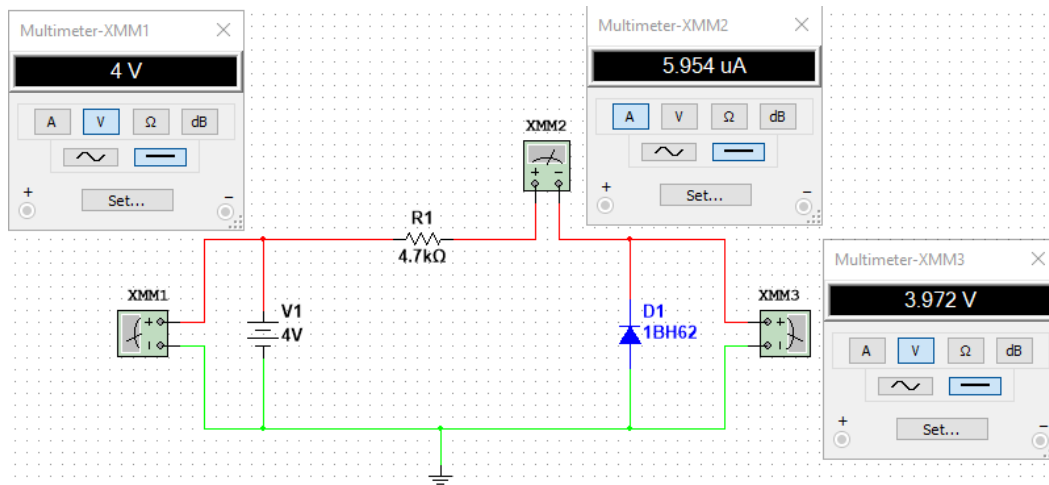


Figure 1.2.7 Inverted polarization of a silicon diode with 4 Volt supply voltage in "Multisim"

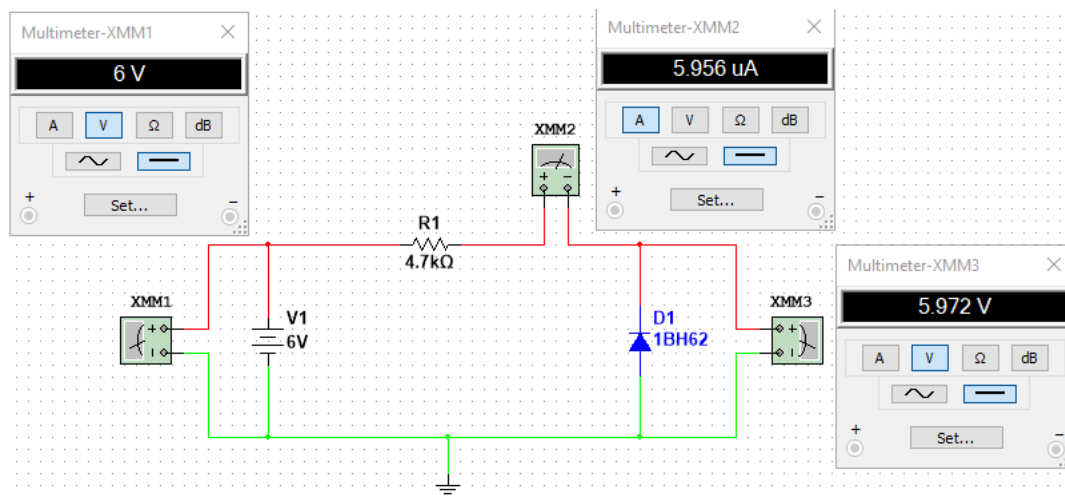


Figure 1.2.8 Inverted polarization of a silicon diode with 6 Volt supply voltage in "Multisim"

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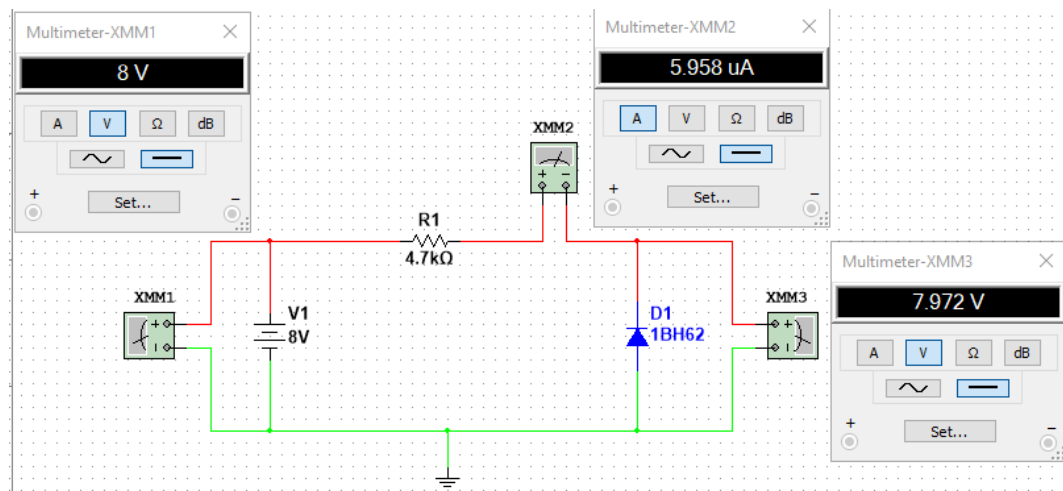


Figure 1.2.9 Inverted polarization of a silicon diode with 8 Volt supply voltage in "Multisim"

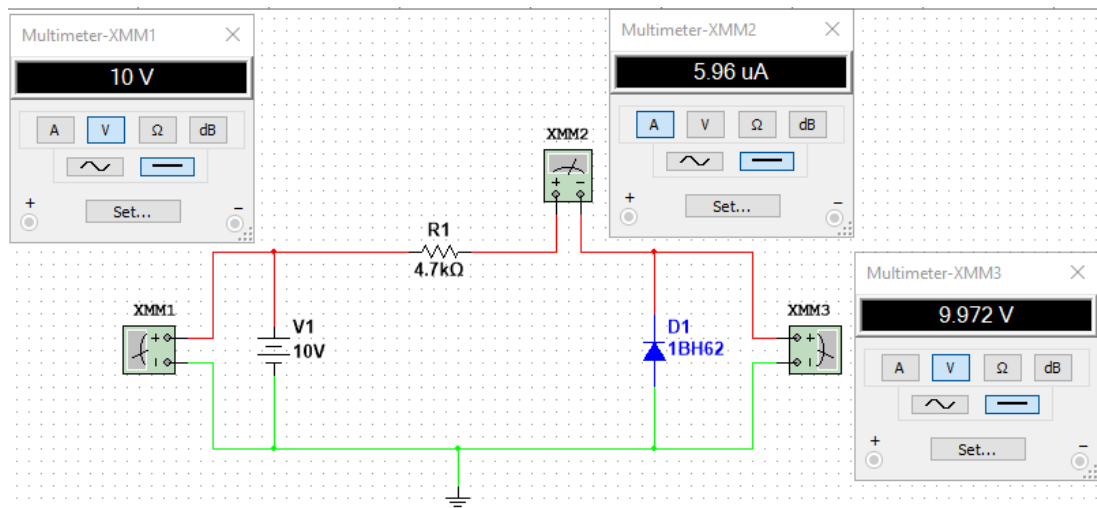


Figure 1.2.10 Inverted polarization of a silicon diode with 10 Volt supply voltage in "Multisim"

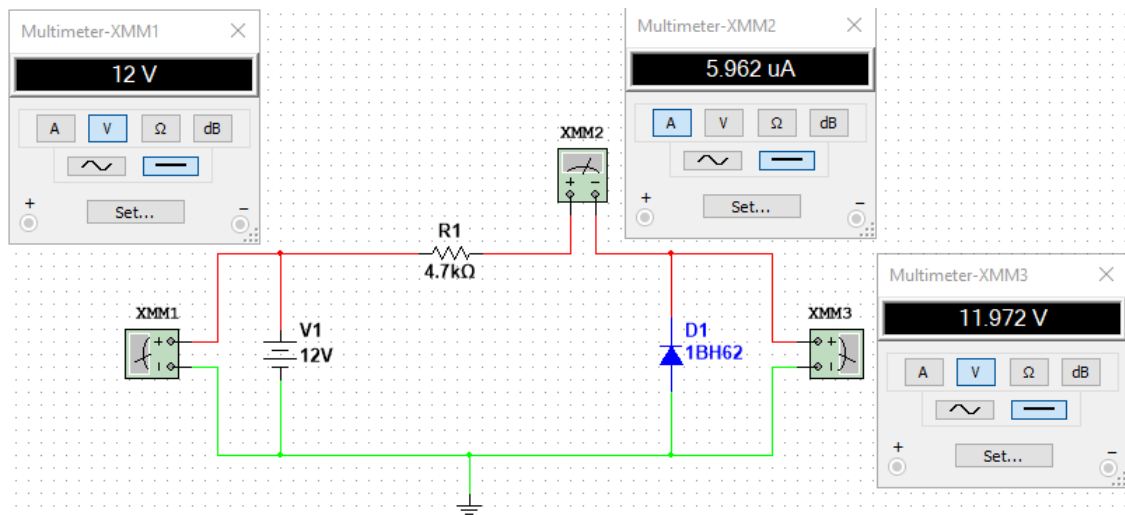


Figure 1.2.11 Inverted polarization of a silicon diode with 12 Volt supply voltage in "Multisim"

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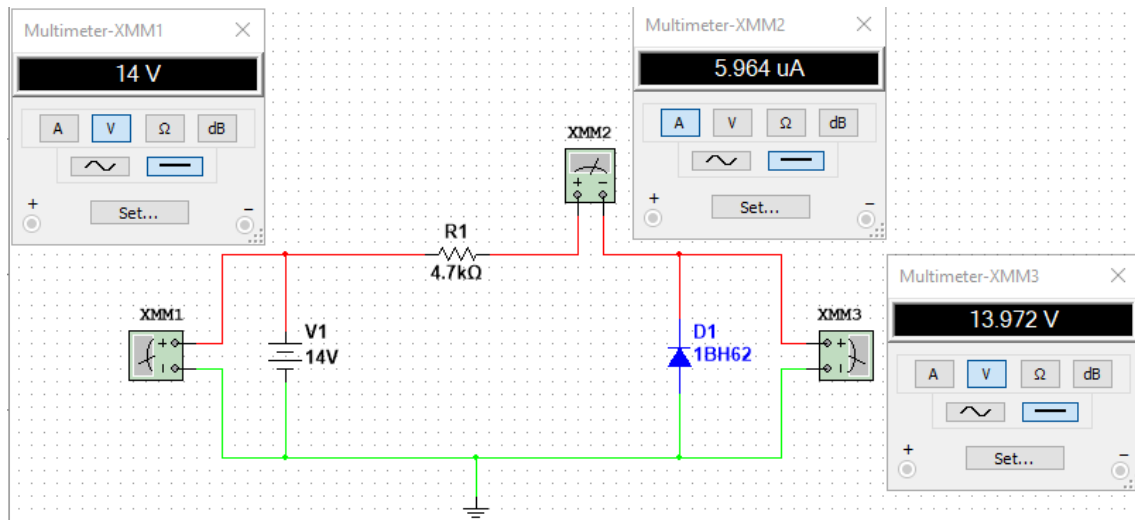


Figure 1.2.12 Inverted polarization of a silicon diode with 14 Volt supply voltage in "Multisim"

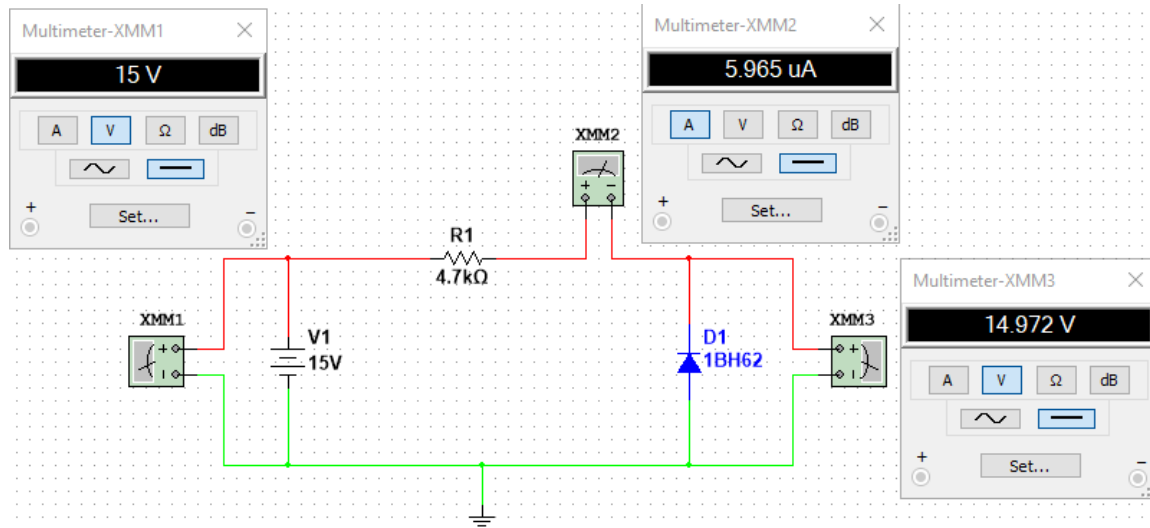


Figure 1.2.13 Inverted polarization of a silicon diode with 15 Volt supply voltage in "Multisim"

Experimental Solution

Figure 1.2.14 shows the illustration of the silicon diode reverse polarization on the "breadboard", where it was carried out in the laboratory.

First, we connect the voltage supply (0-15 Volt) and a digital bench top multimeter set to calculate voltage in Volts, in parallel to the ends of the supply in order to calculate the source voltage. Next, we connect the 4.7 KOhm resistor in series with the source, as well as connect the diode in series with the resistor. The diode will be inversely polarized, which means that the N contact (grey line) will be facing the source, so that the anode voltage (P contact) will be less than the cathode voltage (N contact). We connect the simple digital multimeter set to calculate current intensity in μA, in series with the resistor and before the diode, to calculate the current intensity flowing through the diode and thus the whole circuit. Finally, we connect a digital bench multimeter set to calculate voltage in volts, in parallel to the ends of the diode, in order to calculate the potential difference at the ends of the diode. We also use a wire to connect the earth ground.

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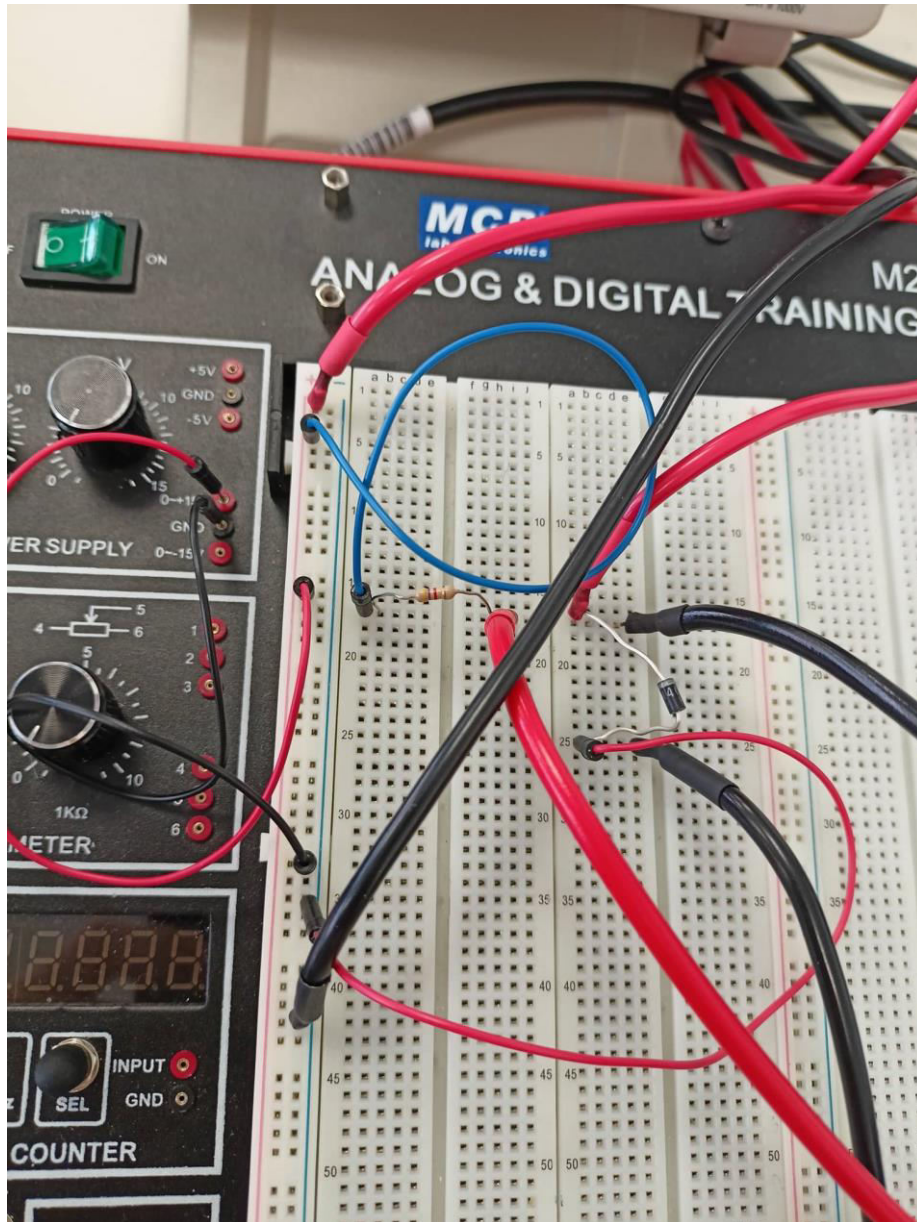


Figure 1.2.14 Experimental illustration of reverse polarization of a silicon diode on the "Breadboard"

In Figures 1.2.15, 1.2.16, 1.2.17, 1.2.18, 1.2.19, 1.2.20, 1.2.21, 1.2.22, 1.2.23 and 1.2.24 the recorded measurements of Table 2 are experimentally verified on the "Breadboard" interface board from the laboratory environment.

ELECTRONICS

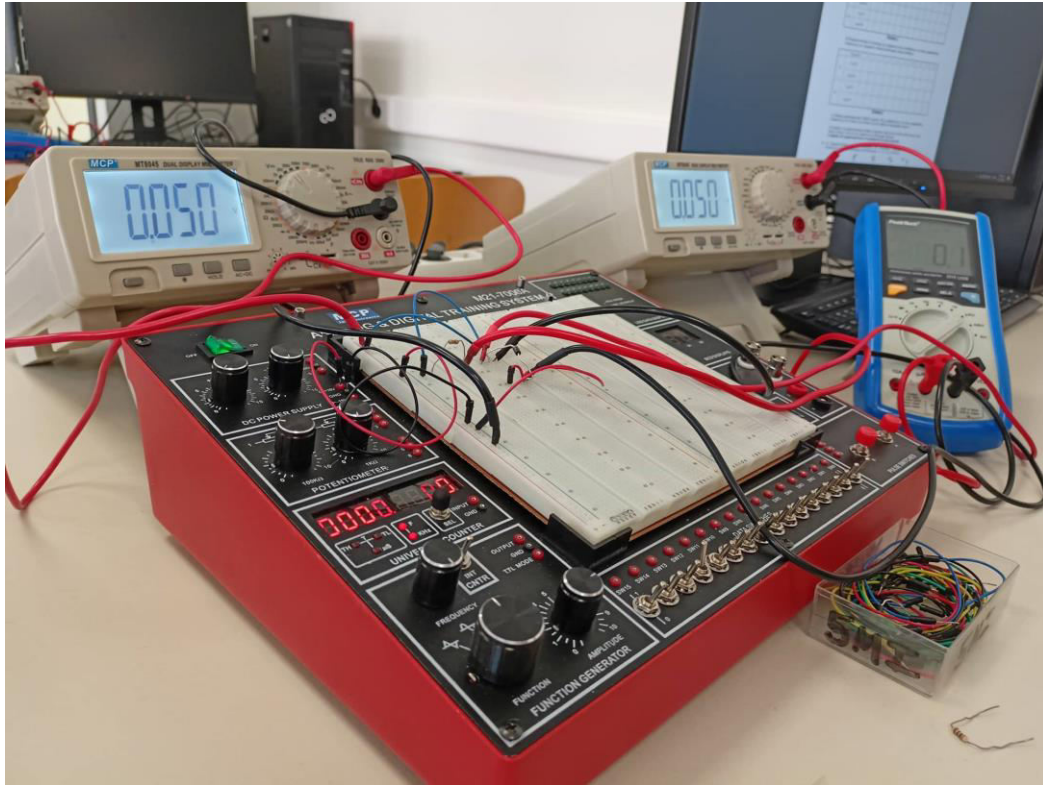


Figure 1.2.15 Inverted polarization of a silicon diode with 0 Volt supply voltage on the "Breadboard"

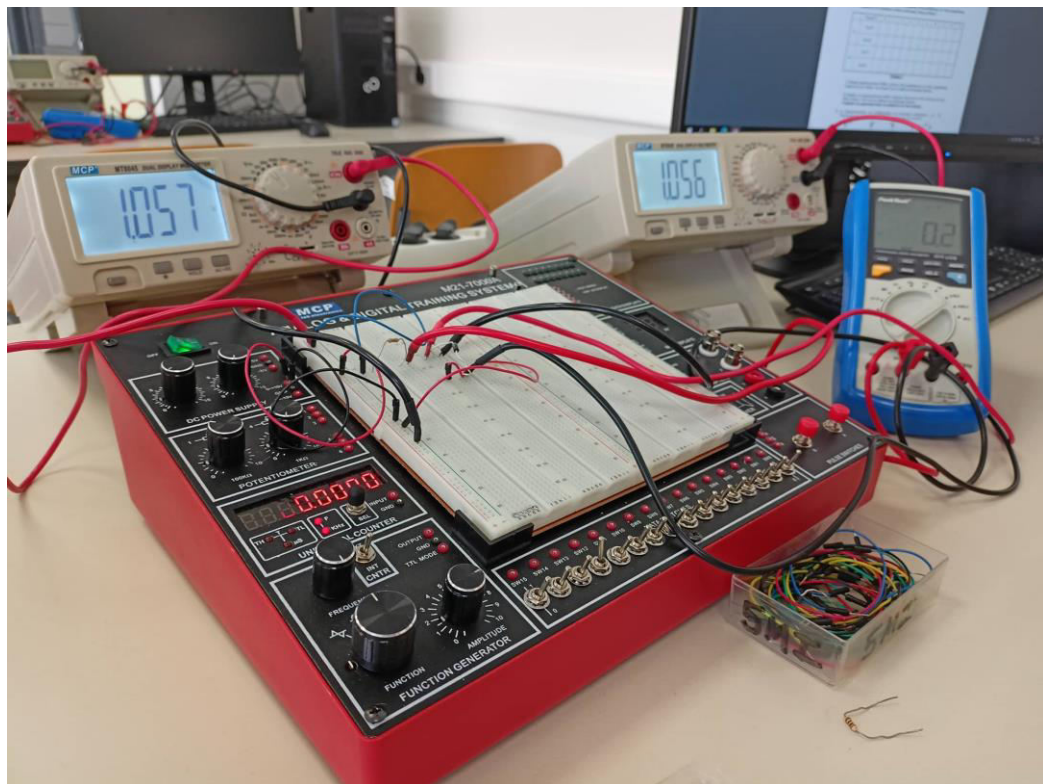


Figure 1.2.16 Inverted polarization of a silicon diode with 1 Volt supply voltage on the "Breadboard"

ELECTRONICS

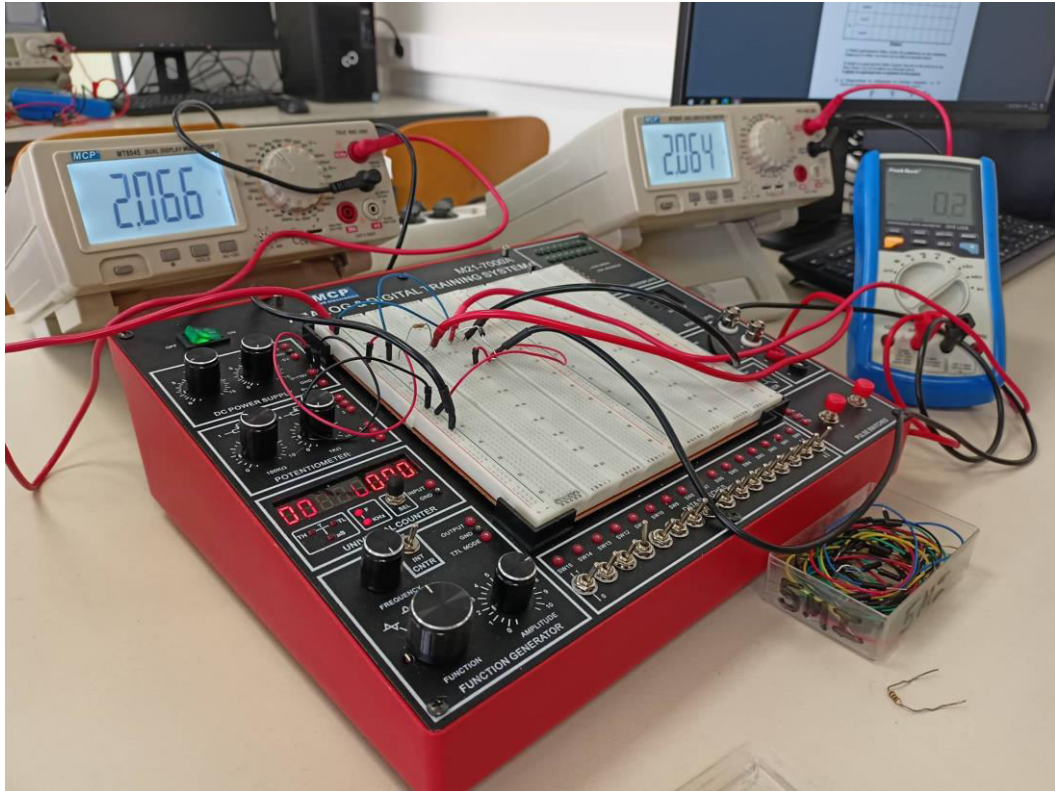


Figure 1.2.17 Inverted polarization of a silicon diode with 2 Volt supply voltage on the "Breadboard"

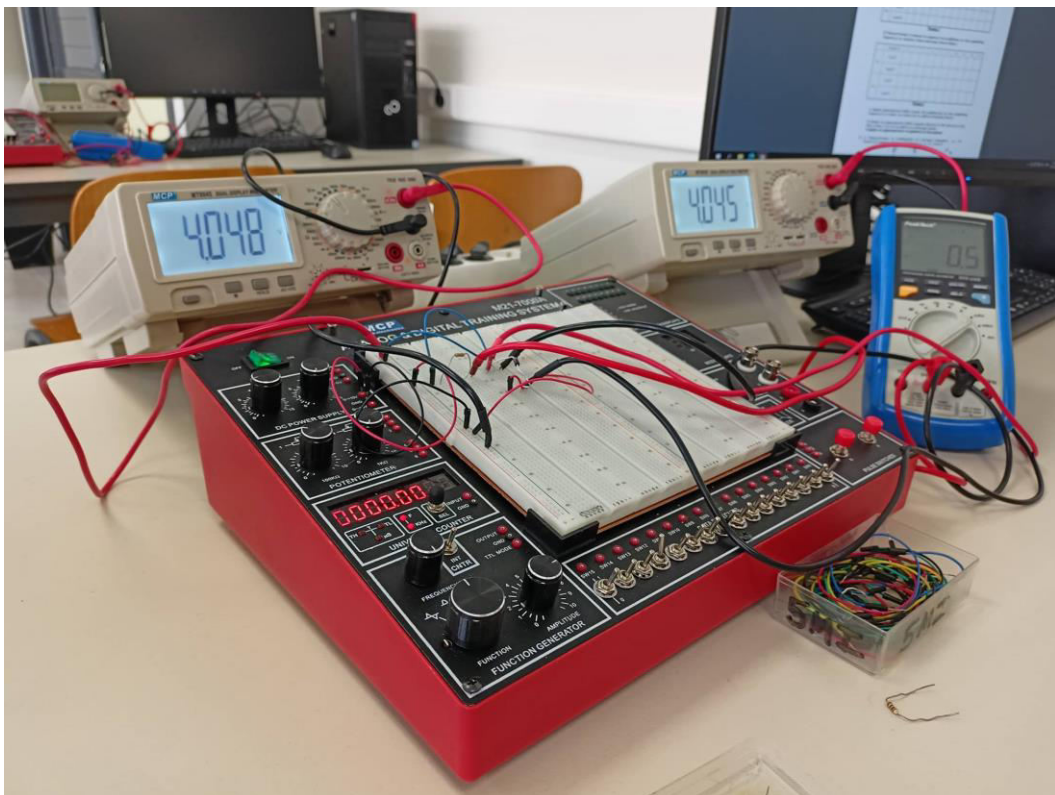


Figure 1.2.18 Inverted polarization of a silicon diode with 4 Volt supply voltage on the "Breadboard"

ELECTRONICS

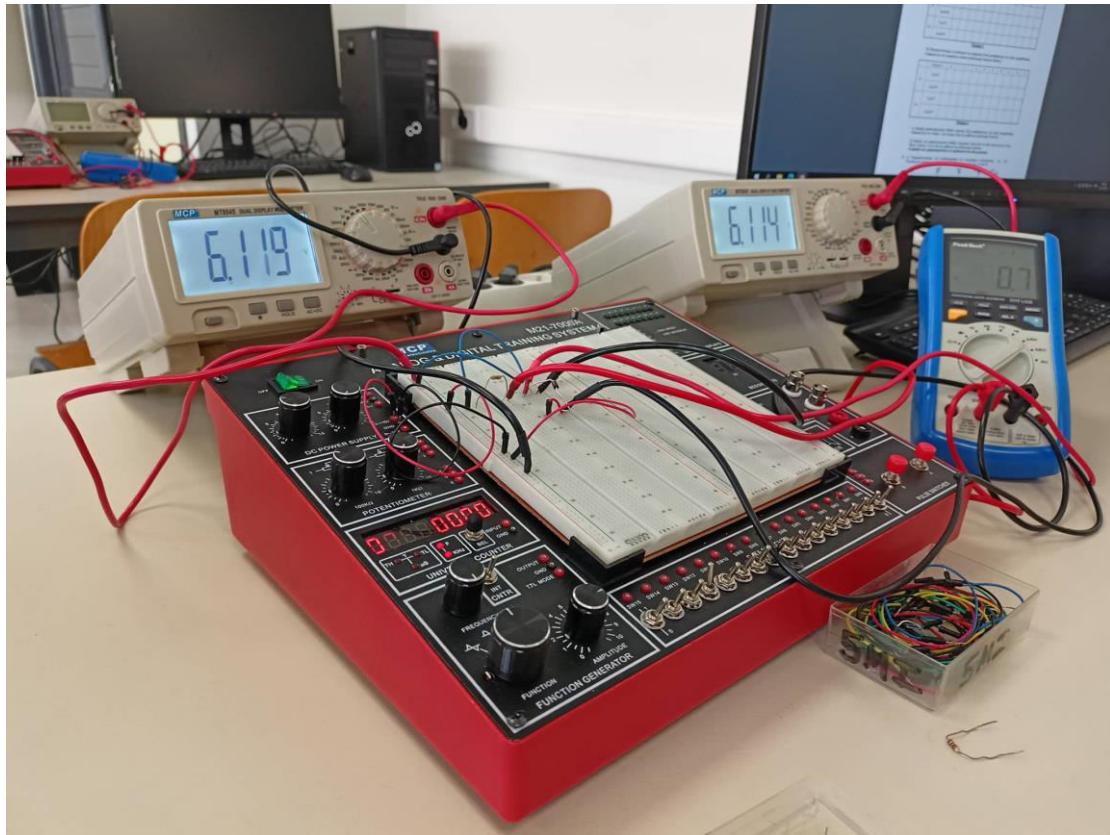
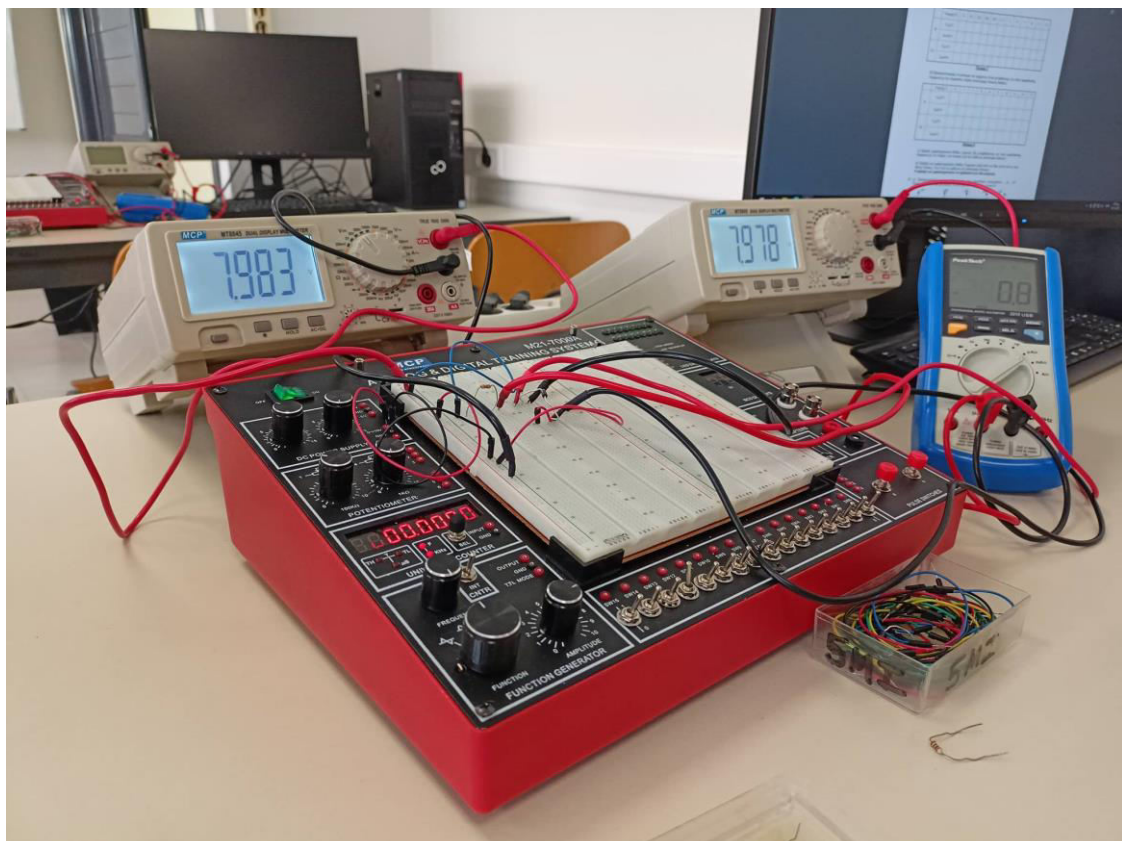


Figure 1.2.19 Inverted polarization of a silicon diode with 6 Volt supply voltage on the "Breadboard"



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Figure 1.2.20 Inverted polarization of a silicon diode with 8 Volt supply voltage on the "Breadboard"

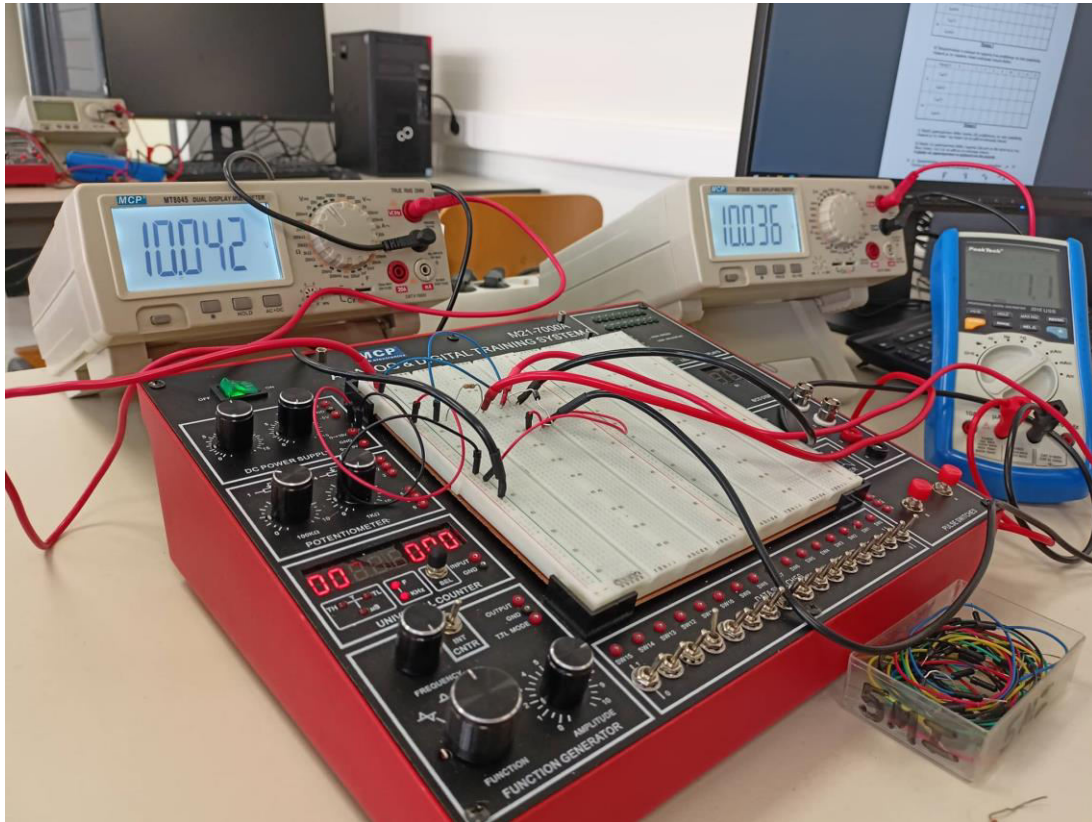


Figure 1.2.21 Inverted polarization of a silicon diode with 10 Volt supply voltage on the "Breadboard"

ELECTRONICS

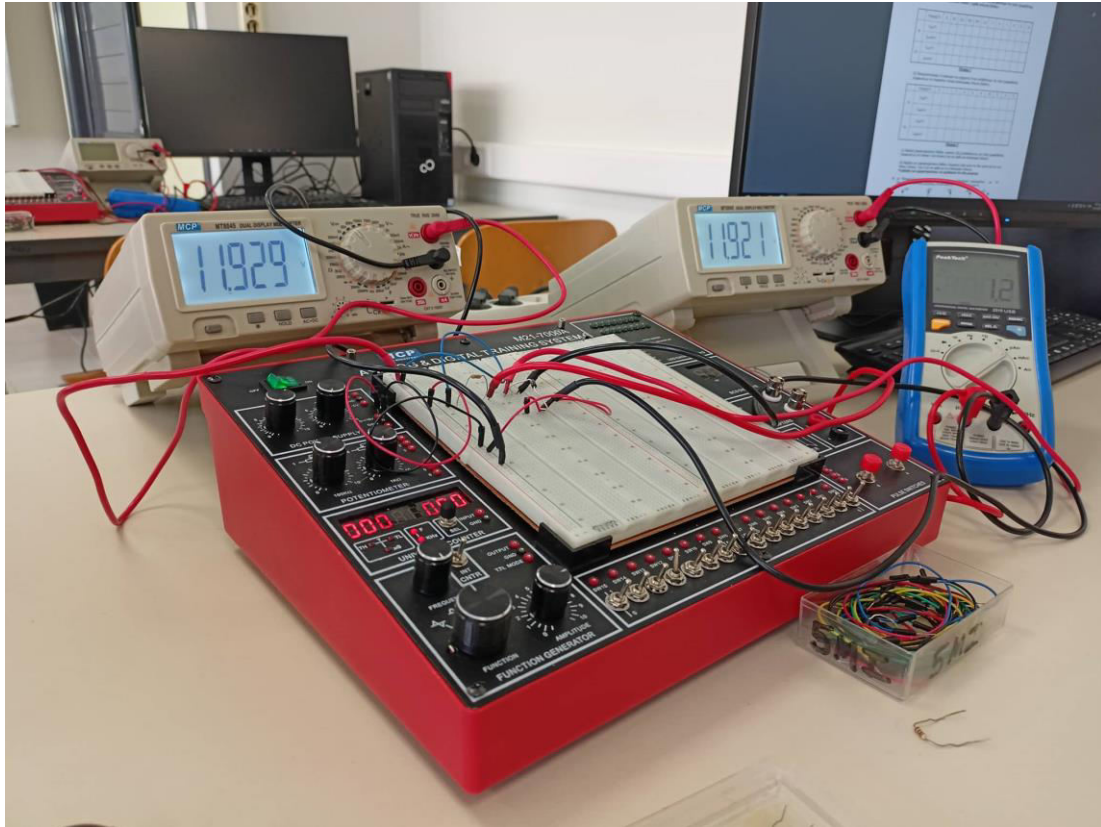


Figure 1.2.22 Inverted polarization of a silicon diode with 12 Volt supply voltage on the "Breadboard"

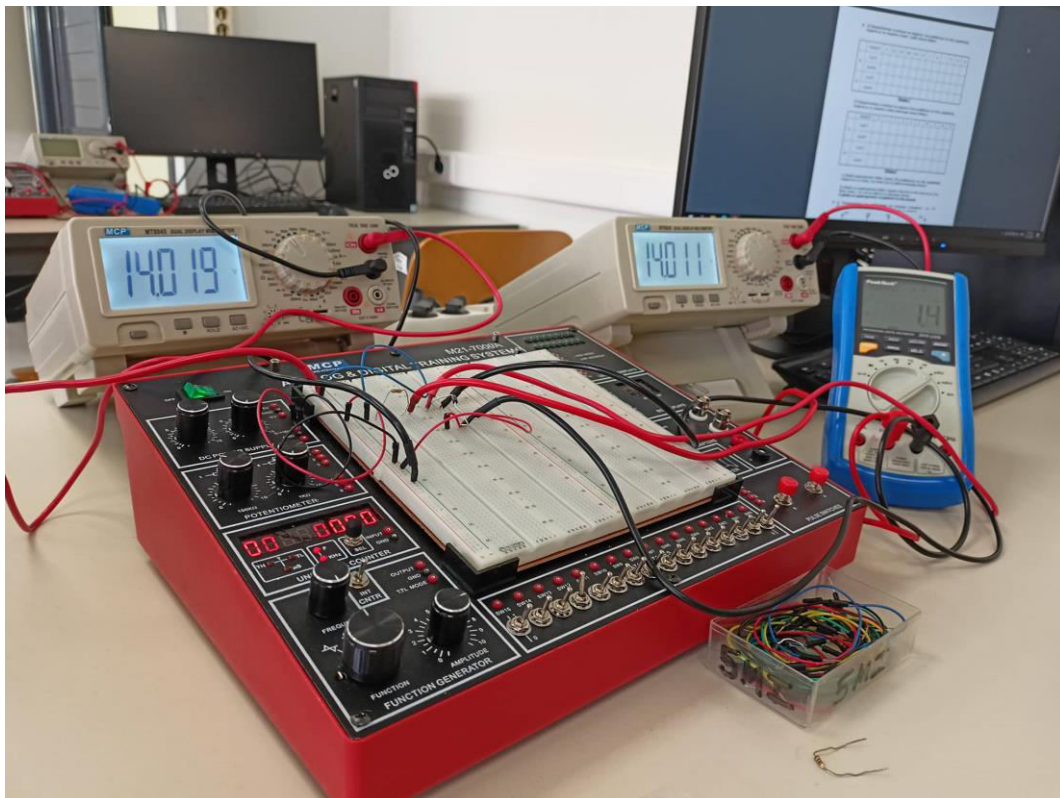


Figure 1.2.23 Inverted polarization of a silicon diode with 14 Volt supply voltage on the "Breadboard"

ELECTRONICS

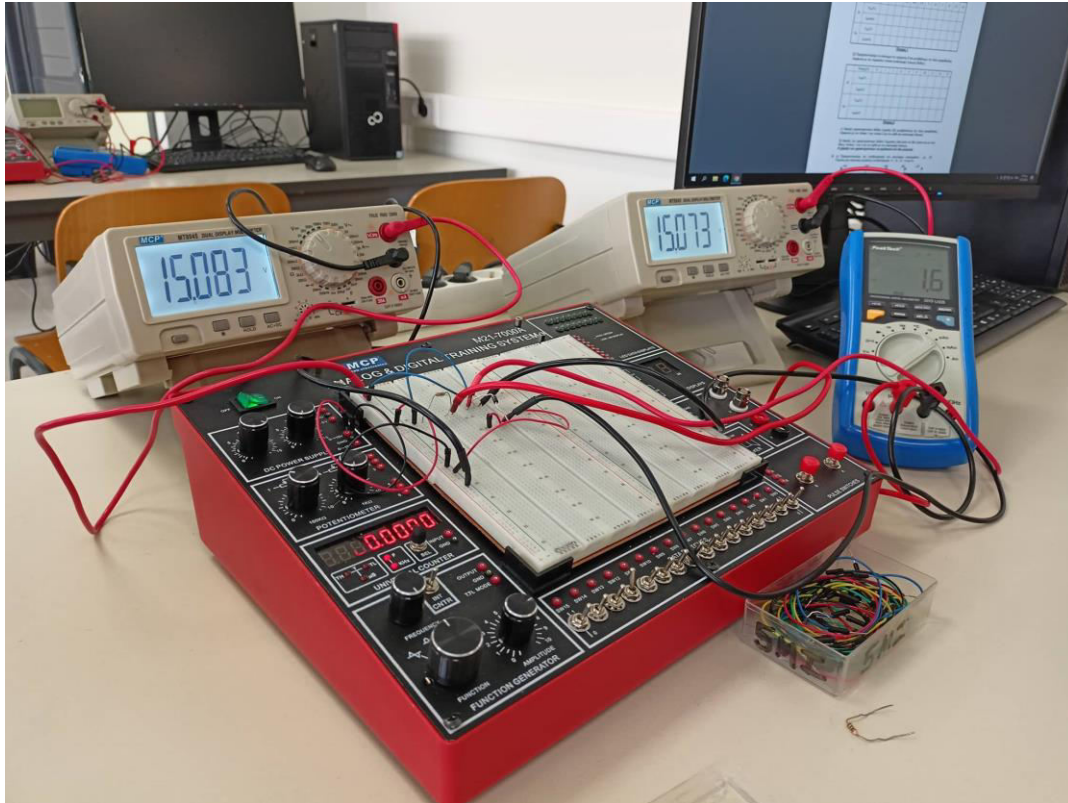


Figure 1.2.24 Inverted polarization of a silicon diode with 25 Volt supply voltage on the "Breadboard"

1.3 STEP 5 OF THE LABORATORY PART

Sub-step A / Recording of voltmeter readings

Figures 1.3.1, 1.3.2, 1.3.3, 1.3.4 and 1.3.5 show the schematics in the drawing software "MS Visio" of some diode circuits. In each circuit a 4.7 KOhm resistor and a 4.7 KOhm resistor are connected in series with the diode, and a voltmeter connected in parallel to respective nodes.

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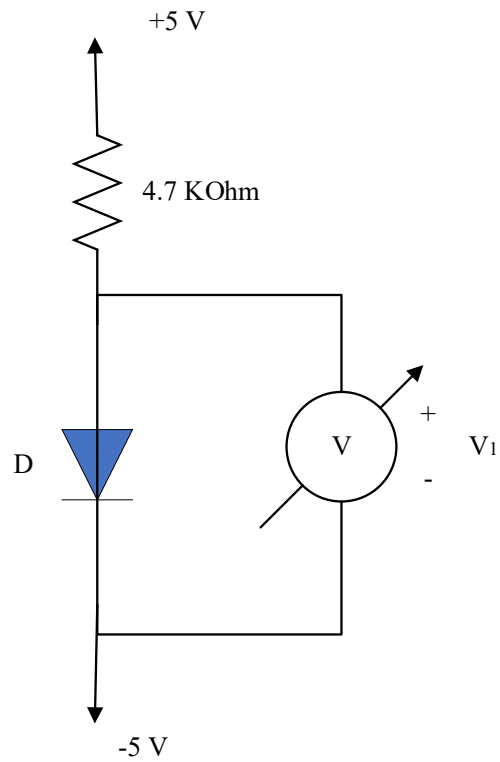


Figure 1.3.1 Schematic representation of the 1st circuit with silicon diode in "MS Visio"

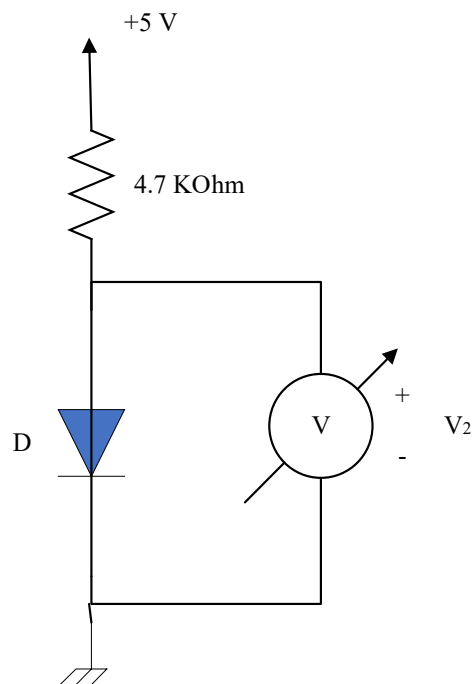


Figure 1.3.2 Schematic representation of the 2nd circuit with silicon diode in "MS Visio"

ELECTRONICS

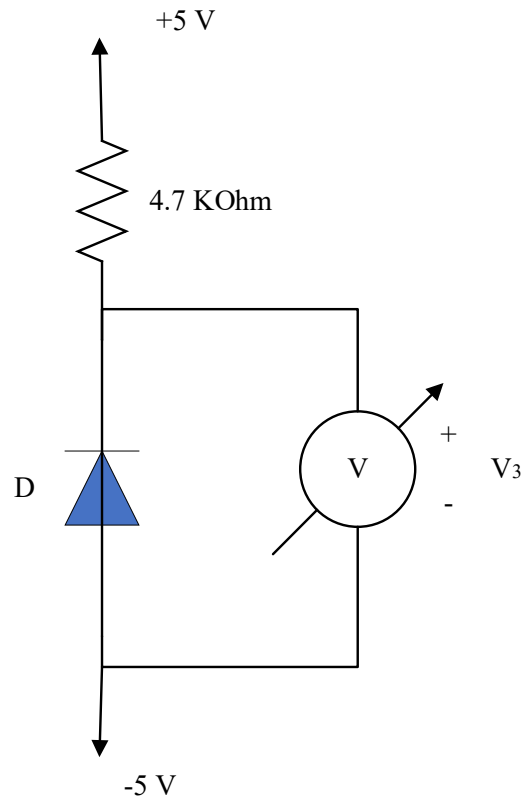


Figure 1.3.3 Schematic representation of the 3^o circuit with silicon diode in "MS Visio"

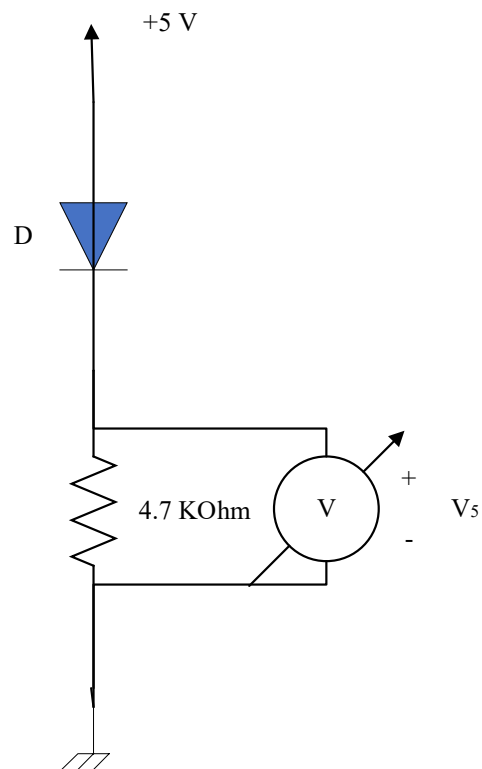


Figure 1.3.4 Schematic representation of the 4^o circuit with silicon diode in "MS Visio"

ELECTRONICS

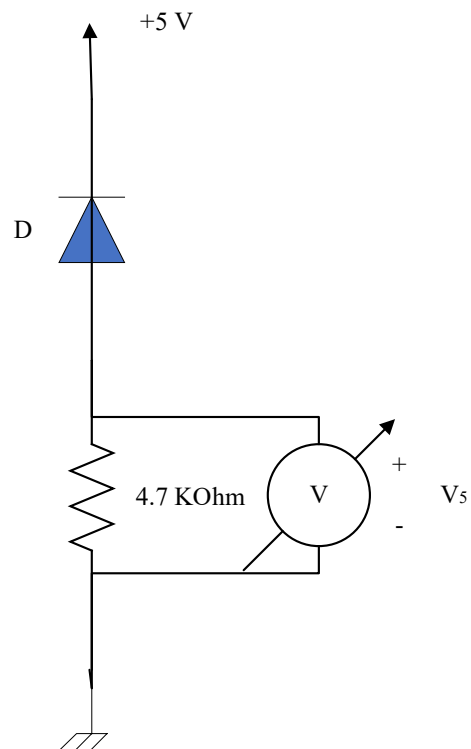


Figure 1.3.5 Schematic representation of the 5^{ov} circuit with silicon diode in "MS Visio"

In Table 3, the voltmeter readings from the respective circuit are recorded according to the topology of each circuit.

Table 3.

V₁ = 0.6 Volt	V₂ = 0.5 Volt	V₃ = 9.97 volts	V₄ = 4.4 volts	V₅ = 0.03 Volt
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Sub-step B / Theoretical and simulated solution

1° Circuit ($V_1 = 0.6$ Volt)

In the figure in Figure 1.3.1, the diode turns off as the voltage at the anode is greater than the voltage at the cathode. The potential difference at the anode is < 5 volts from the supply of the top DC source of 5 volts and from the voltage drop of the 4.7 KOhm resistor. The potential difference at the cathode is -5 volts from the supply of the lower source of -5 volts. The voltmeter reading of 0.6 Volts at the ends of the diode confirms the correct polarization of the diode in the circuit, since the voltage drop at the ends of the diode is marginally zero, as it is a true diode. The blocking voltage of the diode is 0.6 Volt. Figure 1.3.6 shows the simulated solution of the 1^{ou} circuit in the simulation software "Multisim" which verifies the theoretical solution.

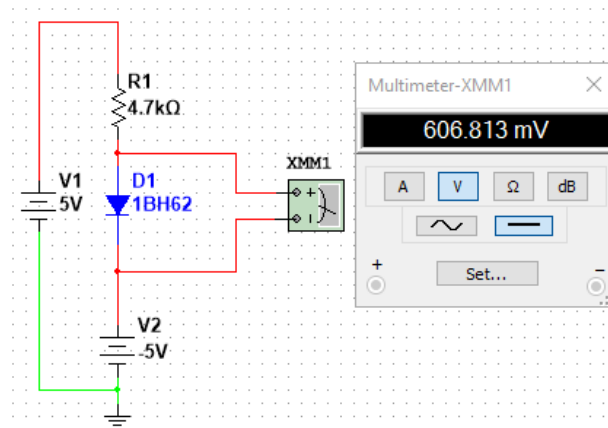


Figure 1.3.6 Simulation of the 1^{ou} circuit with silicon diode in "Multisim"

2° Circuit ($V_2 = 0.5$ Volt)

In the figure in Figure 1.3.2, the diode turns off as the voltage at the anode is greater than the voltage at the cathode. The potential difference at the anode is < 5 volts from the supply of the top DC source of 5 volts and from the voltage drop of the 4.7 KOhm resistor. The potential difference at the cathode is 0 volts from ground. The voltmeter reading at 0.5 Volts at the ends of the diode confirms the correct polarization of the diode in the circuit, as the voltage drop at the ends of the diode is marginally zero, since it is a true diode. Figure 1.3.7 shows the simulated solution of the 2^{ou} circuit in the simulation software "Multisim" which verifies the theoretical solution.

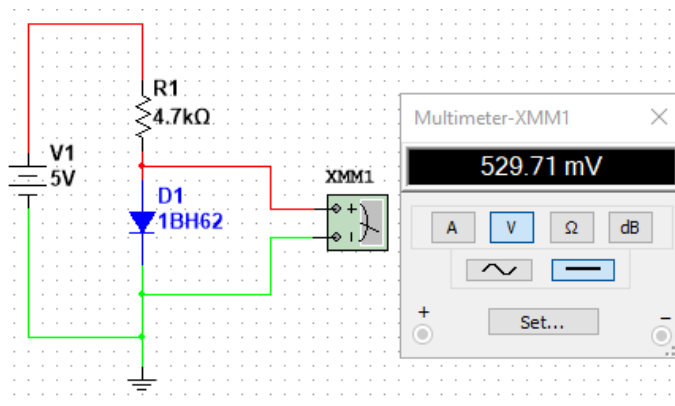


Figure 1.3.7 Simulation of the 2^{ou} circuit with silicon diode in "Multisim"

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3^o Circuit ($V_3 = 9.97$ Volt)

In the figure in Figure 1.3.3, the diode does not turn on as the voltage at the anode is less than the voltage at the cathode. The potential difference at the cathode is < 5 volts from the supply of the top DC source of 5 volts and from the voltage drop of the 4.7 KOhm resistor. The potential difference at the anode is -5 Volts from the supply of the lower source of -5 Volts. The voltmeter reading of 9.97 Volts at the ends of the diode confirms the reverse bias of the diode in the circuit, as there was almost no voltage drop at the ends of the diode. Because, it is a real diode, a very small current is leaking. Figure 1.3.6 shows the simulated solution of the 3^o circuit for silicon diode in the simulation software "Multisim" which verifies the theoretical solution.

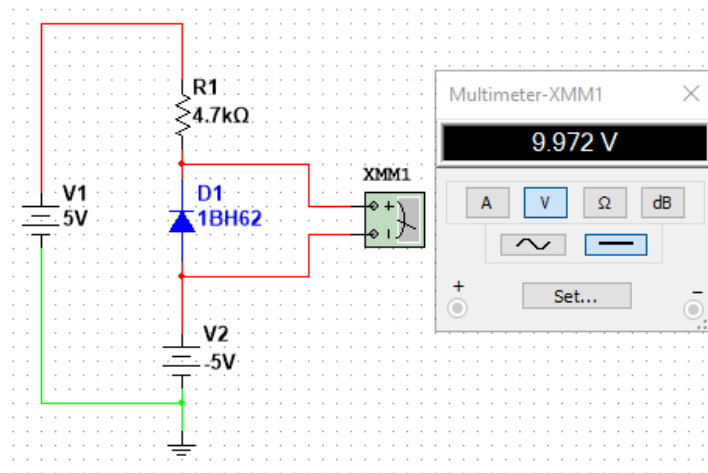


Figure 1.3.7 Simulation of the 3^o circuit with silicon diode in "Multisim"

4^o Circuit ($V_4 = 4.4$ Volt)

In the figure in Figure 1.3.4, the diode turns off as the voltage at the anode is greater than the voltage at the cathode. The potential difference at the cathode is the slight voltage rise experienced by the 4.7 KOhm resistor connected in series with ground. The potential difference at the anode is 5 volts from the DC source supply. The positive terminal of the voltmeter is connected to the node which has a voltage of $5 - 0.6 = 4.4$ Volts, where 0.6 is the diode barrier voltage. The negative terminal is connected to ground. Therefore, the voltmeter reading at the ends of the resistor is $4.4 - 0 = 4.4$ Volts. Figure 1.3.8 shows the simulated solution of the 4^o circuit for silicon diode in the simulation software "Multisim" which verifies the theoretical solution.

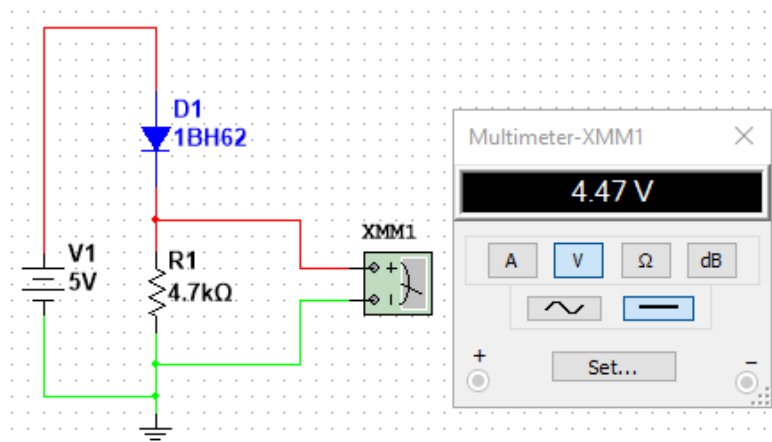


Figure 1.3.8 Simulation of the 4^o circuit with silicon diode in "Multisim"

5^o Circuit ($V_5 = 0.03$ Volt)

ELECTRONICS

In the figure in Figure 1.3.5, the diode does not turn on as the voltage at the anode is less than the voltage at the cathode. The potential difference at the anode is the slight voltage rise experienced by the 4.7 KOhm resistor connected in series with ground. The potential difference at the cathode is 5 volts from the DC source supply. The voltmeter reading of 0.03 Volts at the ends of the 4.7 KOhm resistor confirms the reverse bias of the diode in the circuit, as there is marginally no voltage drop at the ends of the resistor from the diode blocking voltage (0.6 Volts). This is due to the fact that a very small current is flowing through the resistor and a small voltage is produced by the resistor. Figure 1.3.9 shows the simulated solution of the 5^{ov} circuit for silicon diode in the simulation software "Multisim" which verifies the theoretical solution.

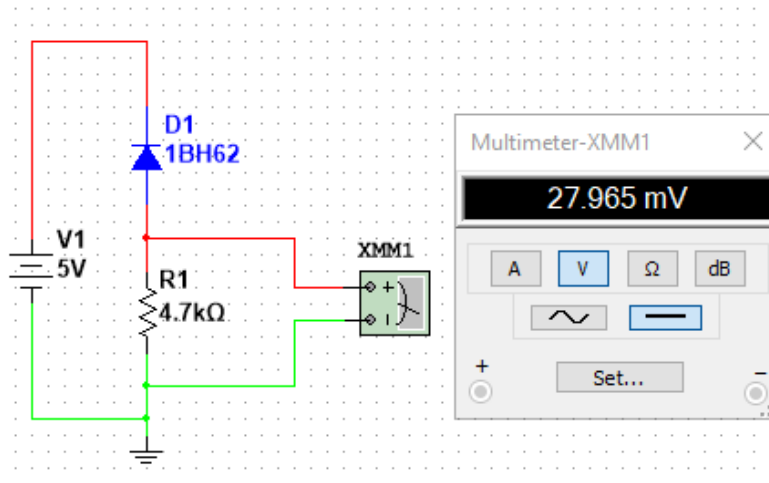


Figure 1.3.8 Simulation of the 5^{ov} circuit with silicon diode in "Multisim"

1.4 QUESTIONS

Question 1.4.1

Explain the shape of the diode characteristics.

The characteristic $I_{DF} - V_{DF}$ of the orthogonally polarized silicon diode is shown in Figure 1.1.2 on page 8. Initially, the current increases gradually for small rectified polarization voltages less than 0.6 Volt, where it takes an exponential slope when the voltage exceeds the diode barrier voltage (0.6 Volt). This slope is due to the small internal ohmic resistance and the fact that the stripping region of the diode is eliminated when the voltage drop across its ends is equal to or greater than the barrier voltage (0.6 Volt for a silicon diode and 0.3 Volt for a germanium diode). A current flows through the diode and it behaves as a conductor.

The characteristic $I_{DR} - V_{DR}$ of the inversely polarized silicon diode is illustrated in Figure 1.1.3 on page 22. The current is nearly constant and very small as the stripping area increases and the internal ohmic resistance becomes very large. Consequently, the characteristic is linear.

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Question 1.4.2

What is the difference between Germanium and Silicon diodes?

Germanium diodes differ from Silicon diodes in terms of blocking voltage. Specifically, in Germanium diodes the barrier voltage is 0.3 Volt, while in Silicon diodes it is 0.6 Volt. This means that the current flow takes place for lower external source voltages in the Germanium diodes which is not the case in the corresponding Silicon diodes.

Question 1.4.3

List the types and uses of the passages you know.

There are different types of diodes, such as PN (germanium, silicon) and Zener diodes, which are used in electronic and digital circuits, for example, in a home appliance or in an integrated circuit.

Question 1.4.4

Also from the values V_{DF} and I_{DF} filled in the previous table 1 (Silicon diode polarization table only) calculate the values of $R_{\sigma\tau\alpha\tau}$ and $R_{\delta uv}$ by filling in the lower table 4. Note that the values $\Delta V(V)$ and $\Delta I(mA)$ are calculated by subtracting two consecutive values $V_{DF}(V)$ and $I_{DF}(mA)$ respectively from Table 1 (page 8).

Table 4.

Spring (V)	0.1	0.3	0.6	0.9	1.5	2	4	7	9	12	15
$R_{\sigma\tau\alpha\tau} = \frac{V_{DF}}{I_{DF}} \text{ (KOhm)}$	11.7	6.7	3.8	3.0	2.0	1.4	0.7	0.4	0.3	0.25	0.2
$\Delta V(V)$	0.07	0.13	0.06	0.04	0.1	0.02	0.08	0.1	0	0	0
$\Delta I(mA)$	0.006	0.024	0.04	0.03	0.1	0.1	0.4	0.7	0	0	0
$R_{\delta uv} = \frac{\Delta V_{DF}}{\Delta I_{DF}} \text{ (KOhm)}$	11.7	5.5	1.5	1.3	1.0	0.2	0.2	0.1	0	0	0

Then from the values in the above table draw the curves of $R_{\delta uv}$ and $R_{\sigma\tau\alpha\tau}$ on milimetric paper. The voltages and currents should be in (Volts) and (mA).

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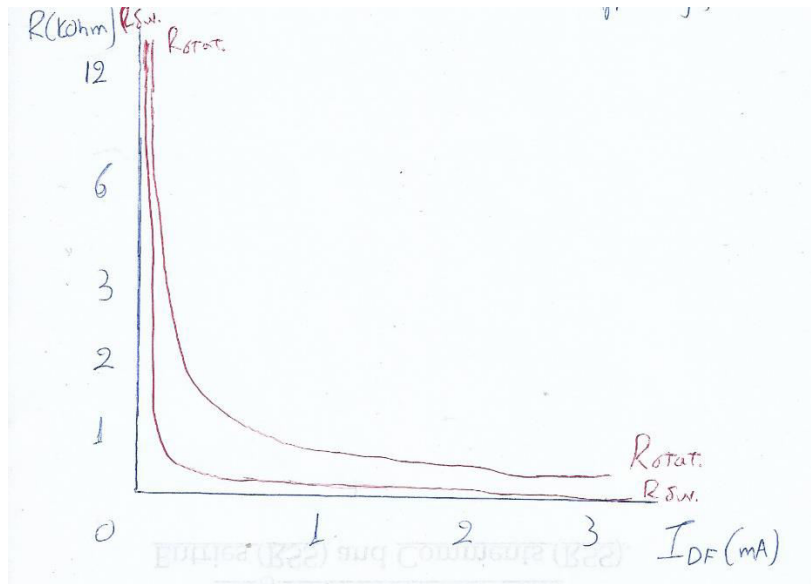


Figure 1.4.1 Curves $R_{\delta v}$ and $R_{\sigma at}$

Question 1.4.5

In the lower circuit calculate the voltages of the voltmeters V_1 and V_2 (D_1 , D_2 silicon real)

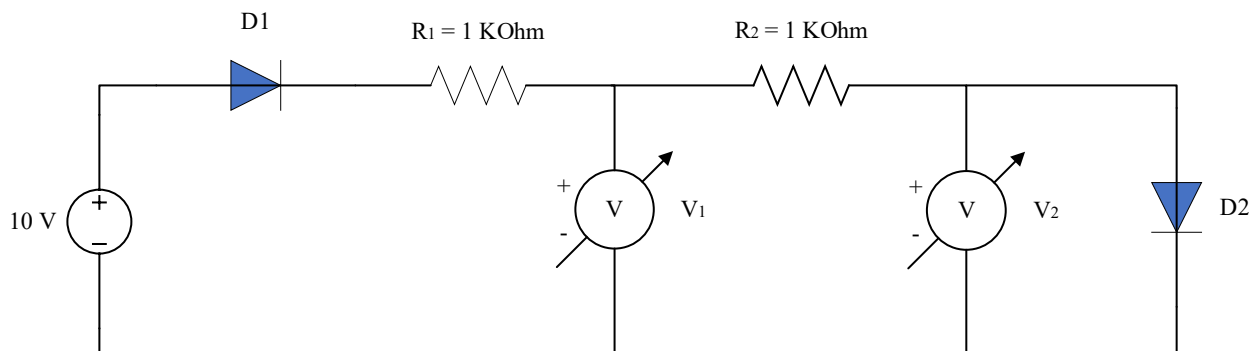


Figure 1.4.2 Schematic representation of the circuit in "MS Visio"

From the voltage divider formula and Ohm's law, for the potential differences at the ends of the resistors R_1 and R_2 we have :

$$V_{R1} = I \times R_1 \rightarrow V_{R1} = \frac{V}{R_1 + R_2} \times R_1 \rightarrow V_{R1} = \frac{R_1}{R_1 + R_2} \times V \rightarrow V_{R1} = \frac{10^3}{10^3 + 10^3} \times 10 \rightarrow V_{R1} = 5 \text{ Volt}$$

$$V_{R2} = I \times R_2 \rightarrow V_{R2} = \frac{V}{R_1 + R_2} \times R_2 \rightarrow V_{R2} = \frac{R_2}{R_1 + R_2} \times V \rightarrow V_{R2} = \frac{10^3}{10^3 + 10^3} \times 10 \rightarrow V_{R2} = 5 \text{ Volt}$$

$V_1 = 5$ volts

Therefore, diode D_1 is correctly polarized, as the voltage at the anode (10 Volts) is higher than the voltage at the cathode (4.3 Volts). The voltage at the cathode analytically is $5 - 0.7 = 4.3$ Volts, where 0.7 Volts is the barrier voltage of the silicon diode. The positive terminal of the voltmeter V_1 is connected to the right end of the resistor R_1 , while the negative terminal is connected to ground. Therefore, the voltmeter reading will show the potential difference at the ends of the resistor, i.e., 5 volts.

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$$V_2 = 0.7 \text{ Volt}$$

Diode D_2 is correctly polarized, as the voltage at the anode (4.3 Volts) is higher than the voltage at the cathode (0 Volts). The voltage at the anode analytically is $5 - 0.7 = 4.3$ Volts, where 0.7 Volts is the barrier voltage of the silicon diode. The terminals of the voltmeter V_2 are connected in parallel to the ends of the diode. Therefore, the voltmeter reading will indicate the potential difference at the ends of the diode, which will be close to the barrier voltage, i.e., 0.7 Volts. Figure 1.4.3 shows the simulated solution of the circuit, which verifies the theoretical solution.

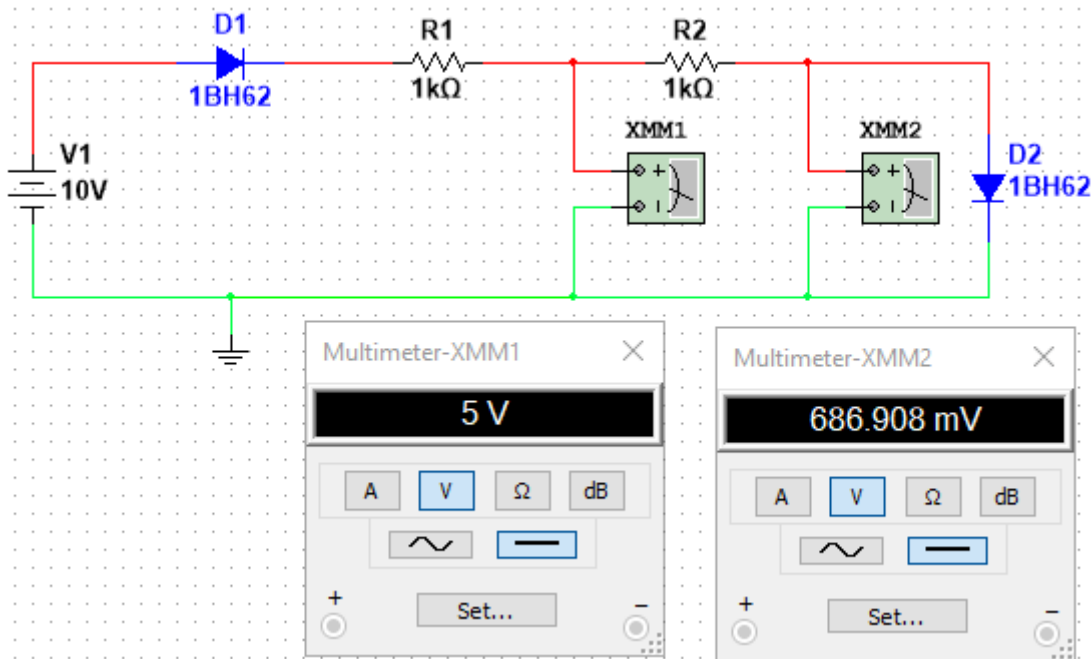


Figure 1.4.3 Simulation of the circuit in "Multisim"

Question 1.4.6

For the following circuits, evaluate the voltmeter readings. Note that the diodes are all Si with $V_{\phi\alpha\gamma\mu\omicron\gamma} (Si) = 0.6$

Figures 1.4.4, 1.4.5, 1.4.6, 1.4.7, 1.4.8, 1.4.9, 1.4.10, 1.4.11, 1.4.12 and 1.4.13 show the schematics of the circuits in the drawing software "MS Visio".

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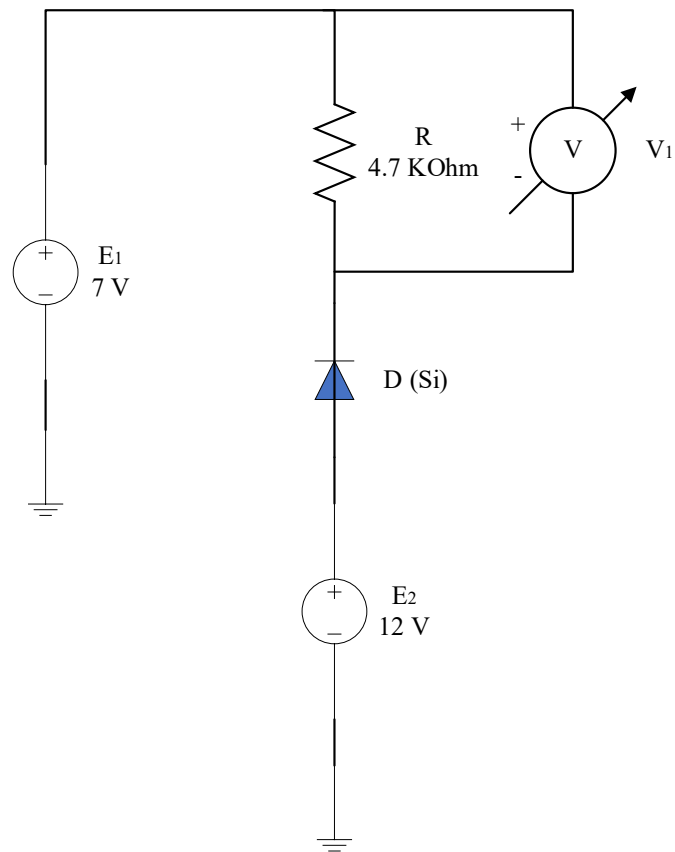
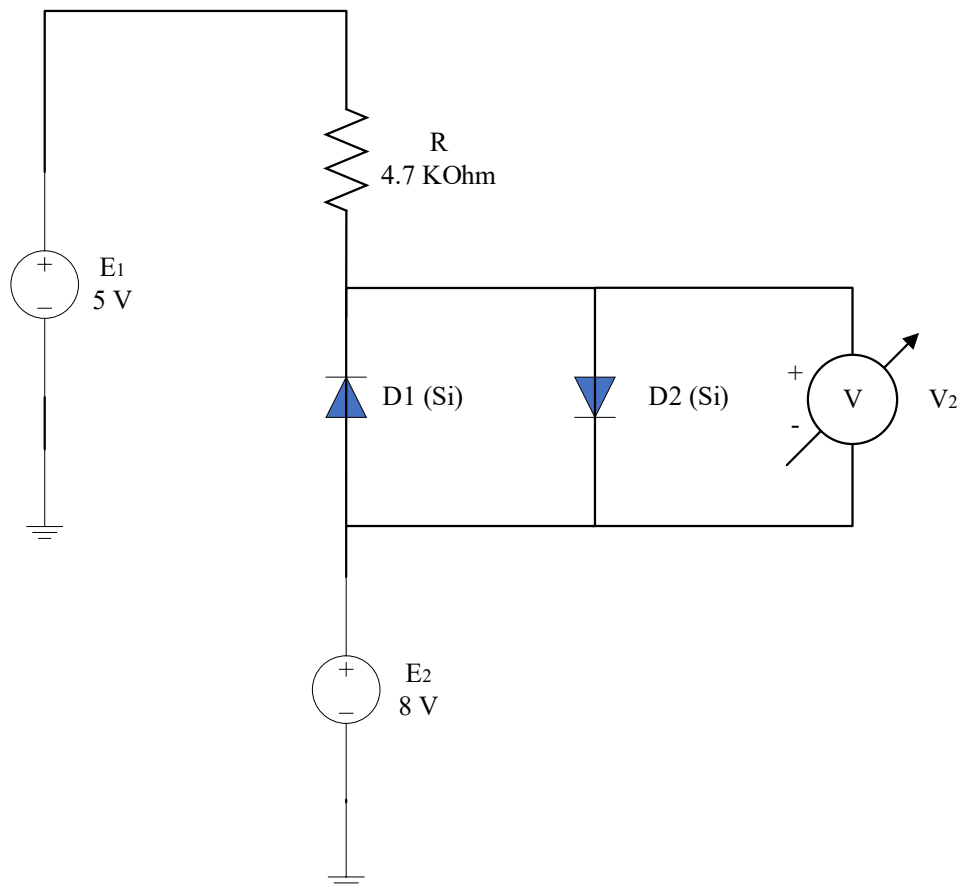


Figure 1.4.4 Schematic representation of the 1^o circuit in "MS Visio"



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Figure 1.4.5 Schematic representation of the 2^o circuit in "MS Visio"

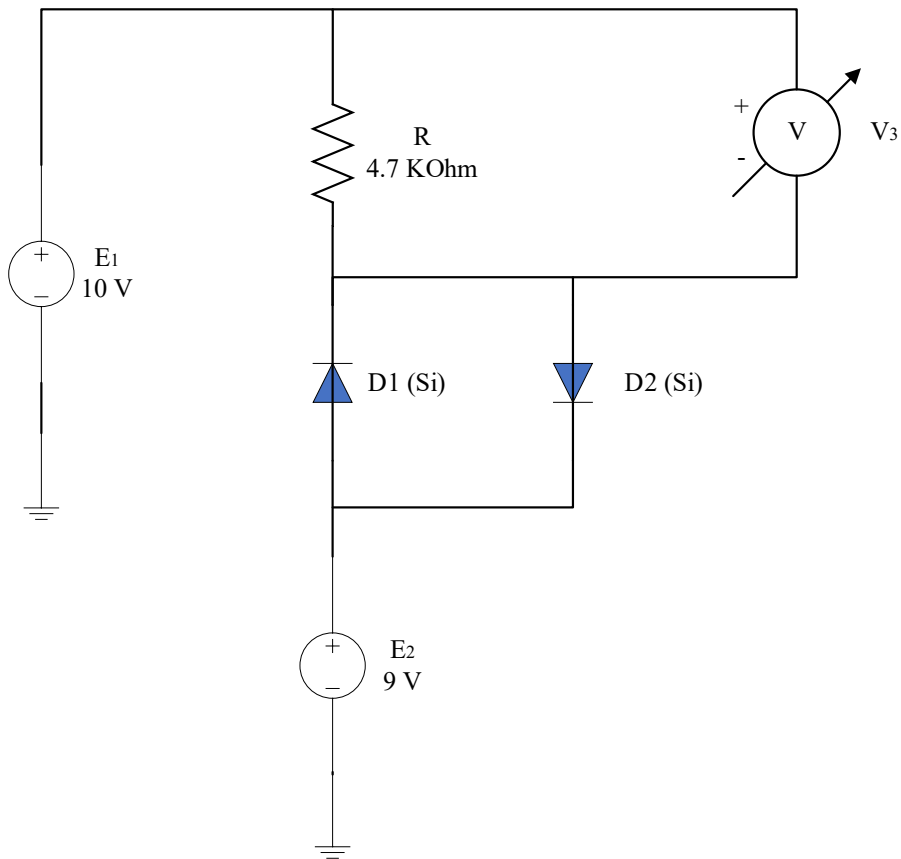


Figure 1.4.6 Schematic representation of the 3^o circuit in "MS Visio"

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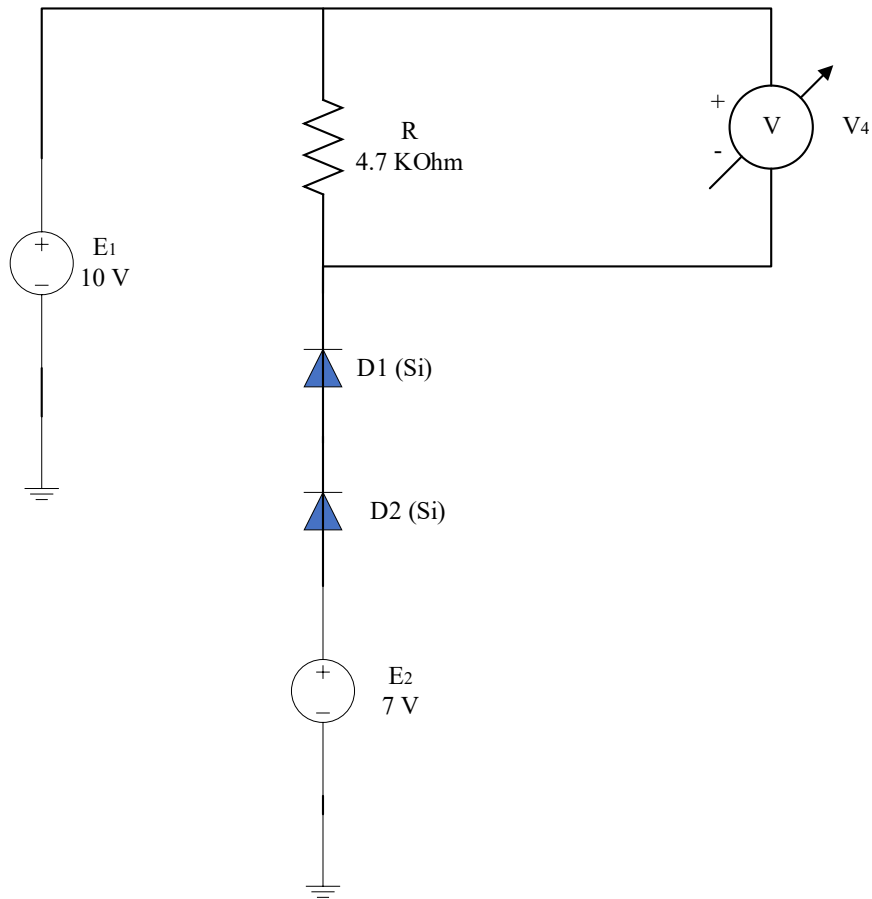


Figure 1.4.7 Schematic representation of the 4th circuit in "MS Visio"

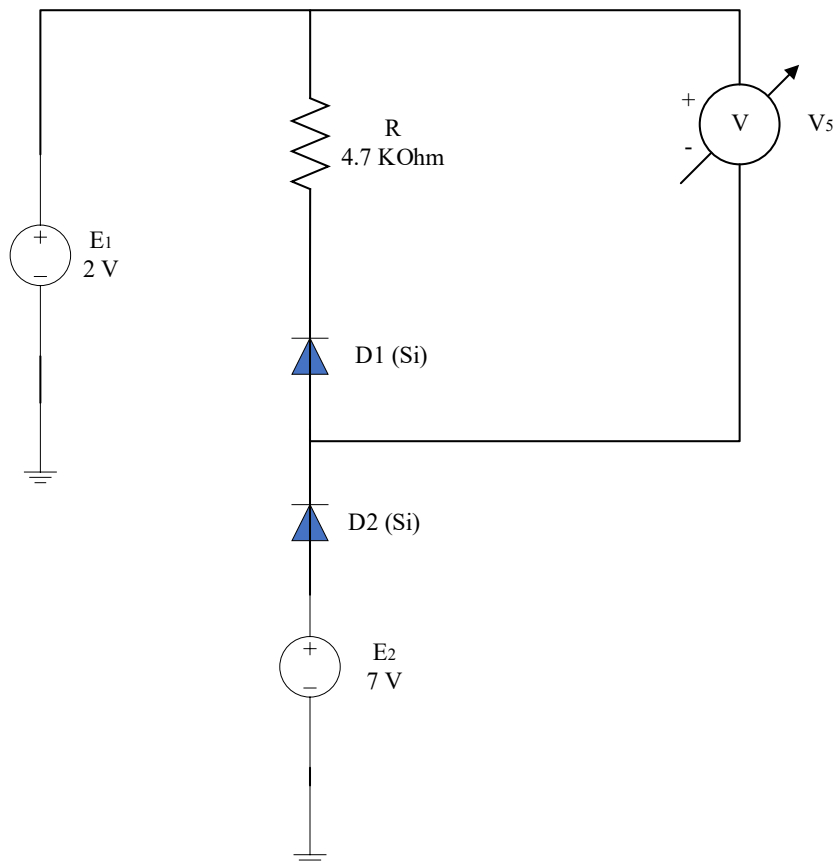


Figure 1.4.8 Schematic representation of the 5th circuit in "MS Visio"

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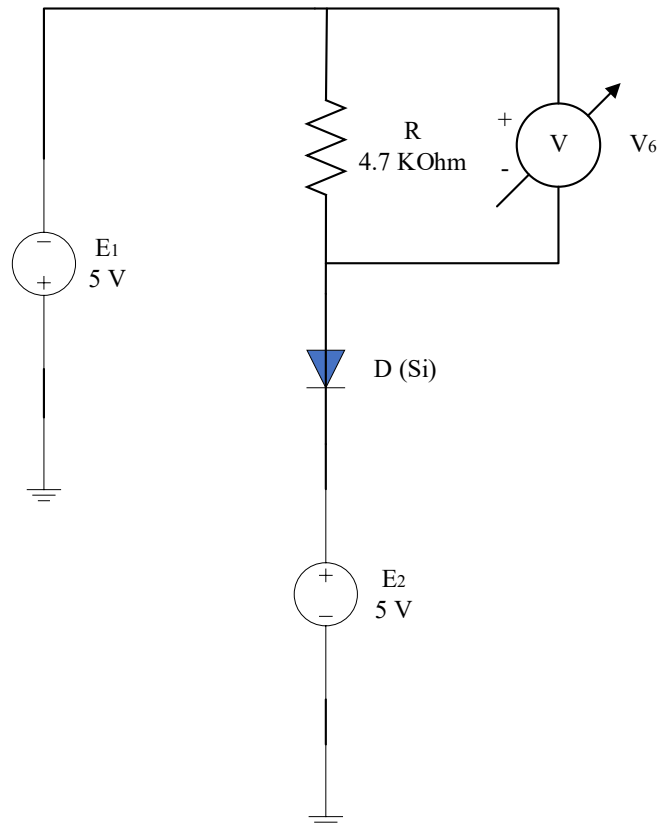


Figure 1.4.9 Schematic representation of the 6th circuit in "MS Visio"

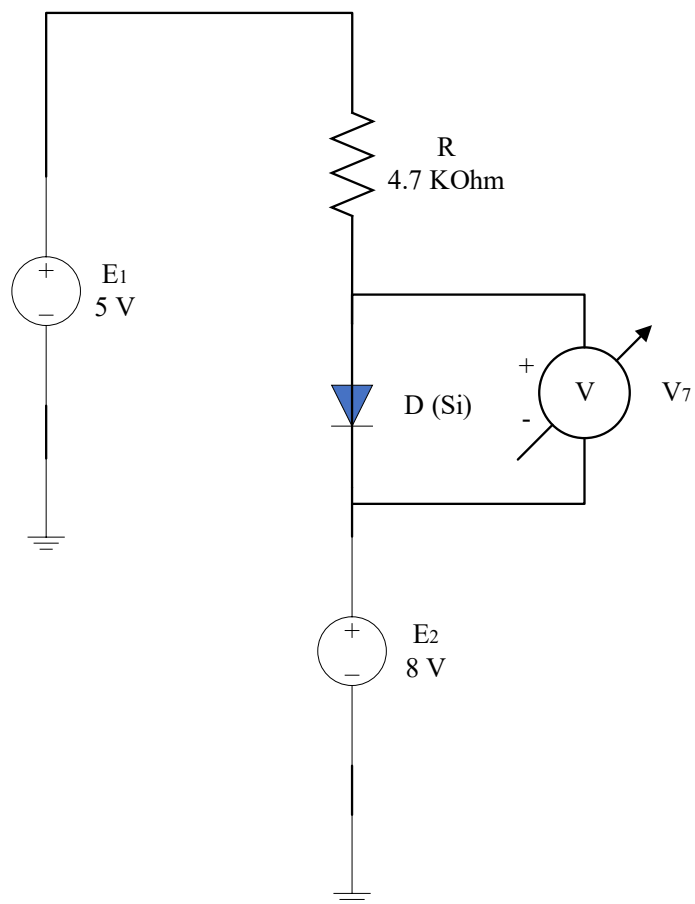


Figure 1.4.10 Schematic representation of the 7th circuit in "MS Visio"

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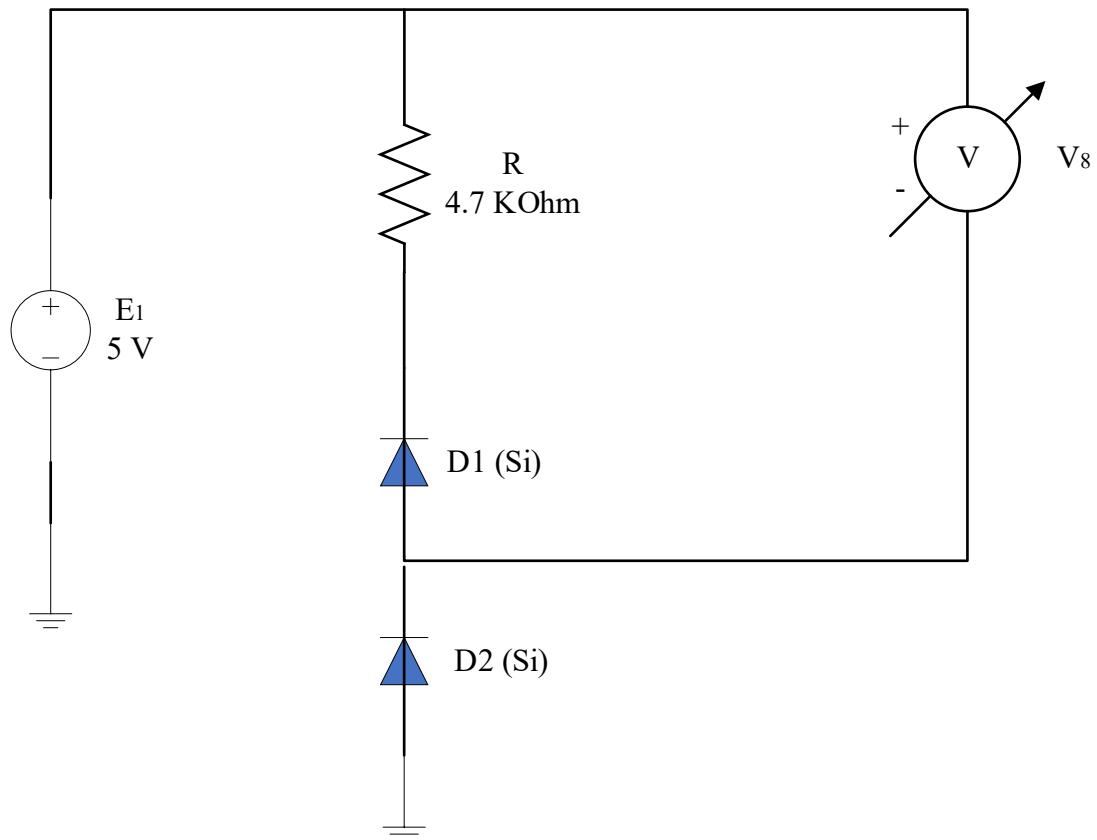


Figure 1.4.11 Schematic representation of the 8th circuit in "MS Visio"

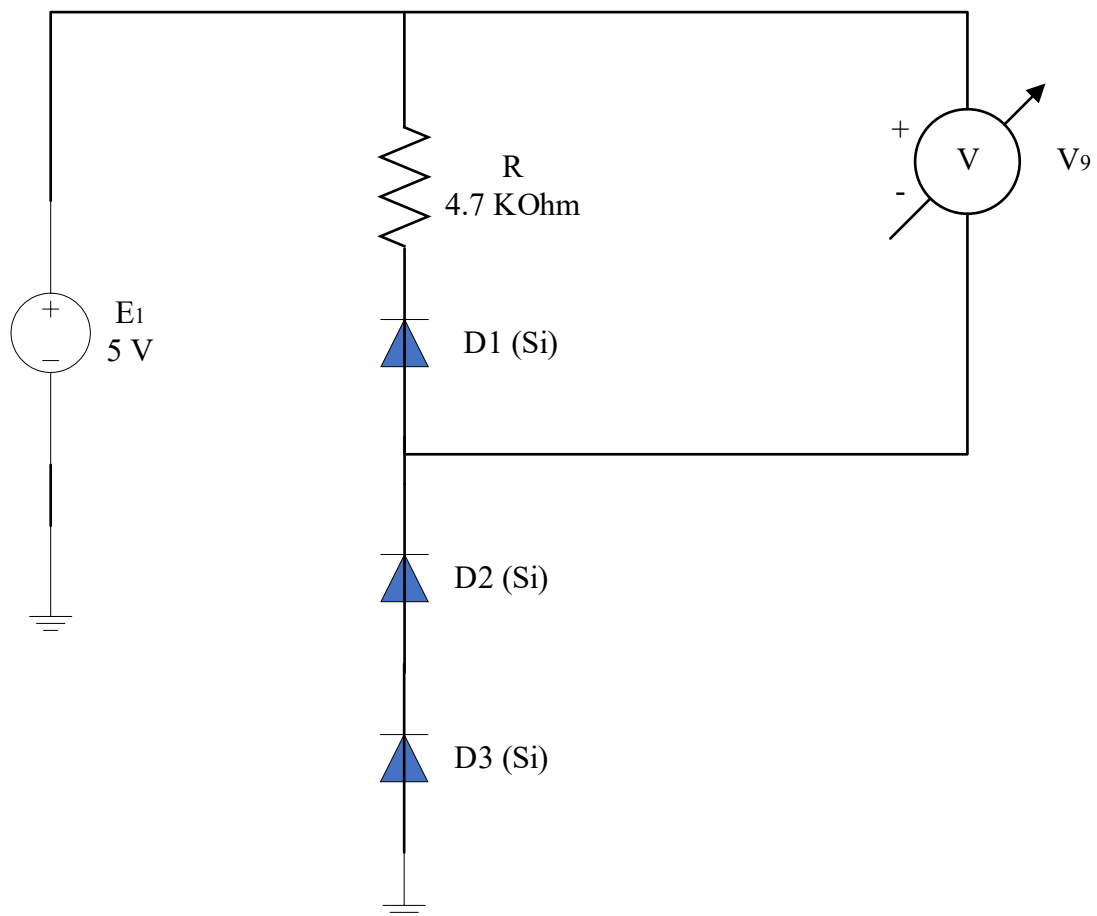


Figure 1.4.12 Schematic representation of the 9th circuit in "MS Visio"

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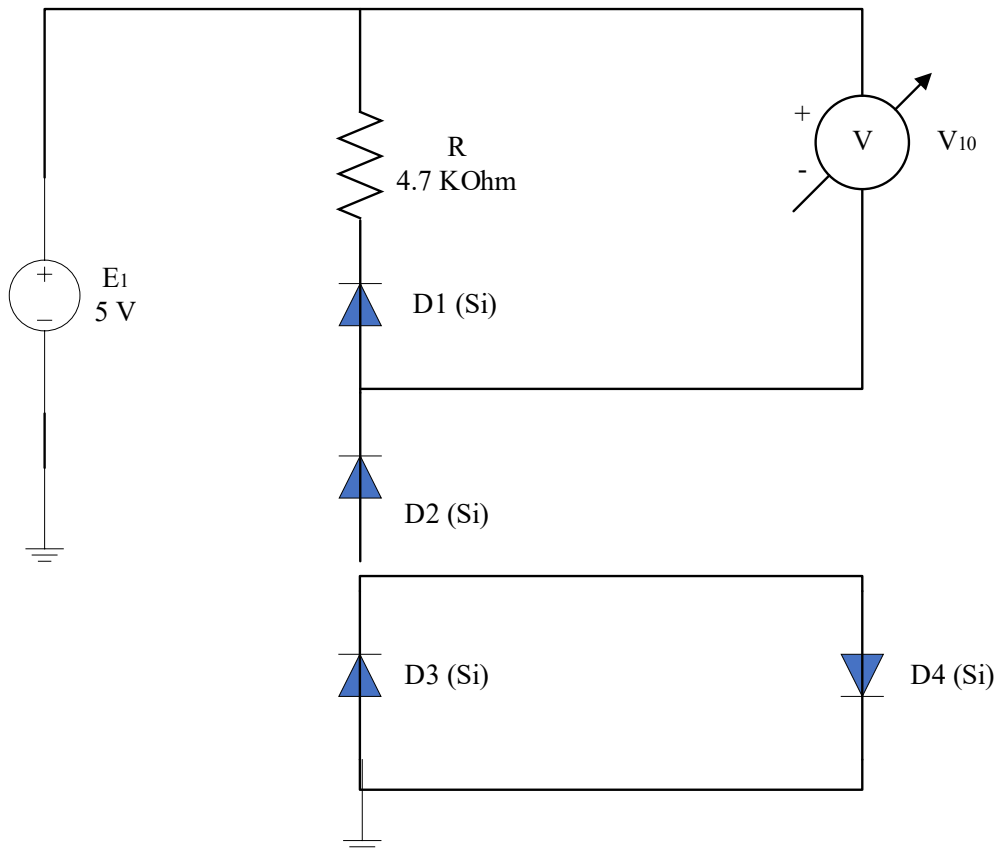


Figure 1.4.13 Schematic representation of the 10th circuit in "MS Visio"

In Table 5, the voltmeter readings from the corresponding circuit are recorded according to the topology of each circuit.

Table 5.

V ₁ (Volt)	V ₂ (Volt)	V ₃ (Volt)	V ₄ (Volt)	V ₅ (Volt)	V ₆ (Volt)	V ₇ (Volt)	V ₈ (Volt)	V ₉ (Volt)	V ₁₀ (Volt)
4.4	0.5	0.6	0.03	4.47	0.03	2.97	0.9	0.9	0.9

1^o Circuit (V₁ = 4.47 Volt)

In the figure in Figure 1.4.4, the diode turns off as the voltage at the anode (12 Volts) is greater than the voltage at the cathode (< 7 Volts). The voltage at the node where the negative terminal of the voltmeter is connected is 7 Volts from the 7 Volt source supply. The voltage at the node where the positive terminal of the voltmeter is connected is 11.4 Volts from the difference between the voltages of the 12 Volt source and the diode block which is 0.6 Volts. The voltmeter reading of $11.4 - 7 = 4.4$ volts at the ends of the resistor confirms the correct polarization of the diode in the circuit, since the voltage drop at the ends of the resistor is due to the diode's blocking voltage proving that current is flowing through it. Figure 1.4.14 shows the simulated solution of the 1^o circuit in the simulation software "Multisim" which verifies the theoretical solution.

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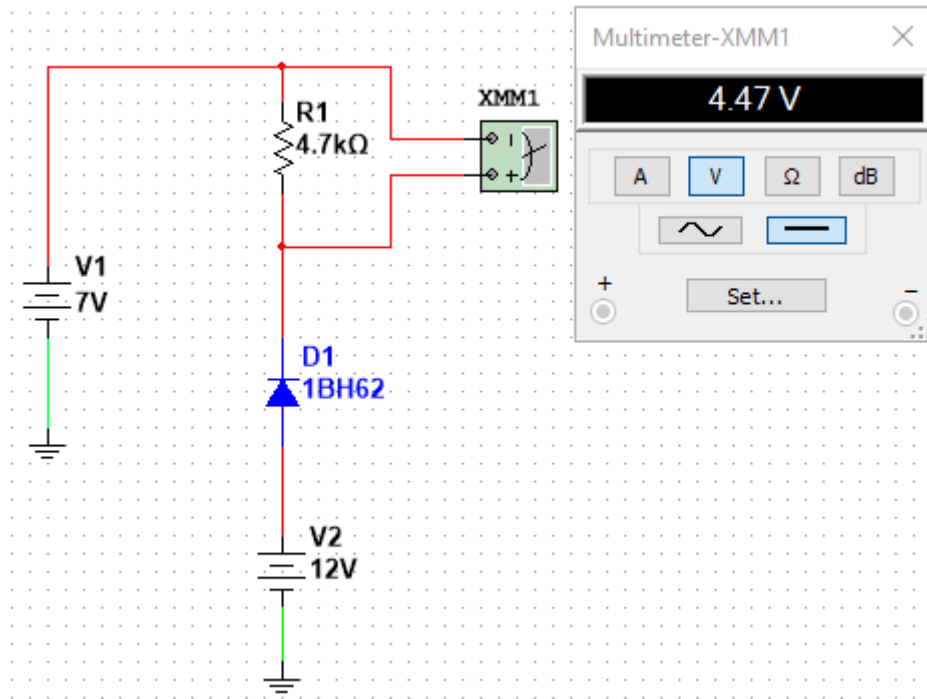


Figure 1.4.14 Simulation of the 1^o circuit in "Multisim"

2^o Circuit ($V_2 = 0.5$ Volt)

In the figure in Figure 1.4.5, diode D1 turns off as the voltage at the anode (8 volts) is greater than the voltage at the cathode (< 5 volts). The potential difference at the cathode < 5 Volt, where the voltage drop is due to the 4.7 KOhm resistor. The voltage at the anode comes from the 8 Volt DC source supply. Therefore, from the diode voltage drop (0.6 Volt) and the 8 Volt DC source, the node above D1 will have a voltage of $8 - 0.6 = 7.4$ Volts.

On the other hand, diode D2 does not turn on, since the voltage at the anode (7.4 Volts) is less than the voltage at the cathode (8 Volts). The voltmeter reading of 0.5 Volt at the ends of diode D2 confirms the correct polarization of diode D1 and the reverse polarization of D2 in the circuit, since the voltage drop at the ends of D2 is due to the blocking voltage of D1 proving that current is flowing. Figure 1.4.15 shows the simulated solution of the 2^o circuit in the simulation software "Multisim" which verifies the theoretical solution.

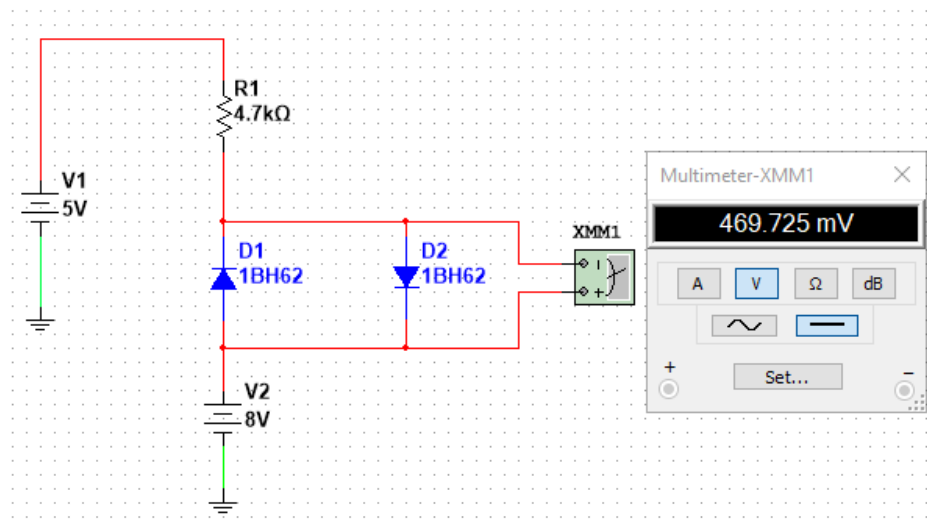


Figure 1.4.15 Simulation of the 2^o circuit in "Multisim"

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3^o Circuit ($V_3 = 0.6$ Volt)

In the figure in Figure 1.4.6, diode D1 does not turn on as the voltage at the anode (9 Volts) is less than the voltage at the cathode (< 10 Volts). The potential difference at the cathode is < 10 Volts, where the voltage drop is due to the 4.7 KOhm resistor. The voltage at the anode comes from the 9 Volt DC source supply.

On the other hand, diode D2 turns off as the voltage at the anode (< 10 Volts) is higher than the voltage at the cathode (9 Volts). That is, the anode-to-cathode voltages of D2 are inversely proportional to those of D1. The positive terminal of the voltmeter is at the top end of the resistor while the negative terminal is at the N contact of D1 and simultaneously at the P contact of D2. Since, the voltage drop across the inversely polarized D1 is almost zero (due to it being a true silicon diode), the voltmeter reading for the potential difference at the ends of the resistor will be at 0.6 volts. The resistor shows a voltage drop from the normally polarized diode D2 and very little from the reverse polarized diode D1. Figure 1.4.16 shows the simulated solution of the 3^o circuit in the simulation software "Multisim" which verifies the theoretical solution.

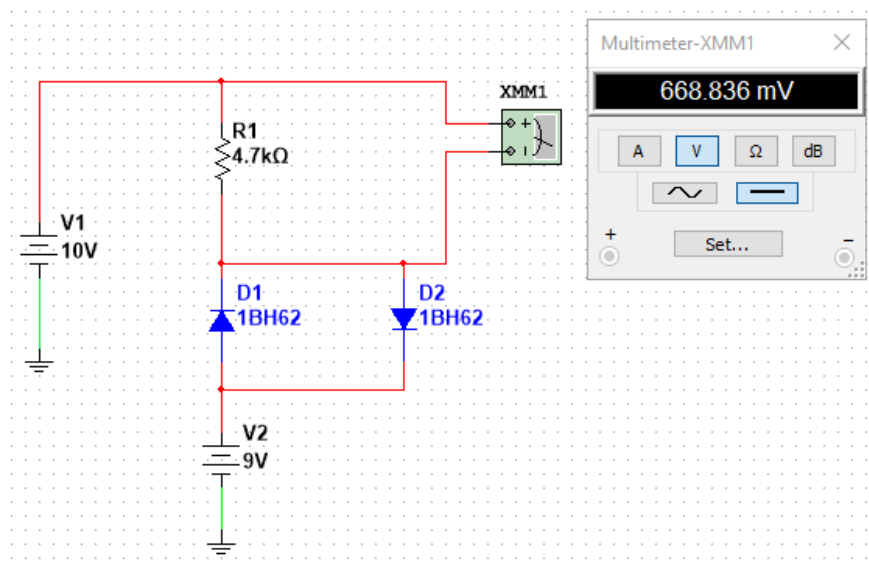


Figure 1.4.16 Simulation of the 3^o circuit in "Multisim"

4^o Circuit ($V_4 = 0.03$ Volt)

In the figure in Figure 1.4.7, neither diode is energized, as the anode voltage (7 volts) is less than the cathode voltage (< 10 volts). Therefore, very little current is flowing in the circuit and the voltmeter reading of 0.03 volts at the ends of the resistor verifies reverse bias on both diodes. Figure 1.4.17 shows the simulated solution of the 4^o circuit in the simulation software "Multisim" which verifies the theoretical solution.

ELECTRONICS

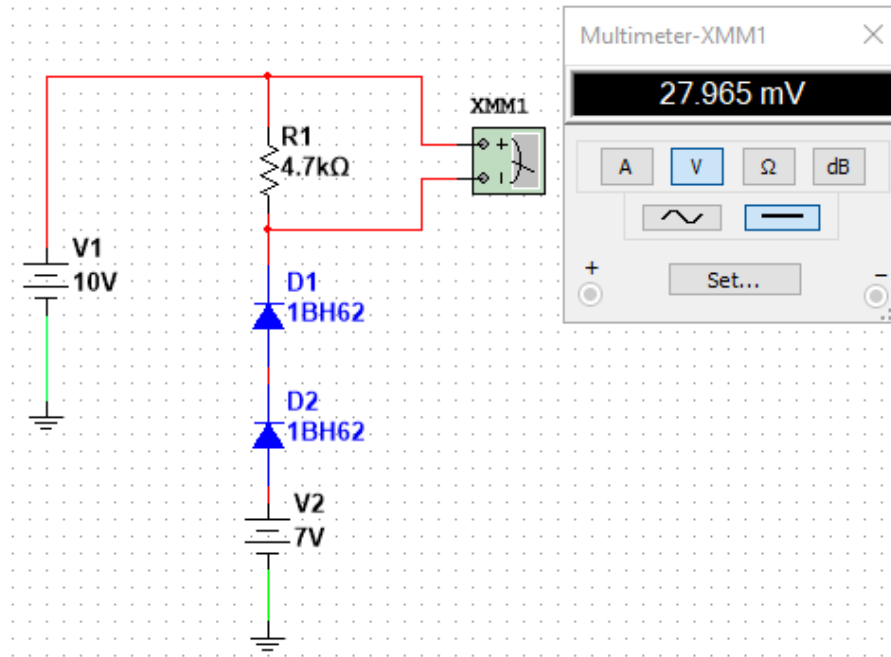


Figure 1.4.17 Simulation of the 4th circuit in "Multisim"

5th Circuit ($V_5 = 4.4$ Volt)

In the figure of Figure 1.4.8 both diodes are open, as the voltage at the anode coming from the 7 Volt DC source supply is higher than the voltage at the cathode coming from the 2 Volt DC source supply and the voltage drop from the 4.7 KOhm resistor. The positive terminal of the voltmeter is connected to the node having a voltage of $7 - 0.6 = 6.4$ Volts, where 0.6 is the diode blocking voltage, while the negative terminal is connected to the node having a voltage of 2 Volts from the 2 Volt DC source supply. Therefore, the voltmeter reading is $6.4 - 2 = 4.4$ Volts. Figure 1.4.18 shows the simulated solution of the 5th circuit in the simulation software "Multisim" which verifies the theoretical solution.

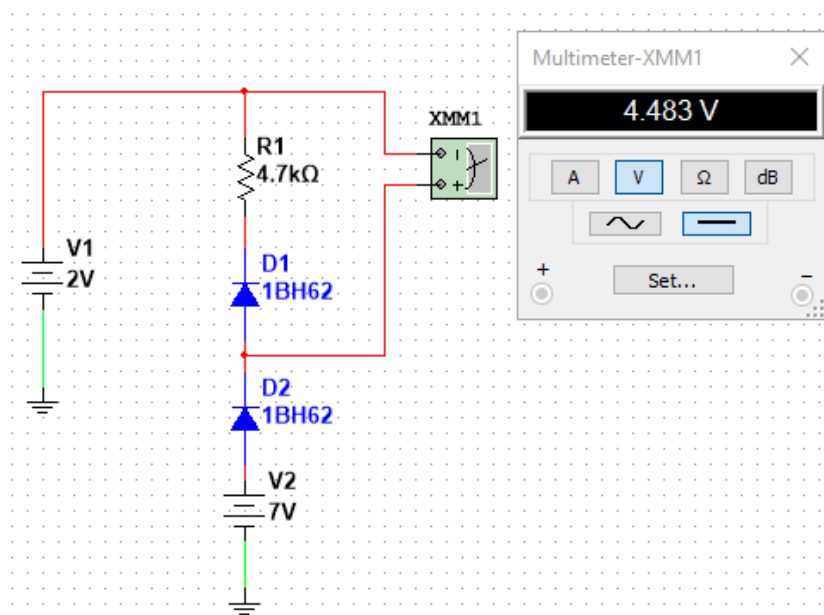


Figure 1.4.18 Simulation of the 5th circuit in "Multisim"

ELECTRONICS

6° Circuit ($V_6 = 0.03$ Volt)

In the figure in Figure 1.4.9, the diode is not energized, as the P contact is connected to the negative pole of the 5 Volt source. Therefore, a very small voltage drop will be experienced at the ends of the 4.7 KOhm resistor from the very small current flowing through the inversely polarized diode. The voltmeter reading at 0.03 volts verifies the above claims. Figure 1.4.19 shows the simulated solution of the 6^{oo} circuit in the simulation software "Multisim" which verifies the theoretical solution.

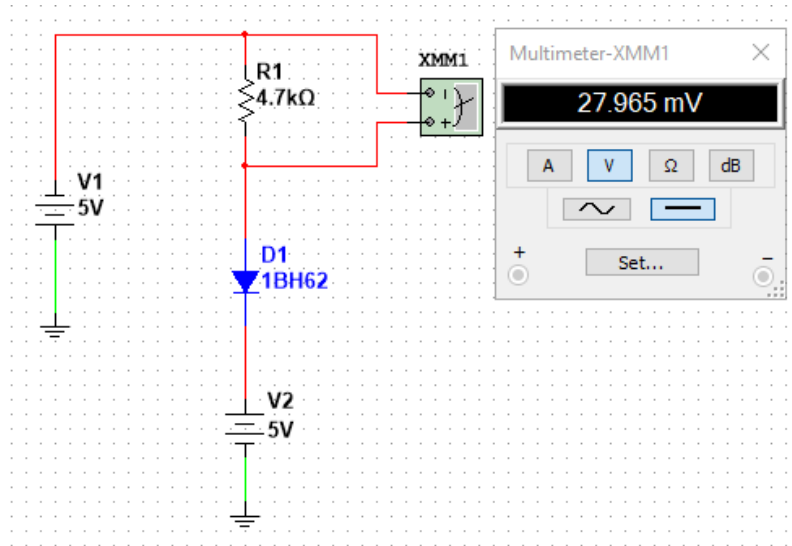


Figure 1.4.19 Simulation of the 6^{oo} circuit in "Multisim"

7° Circuit ($V_7 = 2.97$ Volt)

In the figure of Figure 1.4.10, the diode does not turn on, as the voltage at the anode coming from the 5 Volt DC source supply and the resistance drop at the ends of the 4.7 KOhm resistor is less than the voltage at the cathode coming from the 8 Volt DC source supply. The voltmeter reading at the ends of the reverse polarized diode which is at 2.97 Volts justifies the difference of $8 - 5 = 3$ Volts from the two DC sources combined with the voltage drop at the ends of the 4.7 KOhm resistor (0.03 Volts). Figure 1.4.20 shows the simulated solution of the 7^{oo} circuit in the "Multisim" simulation software which verifies the theoretical solution.

ELECTRONICS

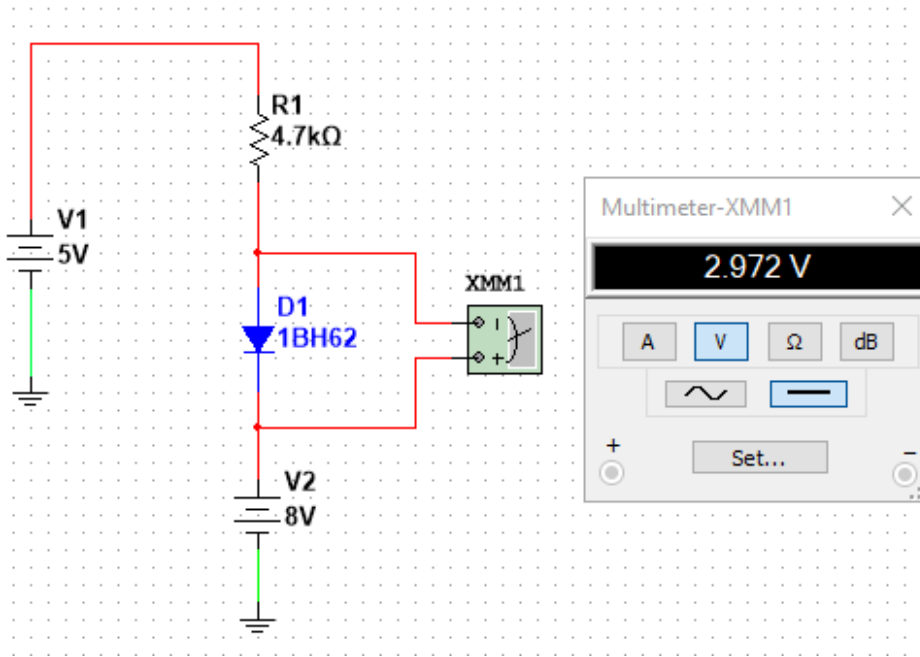


Figure 1.4.20 Simulation of the 7^o circuit in "Multisim"

8^o Circuit ($V_8 = 0.9$ Volt)

In the figure in Figure 1.4.11, the diodes do not open, as the positive pole of the source is connected to the N contacts of the diodes. The voltmeter reading is the potential difference between the node supplied by the 5 volt DC source and the undefined voltage node between the two reverse-polarized diodes. Specifically, the voltmeter reading is at 0.9 volts. Figure 1.4.21 shows the simulated solution of the 8^o circuit in the simulation software "Multisim" which verifies the theoretical solution.

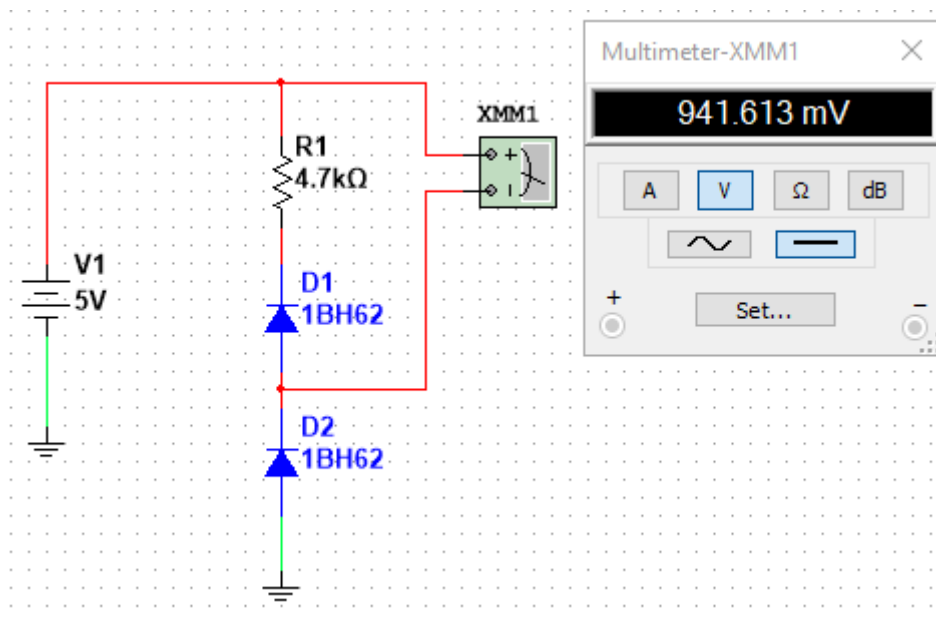


Figure 1.4.21 Simulation of the 8^o circuit in "Multisim"

ELECTRONICS

9° Circuit ($V_9 = 0.9$ Volt)

In the figure in Figure 1.4.12, the diodes do not open, as the positive pole of the source is connected to the N contacts of the diodes. The voltmeter reading is the potential difference between the node supplied by the 5 volt DC source and the undefined voltage node between the two reverse-polarized diodes D1 and D2. Specifically, the voltmeter reading is at 0.9 volts. Figure 1.4.22 shows the simulated solution of the 9^o circuit in the simulation software "Multisim" which verifies the theoretical solution.

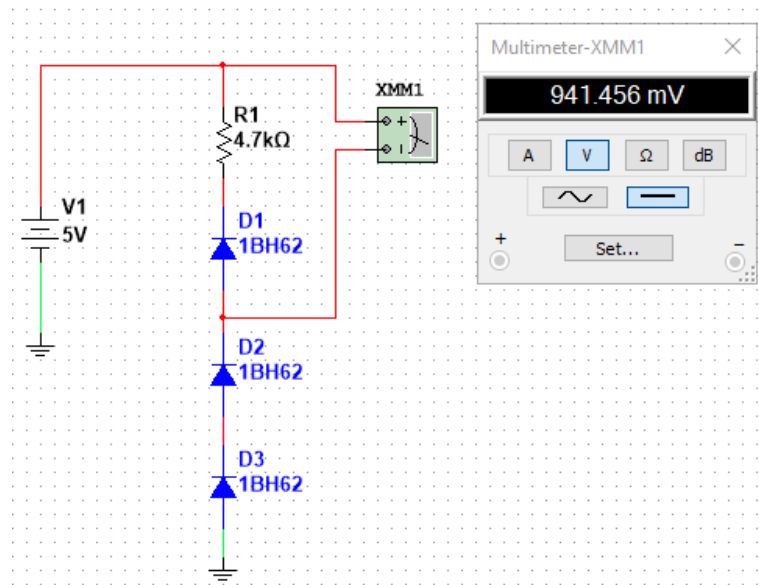


Figure 1.4.22 Simulation of the 9^o circuit in "Multisim"

10° Circuit ($V_{10} = 0.9$ Volt)

In the figure in Figure 1.4.13, diodes D1, D2 and D3 do not open, as the positive pole of the source is connected to the N contacts of the diodes. Diode D4 on the other hand turns on, as the cathode voltage is 0 volts from ground and the anode voltage is the positive undefined voltage between the inversely polarized diodes D2 and D3. The voltmeter reading is the potential difference between the node supplied by the 5 volt DC voltage source and the undefined voltage node between the two inversely polarized diodes D1 and D2. Specifically, the voltmeter reading is at 0.9 volts. Figure 1.4.23 shows the simulated solution of the 10^o circuit in the simulation software "Multisim" which verifies the theoretical solution.

ELECTRONICS

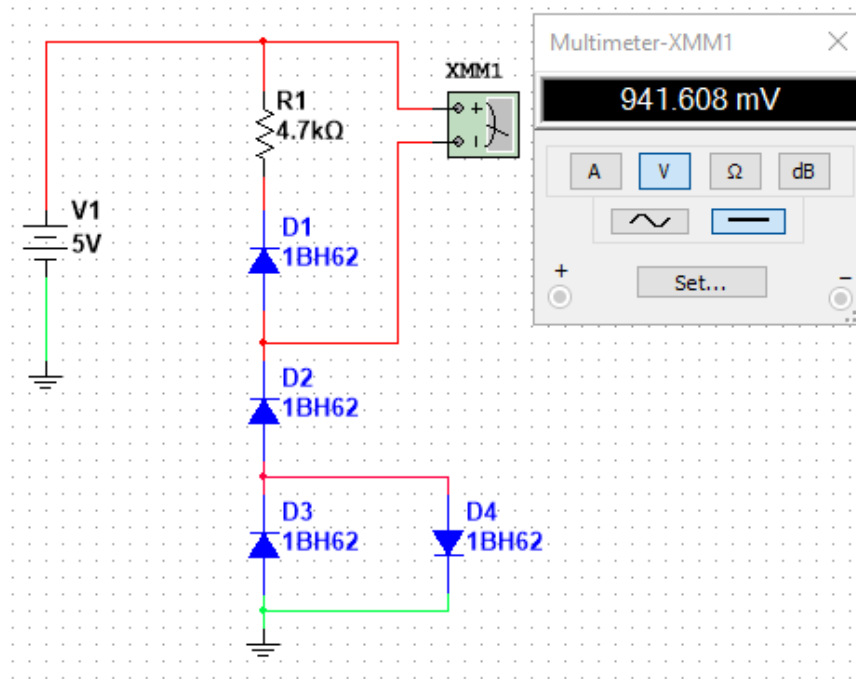


Figure 1.4.23 Simulation of the 10^{oo} circuit in "Multisim"

2. DIODE ZENER

2.1 FORWARD BIAS

General

The chapters "Simulation Solution" and "Experimental Solution" present in detail the components used, the recorded measurements and the observations made during the experiment with snapshots from the simulation software "Multisim" and photographs of the experiment from the laboratory environment respectively, while the chapter "Theoretical Solution" analyses the mathematical proof of the experiment.

Theoretical Solving

Figure 2.1.1 shows the schematic representation of the correct polarization of the Zener diode from the design software "MS Visio".

ELECTRONICS

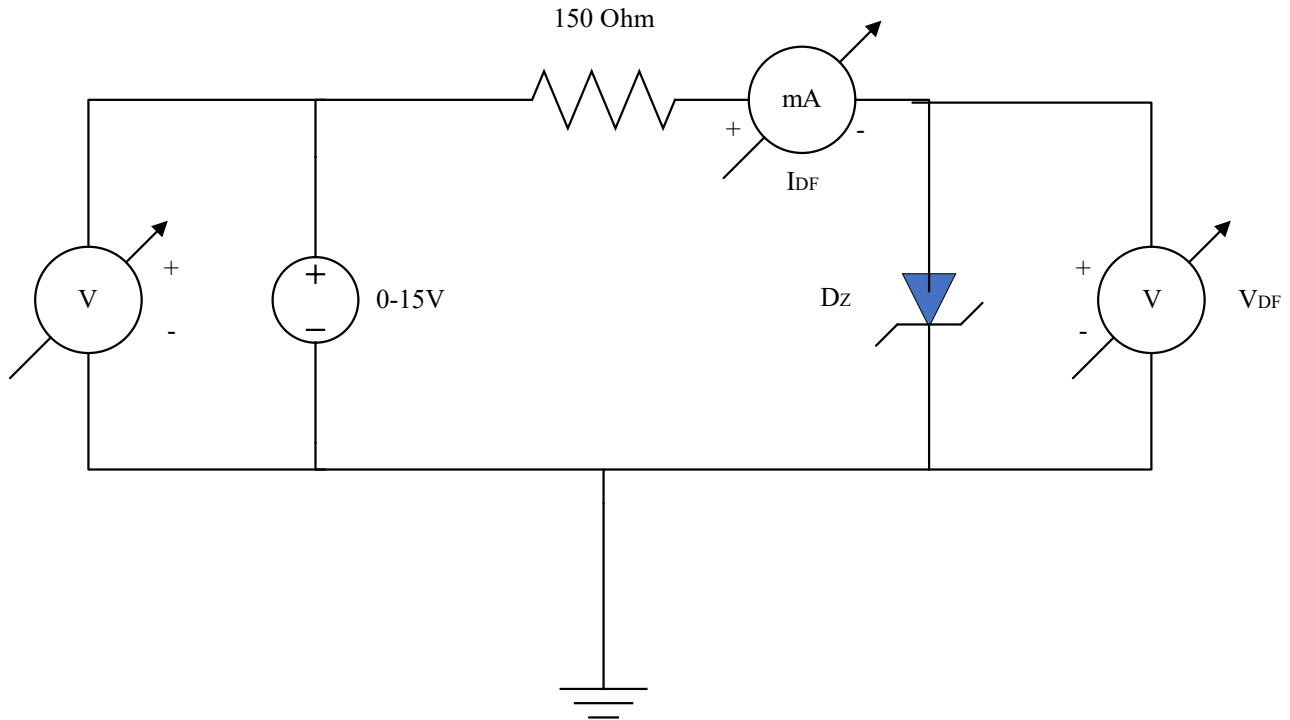


Figure 2.1.1 Schematic representation of correct Zener polarization in "MS Visio"

Table 6 shows the measurements made in the Zener diode right bias experiment, in terms of the current " I_{DF} " flowing through the diode and the potential difference " V_{DF} " between the ends of the diode. The measurements were made for various values of current flowing through the circuit by varying the source supply and a resistor with a resistance of 150 Ohm is connected to the circuit.

Table 6.

I_{DF} (mA)	1	5	10	15	20	25	35	45
V_{DF} (V)	0.7	0.8	0.8	0.8	0.8	0.8	0.9	0.9

The observations recorded according to the data in Table 6 are as follows:

- The potential difference " V_{DF} " at the ends of the Zener diode increases gradually as the current intensity increases from the change in the DC source supply.
- The Zener diode in the right bias behaves like the common diode.

According to Kirchhoff's 2^o rule (law of voltages), the sum of the potential differences in the individual branches of a loop is equal to zero. By Ohm's law, it is true that the potential difference at the ends of a branch is proportional to the intensity of the current flowing through the branch. The measurements recorded in Table 6 are verified from the above laws. For example at 20 mA current with a supply voltage at 3.8 Volts (Figure 2.1.8) we have:

$$V_{\pi\eta\gamma\eta} + V_R + V_{DF} = 0 \rightarrow 3.8 \text{ Volt} - (I_{DF} \times R) - 0.8 \text{ Volt} = 0 \rightarrow$$

$$3.8 \text{ Volt} - (0.02 \text{ Ampere} \times 150 \text{ Ohm}) - 0.8 \text{ Volt} = 0 \rightarrow$$

$$3.8 \text{ Volt} - 3 \text{ Volt} - 0.8 \text{ Volt} = 0 \rightarrow 0 \cong 0$$

ELECTRONICS

The diode behaves as a conductor. Figure 2.1.2 shows the characteristic of a Zener diode during right-hand polarization, varying the current flowing through it according to Table 6.

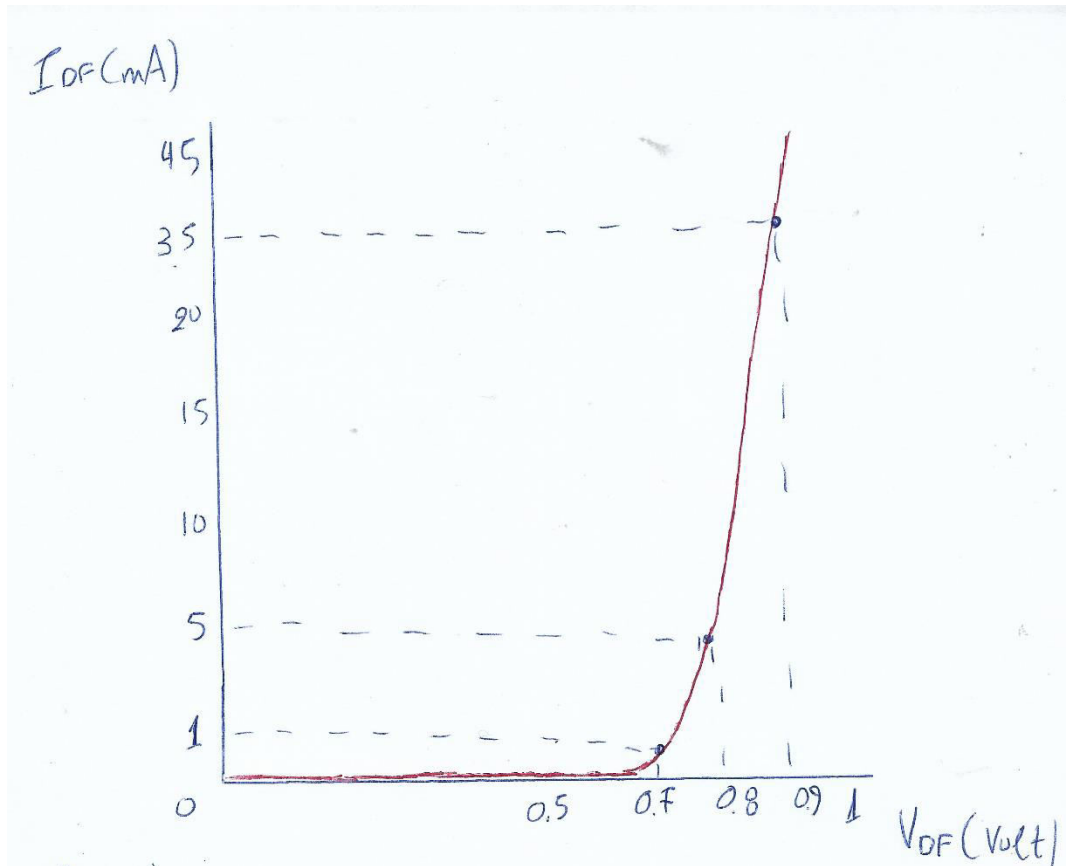
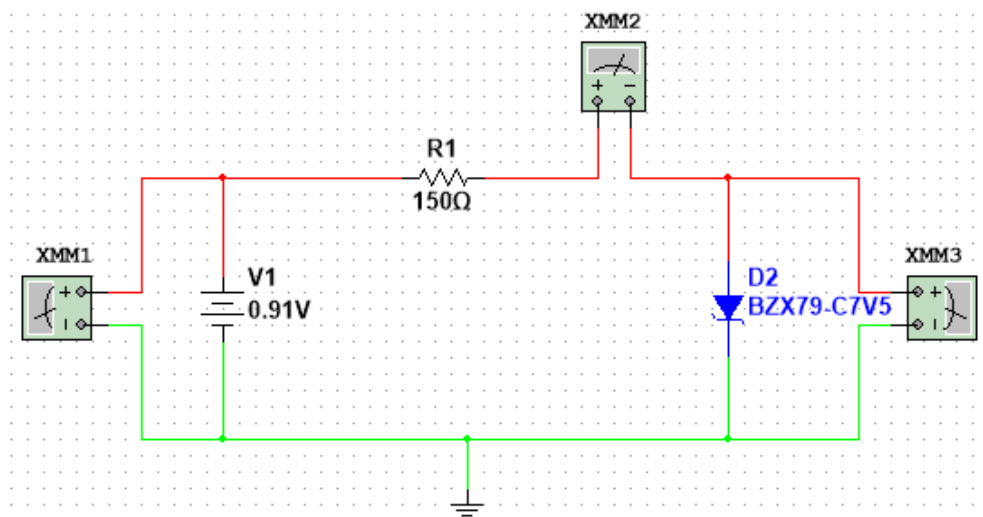


Figure 2.1.2 Zener I polarization characteristic $I_{DF} - V_{DF}$

Simulative Resolution

Figure 2.1.3 shows the simulation of the right bias Zener diode in the simulation software "Multisim".



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Figure 2.1.3 Simulation of the correct polarization of a Zener diode in "Multisim"

In Figures 2.1.4, 2.1.5, 2.1.6, 2.1.7, 2.1.8, 2.1.9, 2.1.10 and 2.1.11 the recorded measurements of Table 6 are simulated in the simulation software "Multisim".

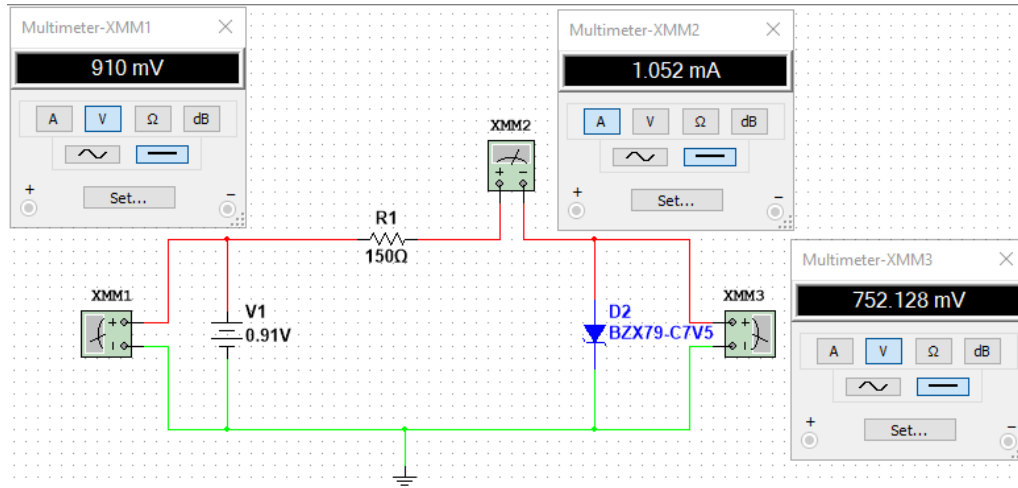


Figure 2.1.4 Correct polarization of a Zener diode with a current of 1 mA in "Multisim"

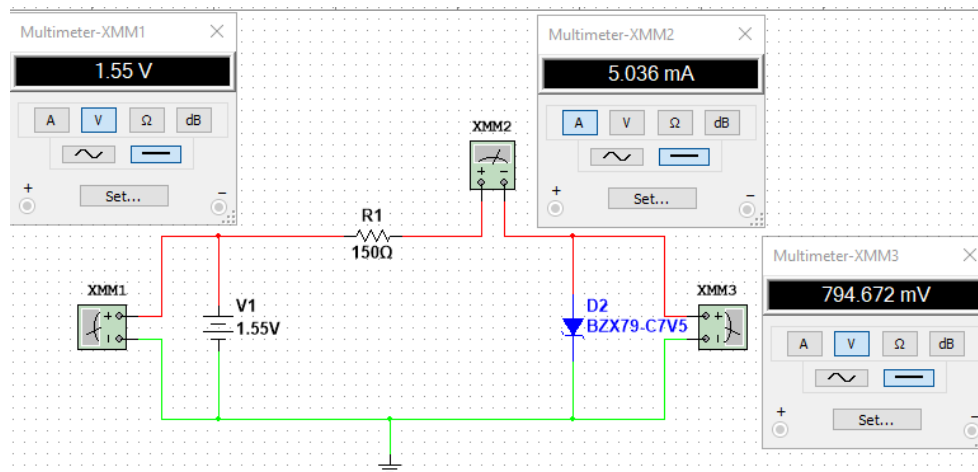


Figure 2.1.5 Correct polarization of a Zener diode with 5 mA current in "Multisim"

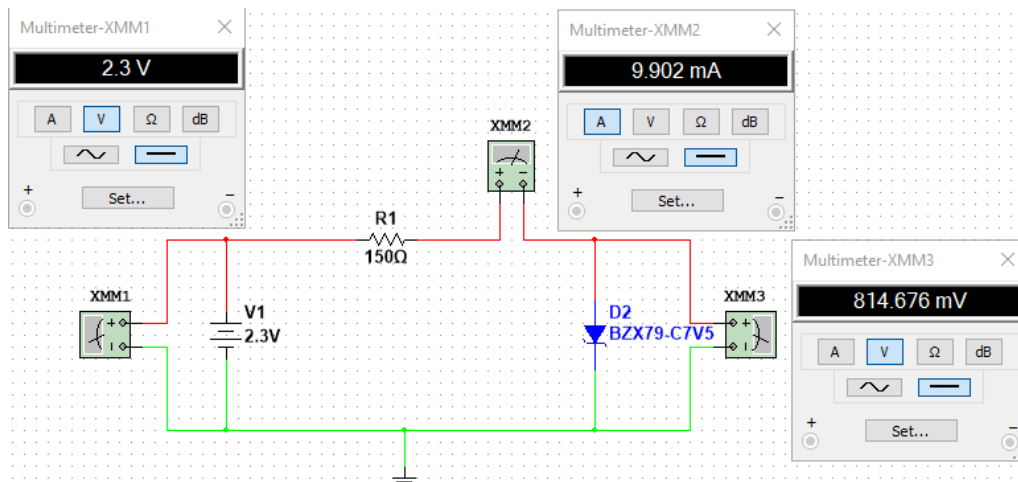


Figure 2.1.6 Correct polarization of a Zener diode with 10 mA current in "Multisim"

ELECTRONICS

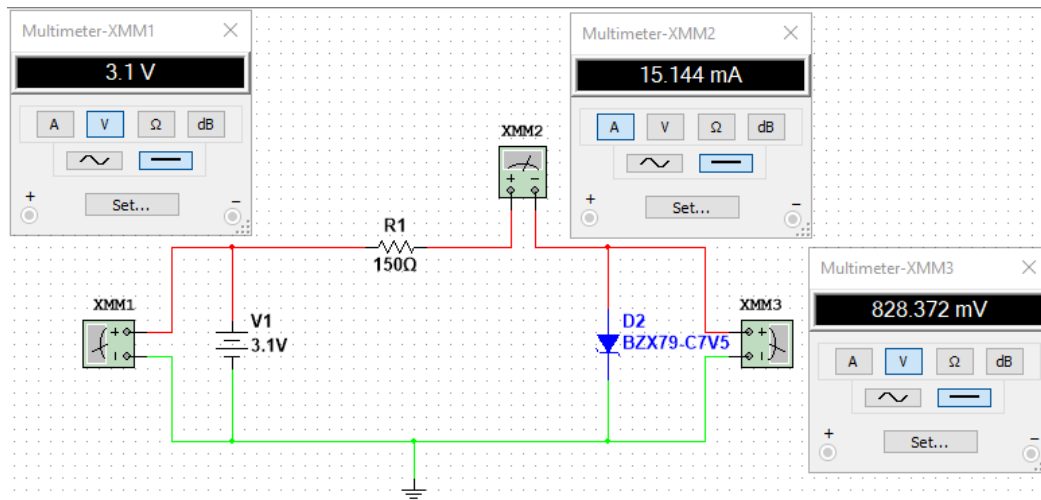


Figure 2.1.7 Correct polarization of a Zener diode with 15 mA current in "Multisim"

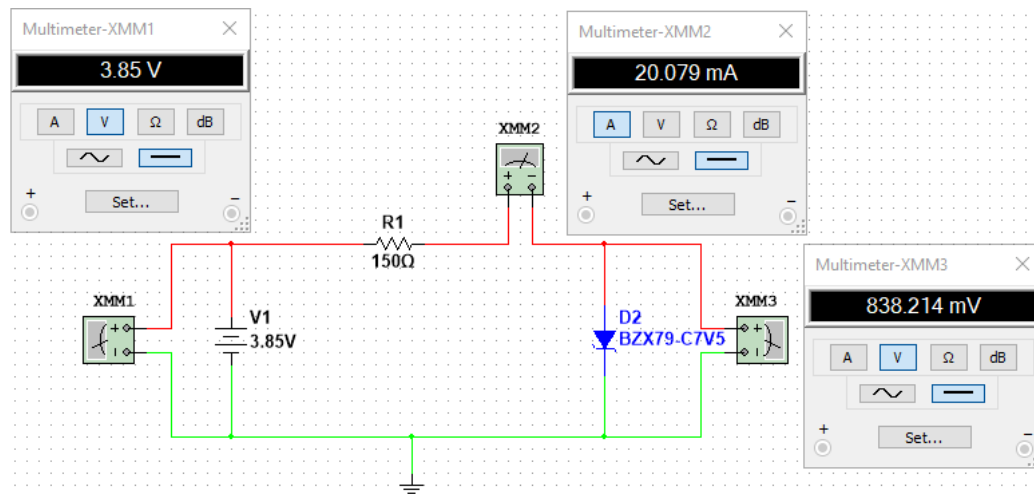


Figure 2.1.8 Correct polarization of a Zener diode with 20 mA current in "Multisim"

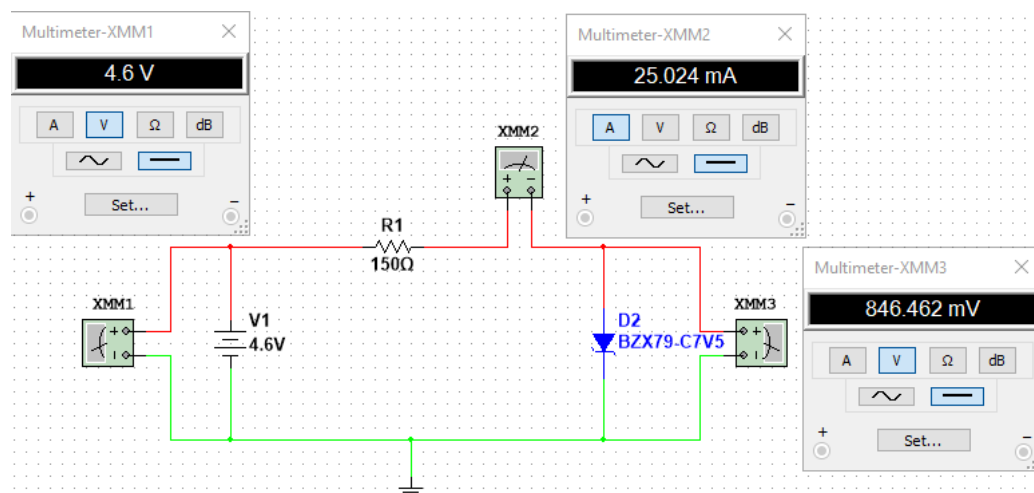


Figure 2.1.9 Correct polarization of a Zener diode with 25 mA current in "Multisim"

ELECTRONICS

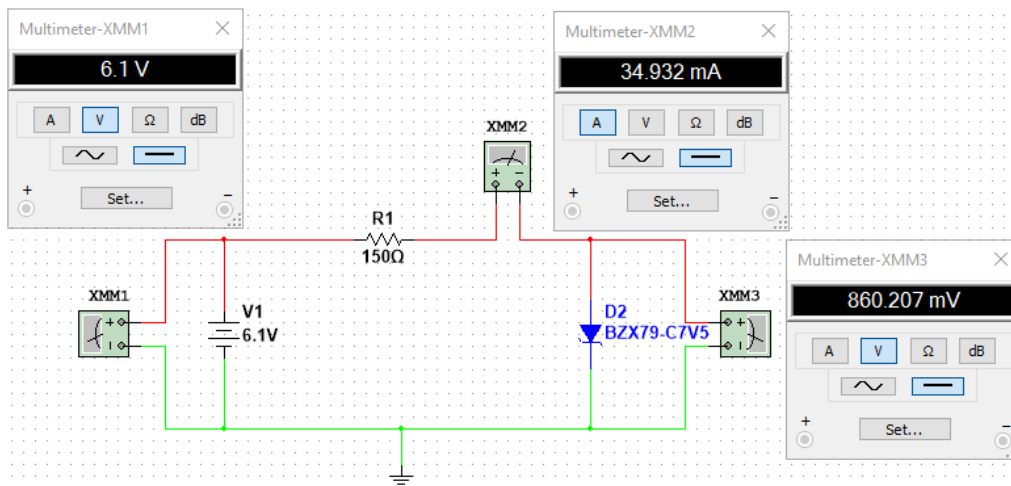


Figure 2.1.10 Correct polarization of a Zener diode with a current of 35 mA in "Multisim"

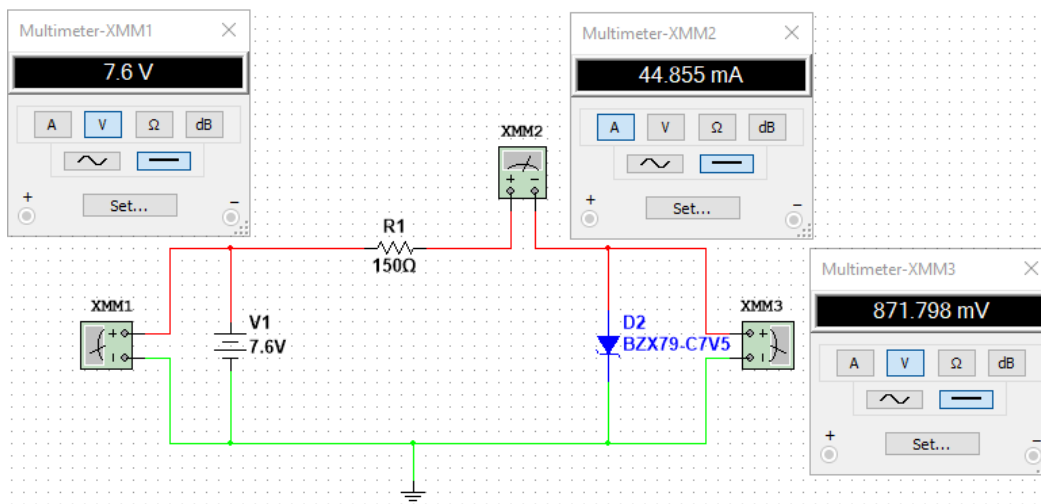


Figure 2.1.11 Correct polarization of a Zener diode with 45 mA current in "Multisim"

Experimental Solution

Figure 2.1.12 shows the illustration of the correct polarization of the Zener diode on the "breadboard", where it was carried out in the laboratory.

First, we connect the voltage supply (0-15 Volt) and a digital bench top multimeter set to calculate voltage in Volts, in parallel to the ends of the supply in order to calculate the source voltage. Next, we connect the 150 Ohm resistor in series with the source, as well as connect the diode in series with the resistor. The diode will be correctly polarized, which means that the N contact (black line) will be turned to ground, so that the anode voltage (P contact) will be greater than the cathode voltage (N contact). We connect the simple digital multimeter set to calculate current intensity in mA, in series with the resistor and before the diode, in order to calculate the intensity of the current flowing through the diode and thus the whole circuit. Finally, we connect a digital bench multimeter set to calculate voltage in volts, in parallel to the ends of the diode, in order to calculate the potential difference at the ends of the diode. We also use a wire to connect the earth ground.

ELECTRONICS

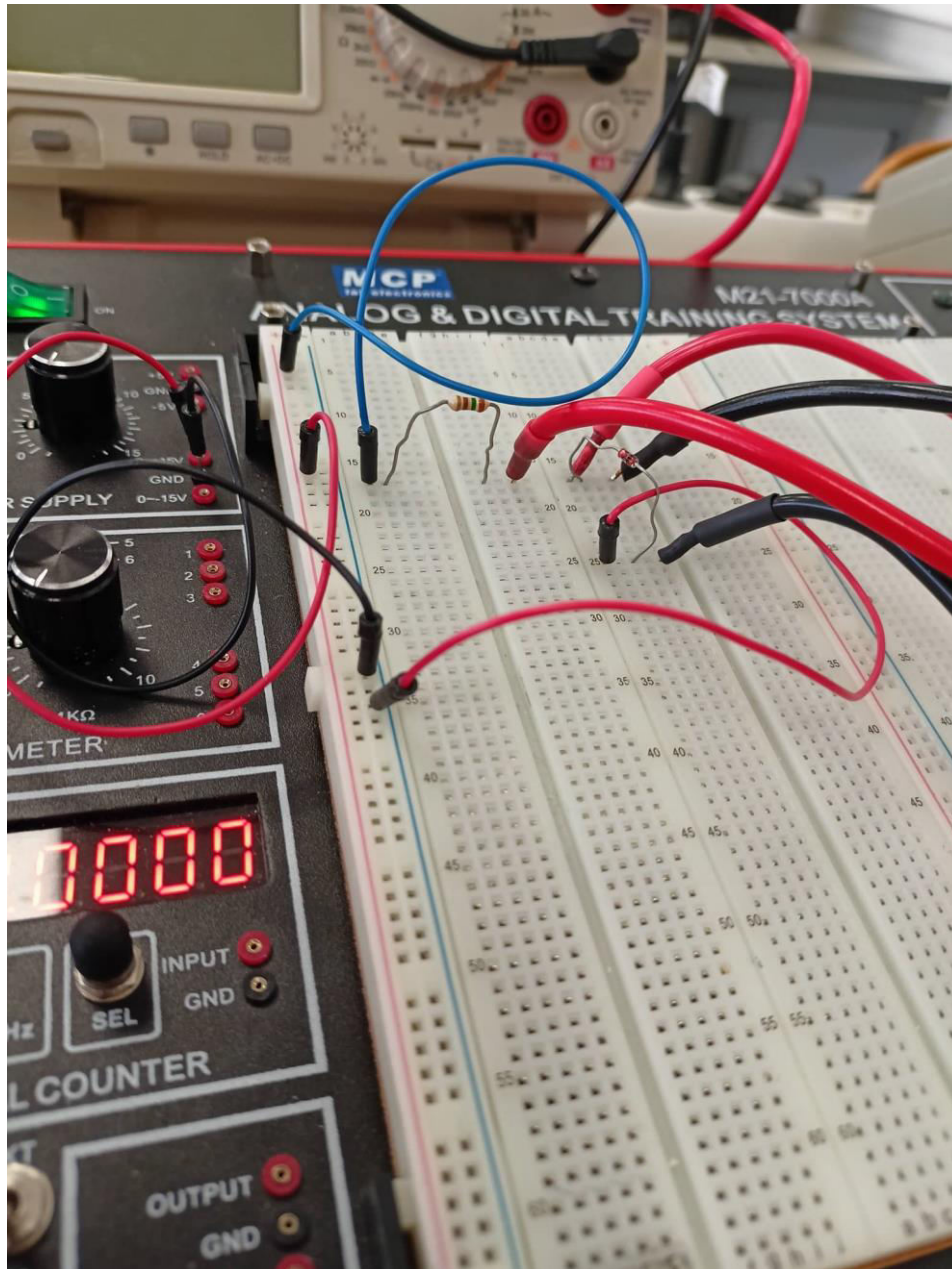


Figure 2.1.12 Experimental illustration of the correct polarization of a Zener diode on the "Breadboard"

In Figures 2.1.13, 2.1.14, 2.1.15, 2.1.16, 2.1.17, 2.1.18, 2.1.19 and 2.1.20 the recorded measurements of Table 6 are experimentally verified on the "Breadboard" interface board from the laboratory environment.

ELECTRONICS

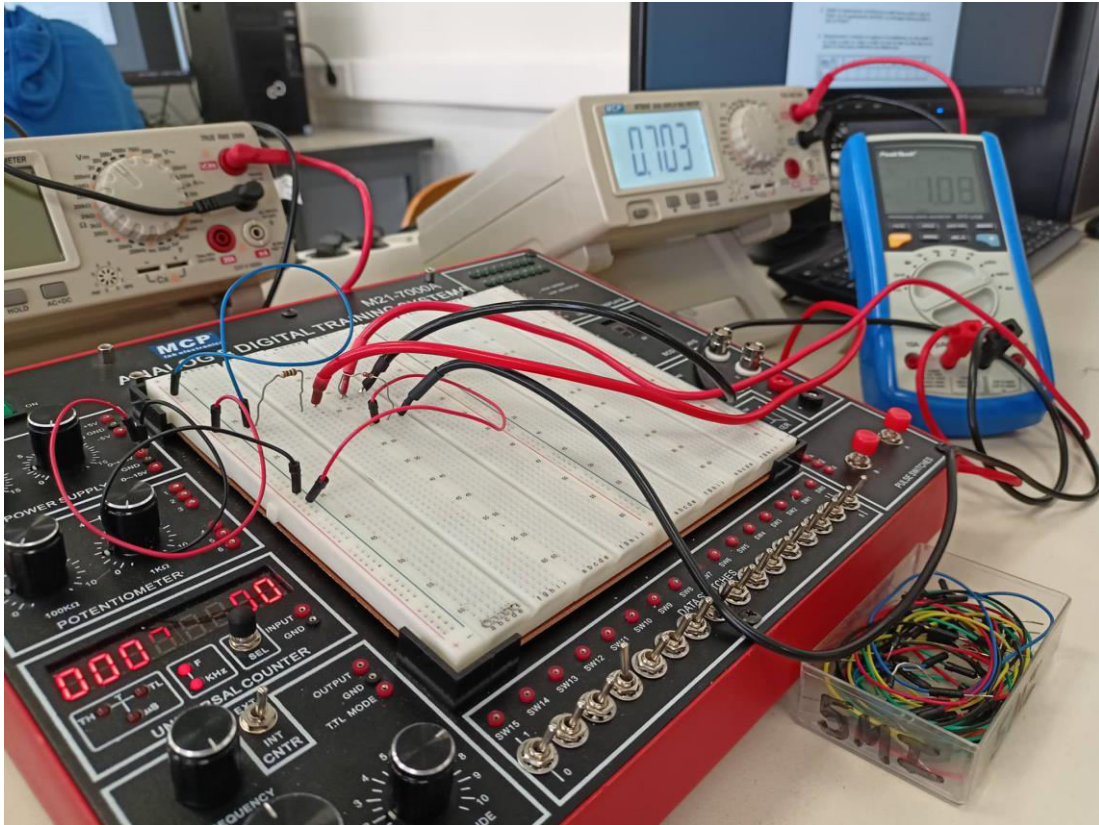


Figure 2.1.13 Correct polarization of a Zener diode with a current of 1 mA on the "Breadboard"

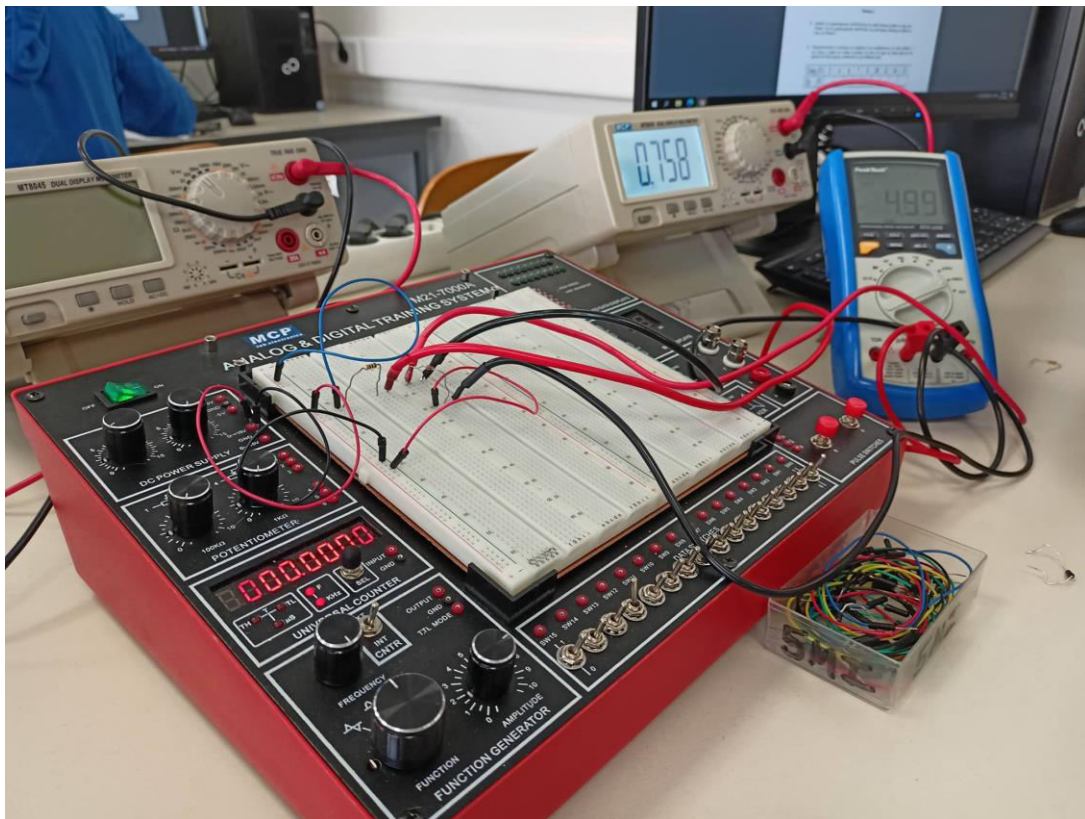


Figure 2.1.14 Correct polarization of a Zener diode with a 5 mA supply voltage on the "Breadboard"

ELECTRONICS

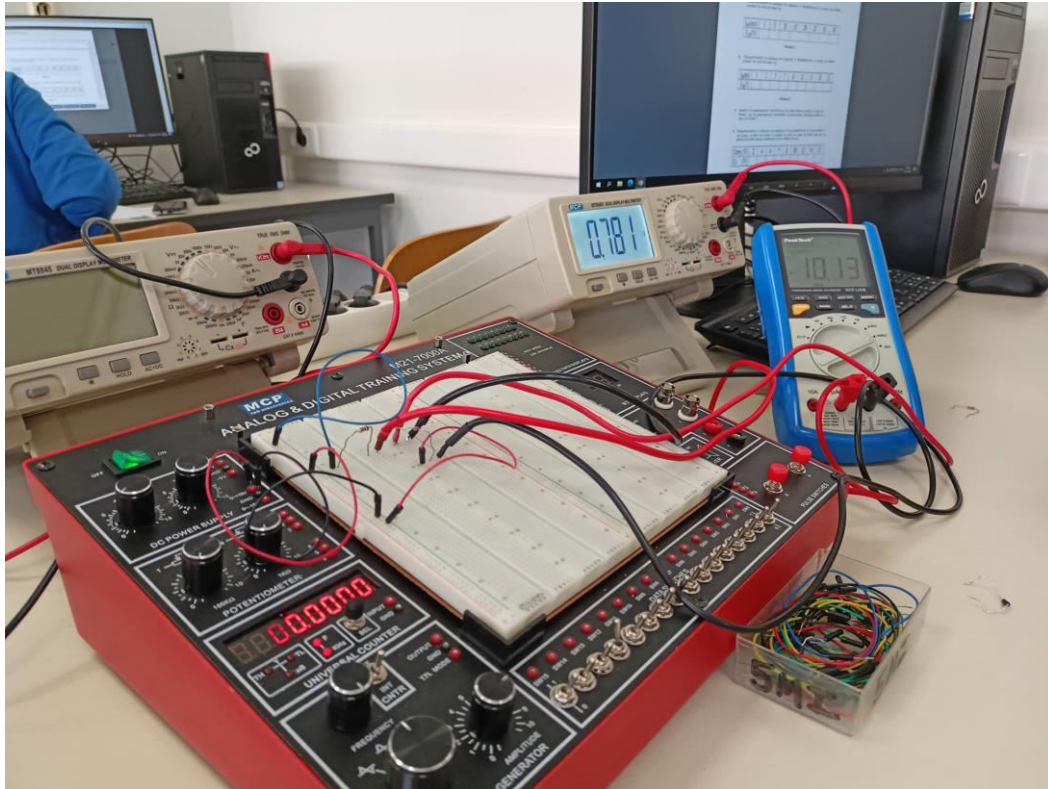


Figure 2.1.15 Correct polarization of a Zener diode with a 10 mA supply voltage on the "Breadboard"

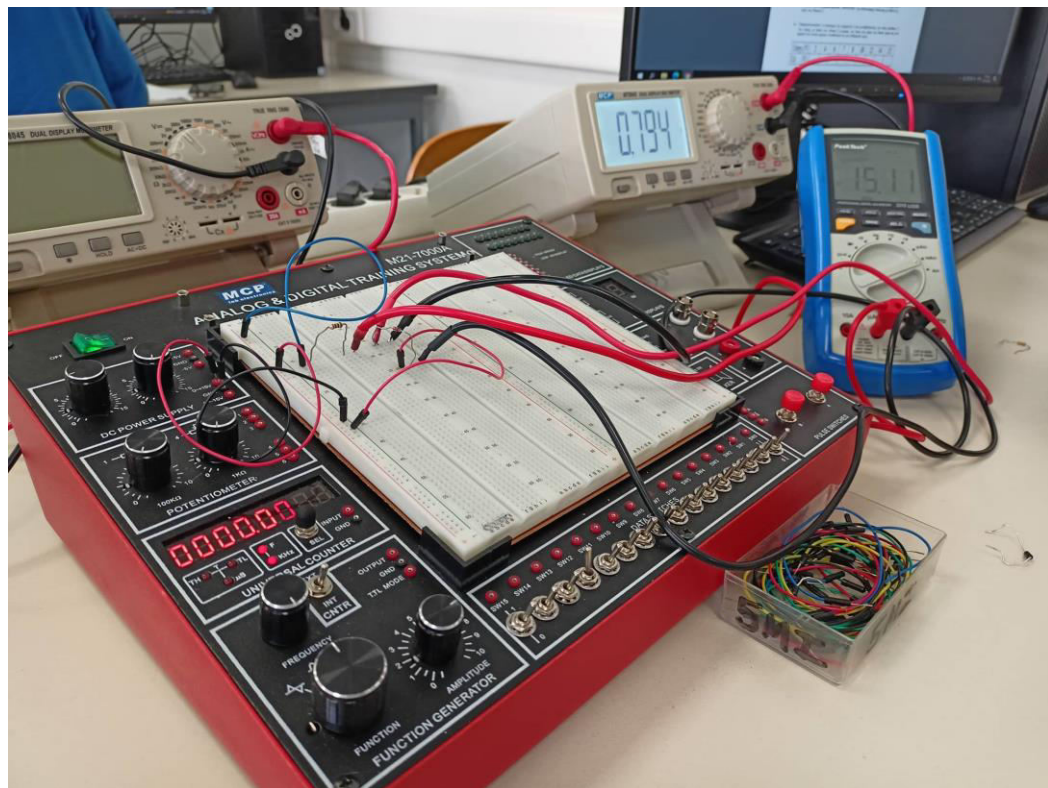


Figure 2.1.16 Correct polarization of a Zener diode with a 15 mA supply voltage on the "Breadboard"

ELECTRONICS

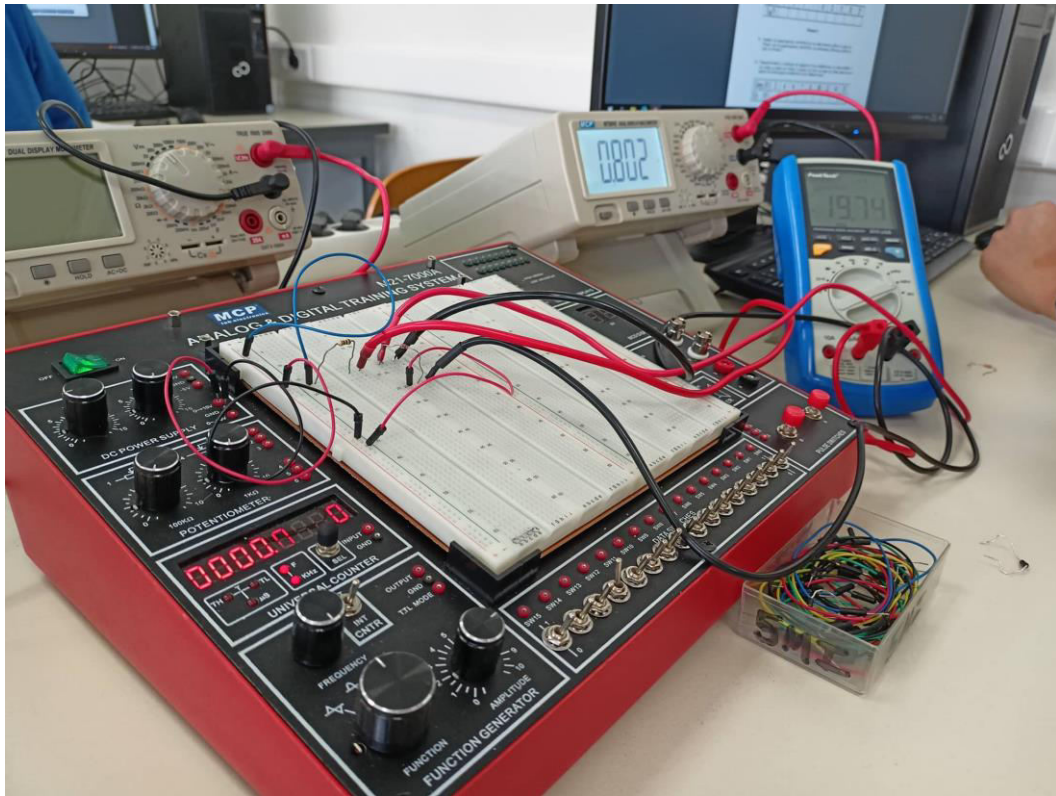


Figure 2.1.17 Correct polarization of a Zener diode with a 20 mA supply voltage on the "Breadboard"

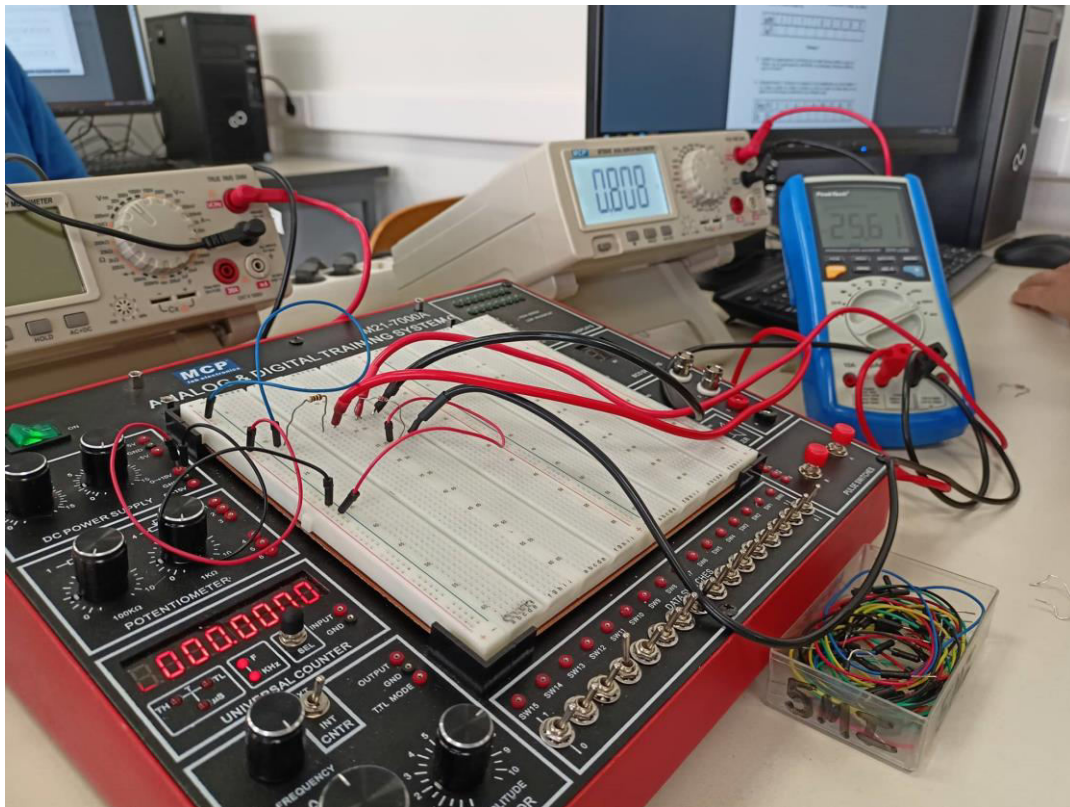


Figure 2.1.18 Correct polarization of a Zener diode with a supply voltage of 25 mA on the "Breadboard"

ELECTRONICS

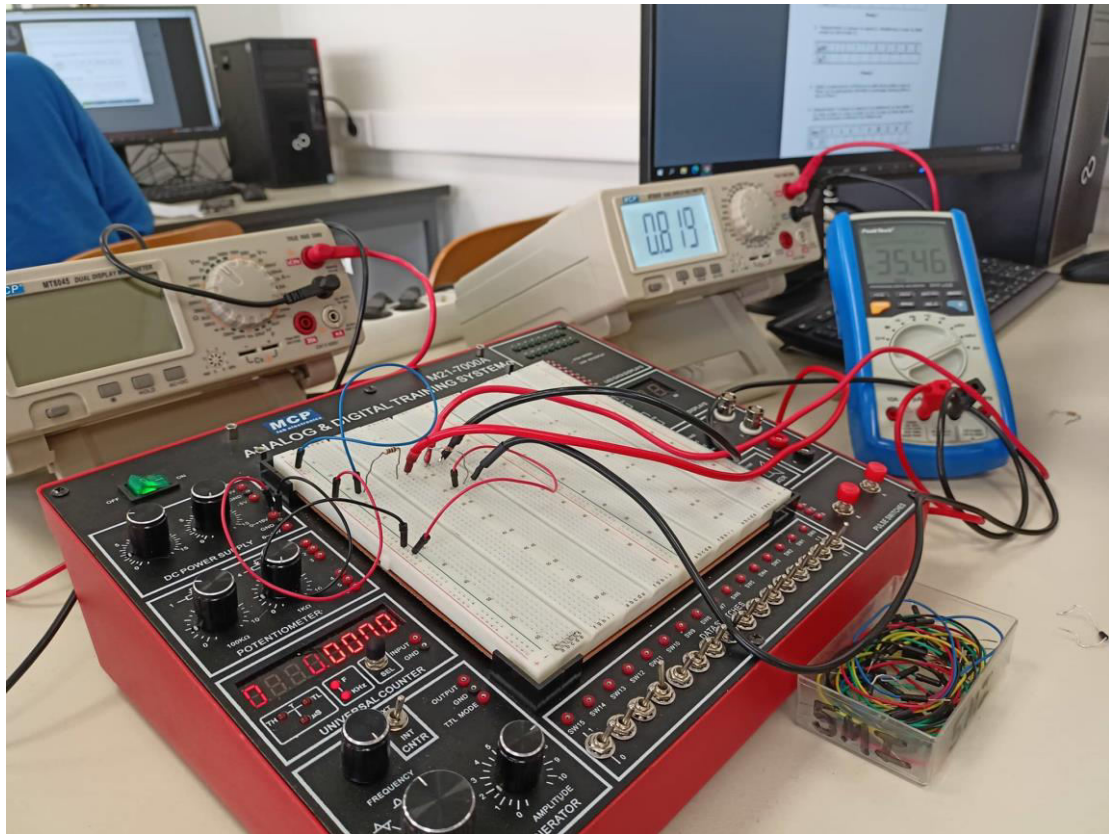


Figure 2.1.19 Correct polarization of a Zener diode with 35 mA supply voltage on the "Breadboard"

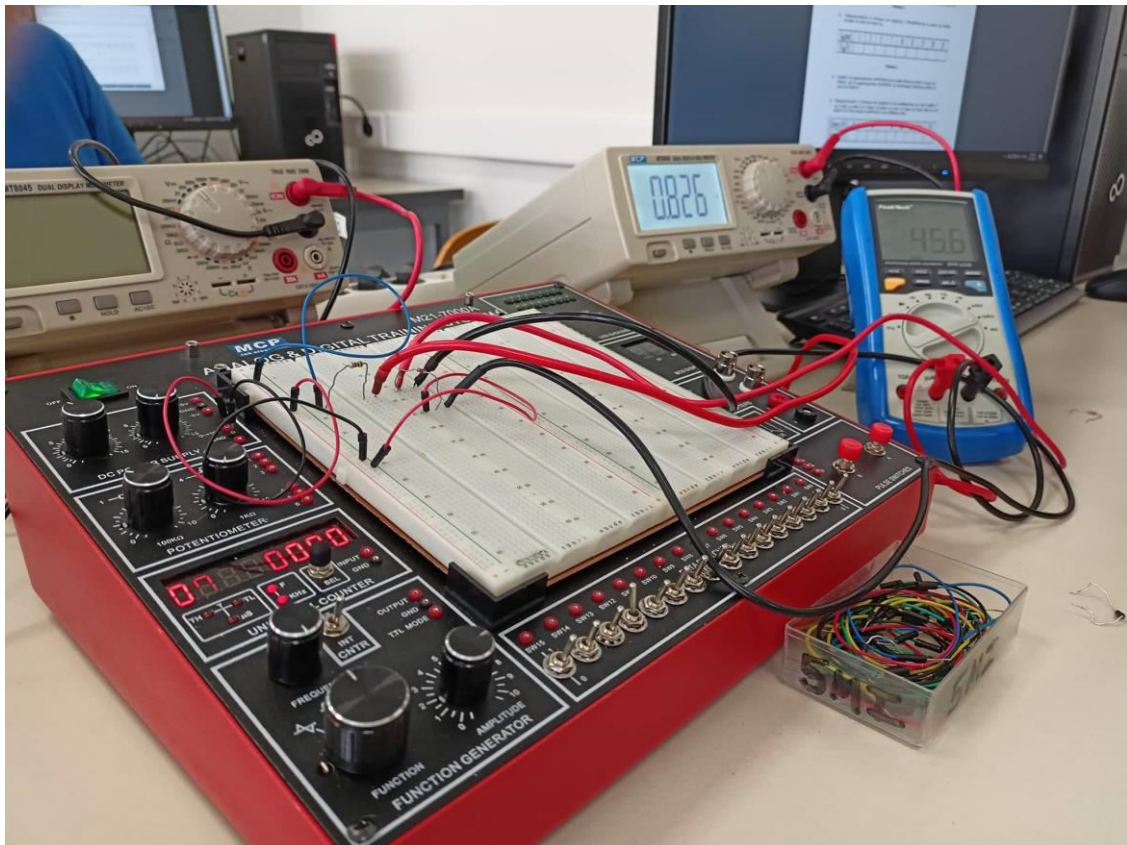


Figure 2.1.20 Correct polarization of a Zener diode with a supply voltage of 45 mA on the "Breadboard"

ELECTRONICS

2.2 INVERSE BIAS

General

The chapters "Simulation Solution" and "Experimental Solution" present in detail the components used, the recorded measurements and the observations made during the experiment with snapshots from the simulation software "Multisim" and photographs of the experiment from the laboratory environment respectively, while the chapter "Theoretical Solution" analyses the mathematical proof of the experiment.

Theoretical Solving

Figure 2.2.1 shows the schematic representation of the Zener diode reverse bias from the design software "MS Visio".

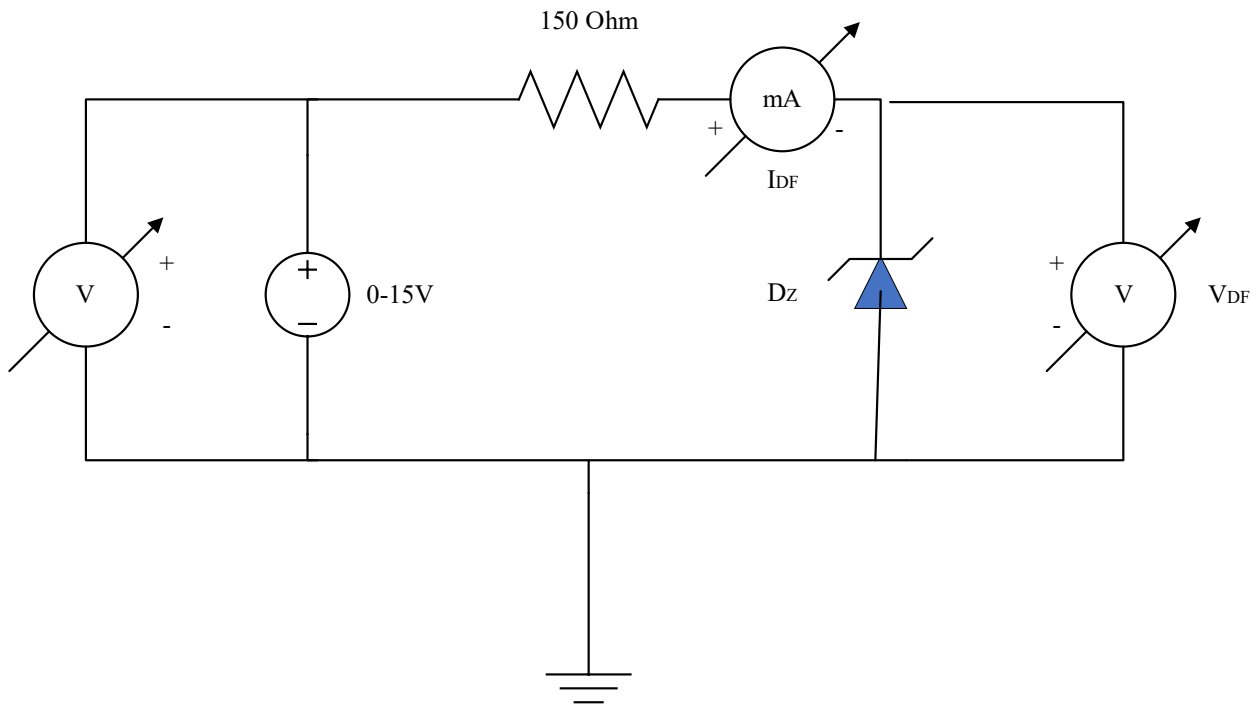


Figure 2.2.2.1 Schematic illustration of Zener inverse polarization in "MS Visio"

Table 7 presents the measurements made in the Zener diode reverse bias experiment, in terms of the current " I_{DF} " flowing through the diode and the potential difference " V_{DF} " between the ends of the diode. Measurements were made for various values of current flowing through the circuit by varying the source supply and a resistor with a resistance of 150 Ohm is connected to the circuit.

Table 7.[illegible]

ELECTRONICS

The observations recorded according to the data in Table 7 are as follows:

- The potential difference " V_{DR} " at the ends of the silicon diode remains approximately the same (due to the voltage drop from the resistor) as the supply voltage every time the current increases.
- No matter how much the supply voltage of the source is increased, the current will increase rapidly, but the potential difference at the ends of the diode will remain almost unchanged.

According to Kirchhoff's 2^o rule (law of voltages), the sum of the potential differences in the individual branches of a loop is equal to zero. By Ohm's law, it is true that the potential difference at the ends of a branch is proportional to the intensity of the current flowing through the branch. From the above laws, the measurements recorded in Table 7 are verified. For example at 10 mA current with a supply voltage at 9.1 Volts (Figure 2.2.) we have:

$$V_{\text{source}} + V_R + V_{DR} = 0 \rightarrow 9.1 \text{ Volt} + (I_{DF} \times R) - 7.5 \text{ Volt} = 0 \rightarrow$$

$$9.1 \text{ Volt} - (0.01 \text{ Ampere} \times 150 \text{ Ohm}) - 7.5 \text{ Volt} = 0 \rightarrow$$

$$9.1 \text{ Volt} - 1.5 \text{ Volt} - 7.5 \text{ Volt} = 0 \rightarrow 0 \cong 0$$

The voltage at the ends of the Zener diode remains approximately the same. The internal resistance is very high, which explains the very low current flowing through the diode (Ohm's law). The diode behaves like an insulator except that for large power supplies that result in sudden increases in current, it will not be destroyed like the silicon diode. Figure 2.2.2 shows the Zener diode characteristic during reverse biasing, varying the supply voltage according to Table 7.

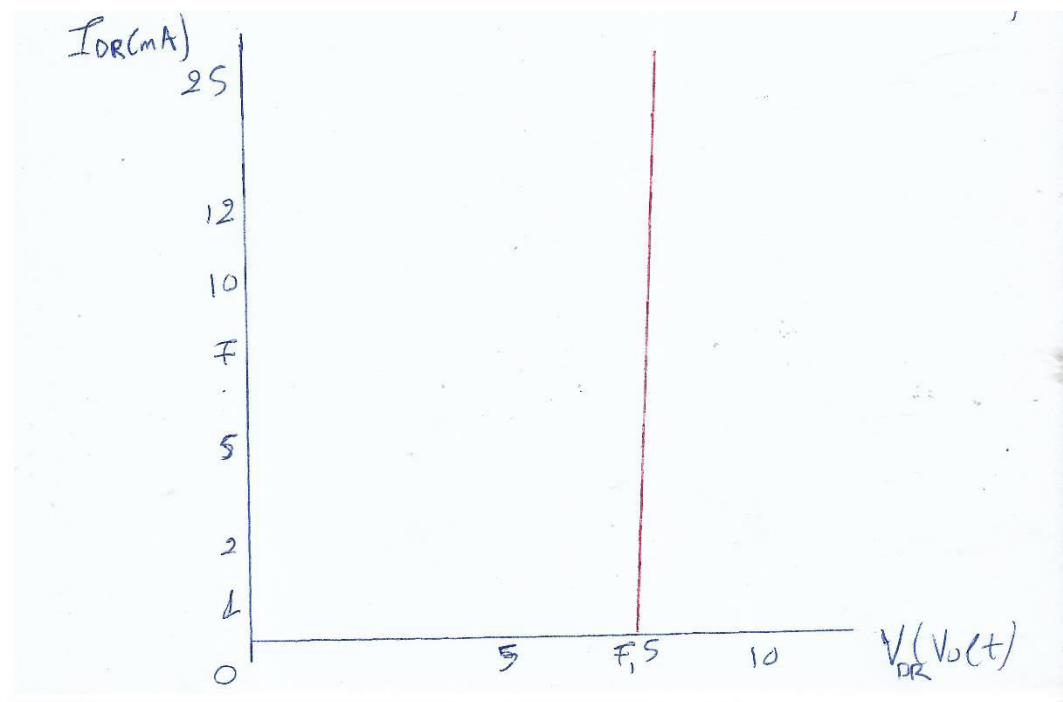


Figure 2.2.2.2 Zener I reverse bias characteristic $I_{DF} - V_{DF}$

ELECTRONICS

Simulative Resolution

Figure 2.2.3 shows the simulation of the Zener diode reverse bias in the simulation software "Multisim".

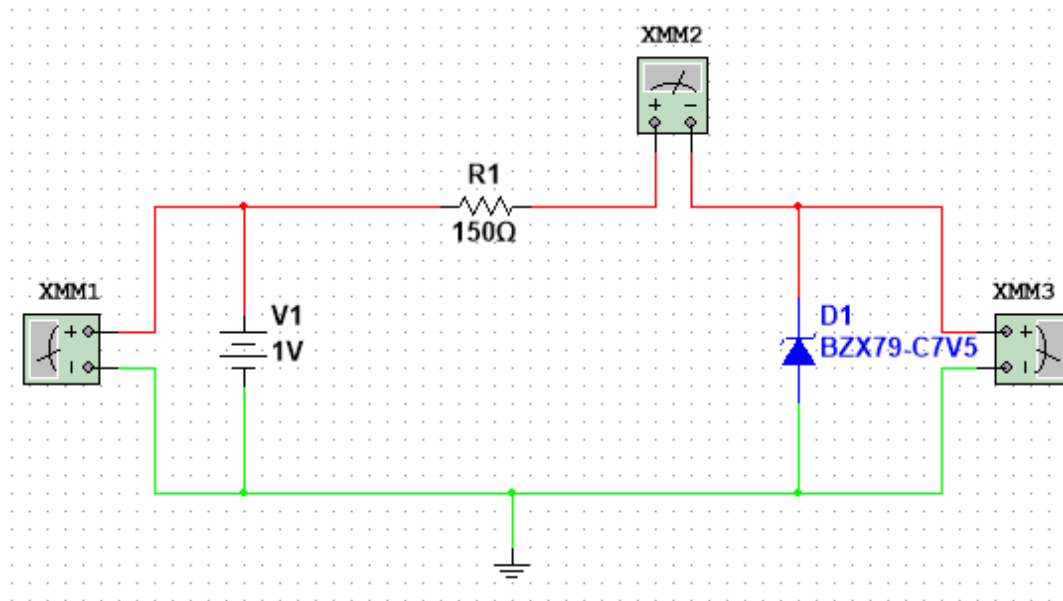


Figure 2.2.3 Simulation of Zener diode reverse bias in "Multisim"

In Figures 2.2.4, 2.2.5, 2.2.6, 2.2.7, 2.2.8, 2.2.9, 2.2.10, 2.2.11 and 2.2.12 the recorded measurements of Table 7 are simulated in the simulation software "Multisim".

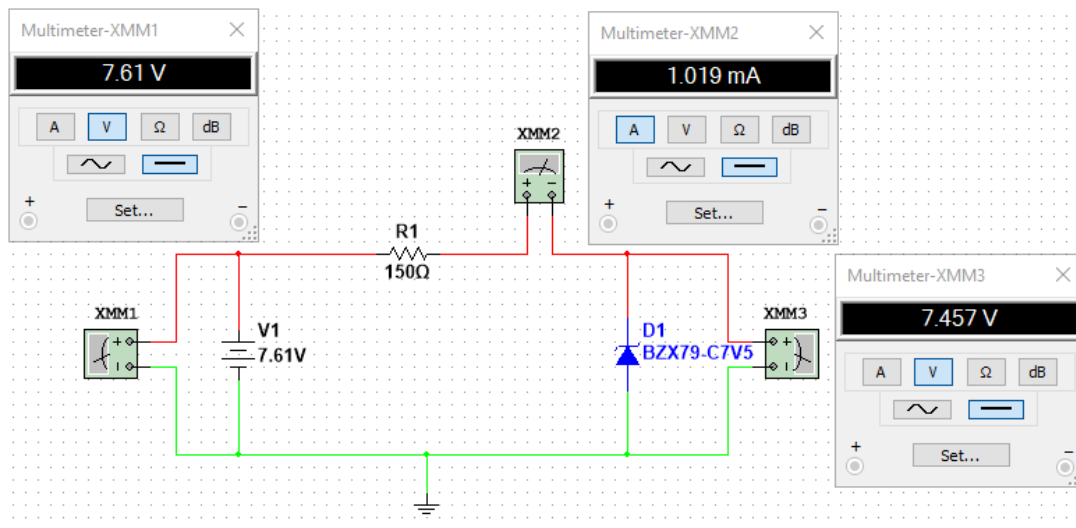


Figure 2.2.4 Reverse polarization of a Zener diode with 1 mA current in "Multisim"

ELECTRONICS

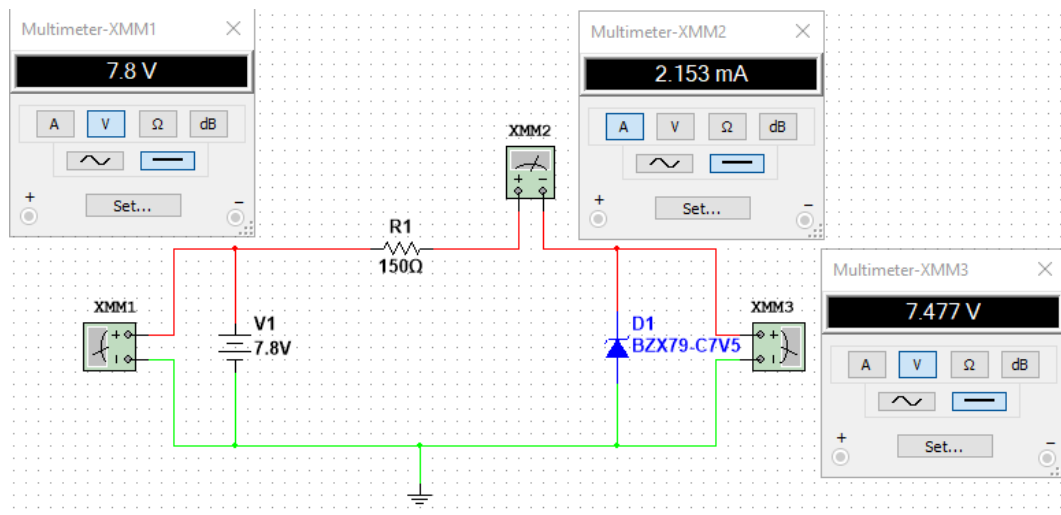


Figure 2.2.5 Reverse biasing of a 2 mA Zener diode in "Multisim"

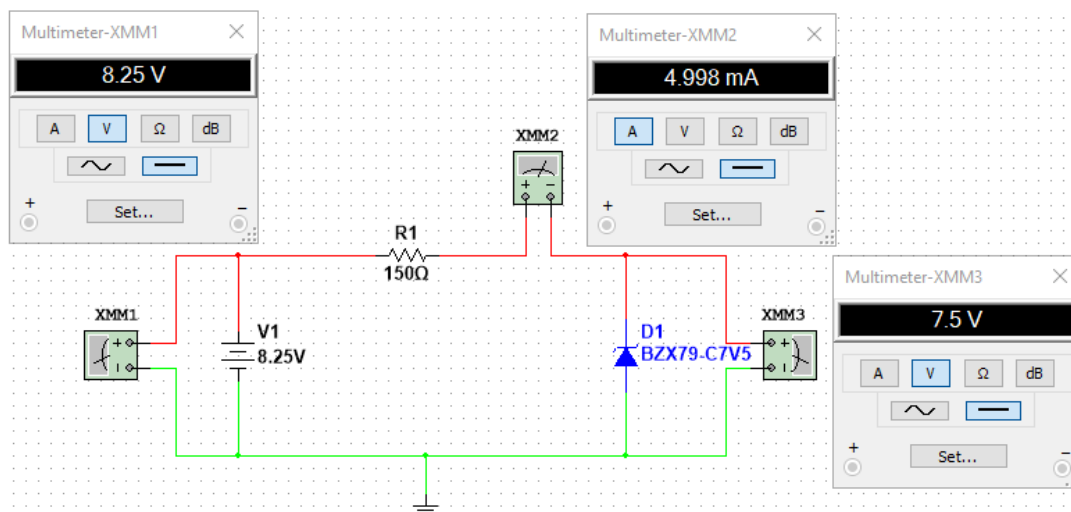


Figure 2.2.2.6 Reverse biasing of a 5 mA Zener diode in "Multisim"

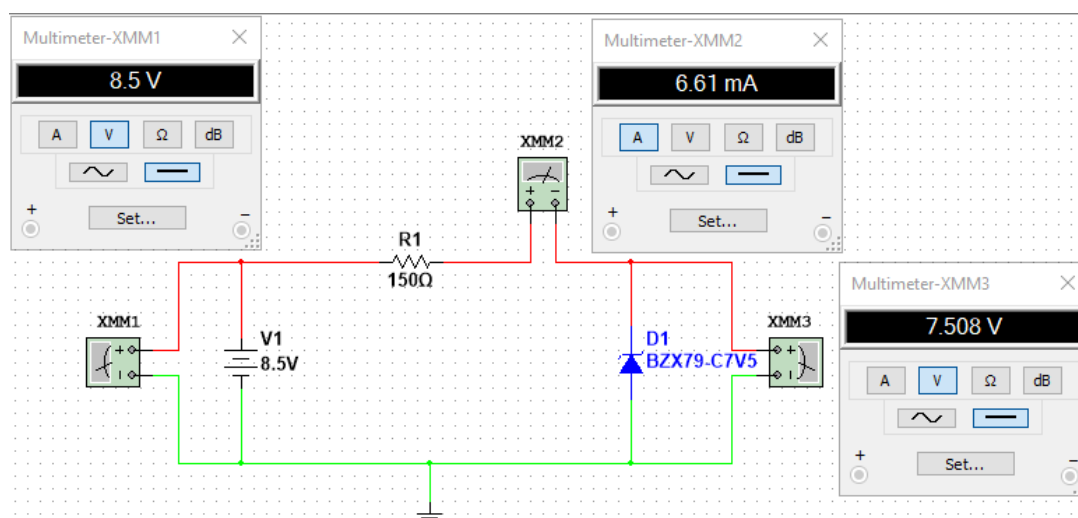


Figure 2.2.2.7 Reverse biasing of a Zener diode with a current of 7 mA in "Multisim"

ELECTRONICS

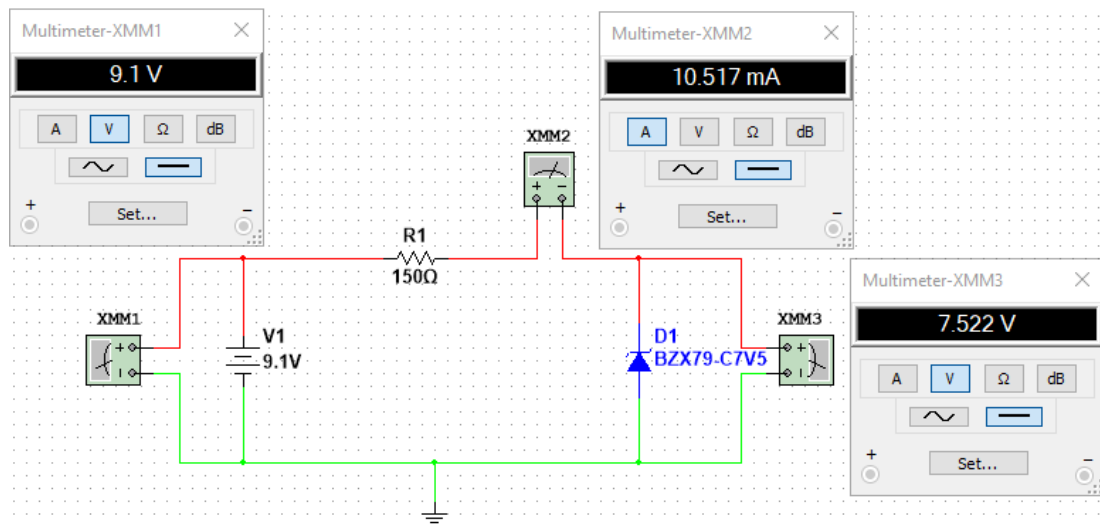


Figure 2.2.8 Reverse polarization of a 10 mA Zener diode in "Multisim"

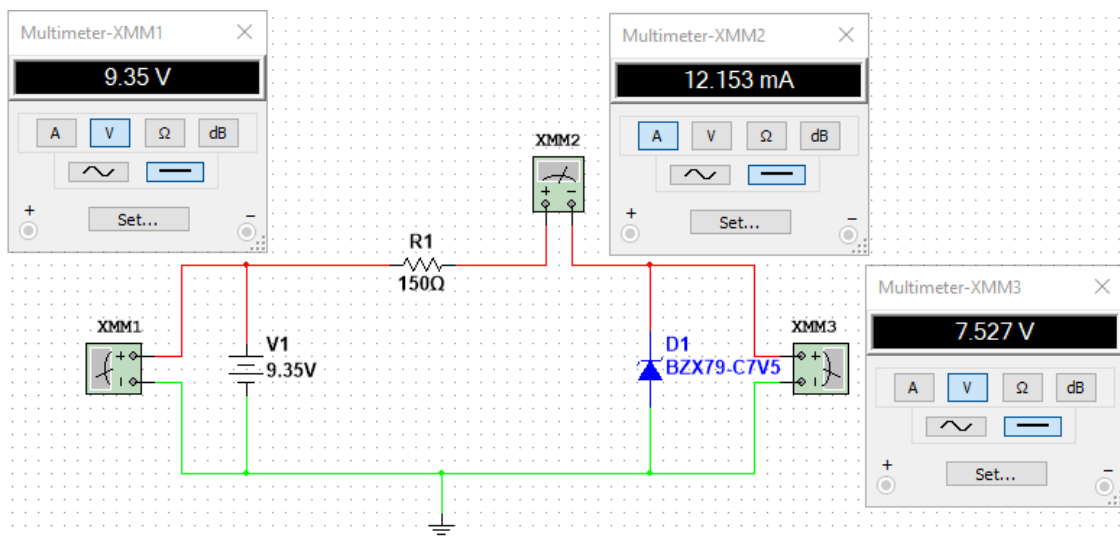


Figure 2.2.9 Reverse biasing of a 12 mA Zener diode in "Multisim"

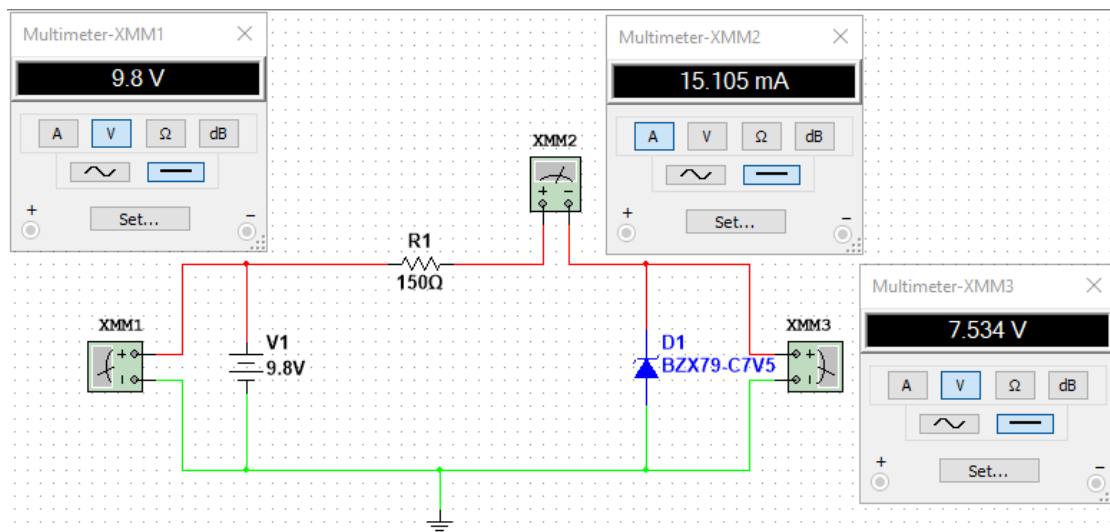


Figure 2.2.10 Reverse biasing of a 15 mA Zener diode in "Multisim"

ELECTRONICS

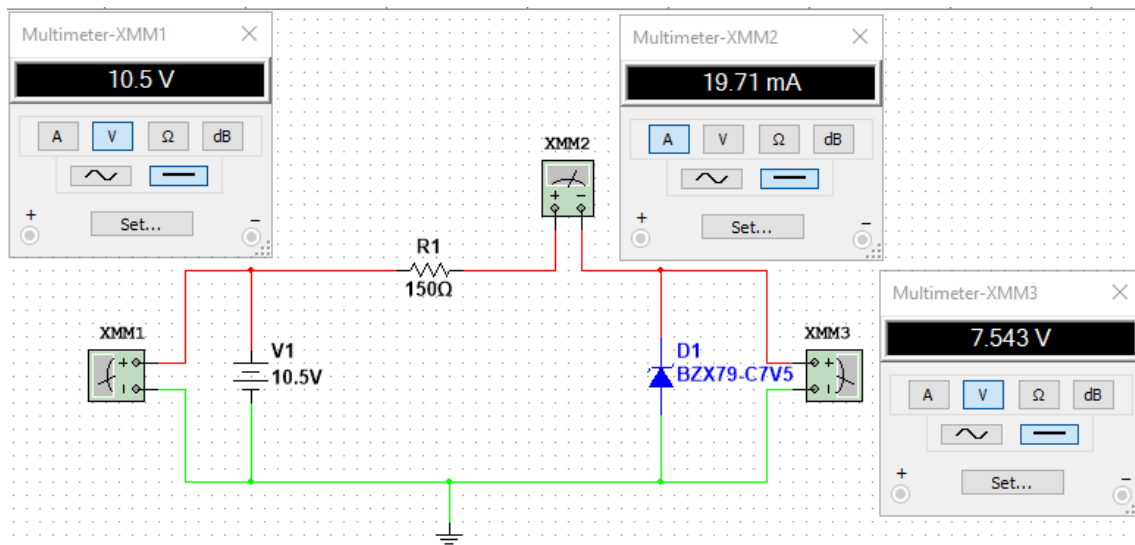


Figure 2.2.11 Reverse polarization of a 20 mA Zener diode in "Multisim"

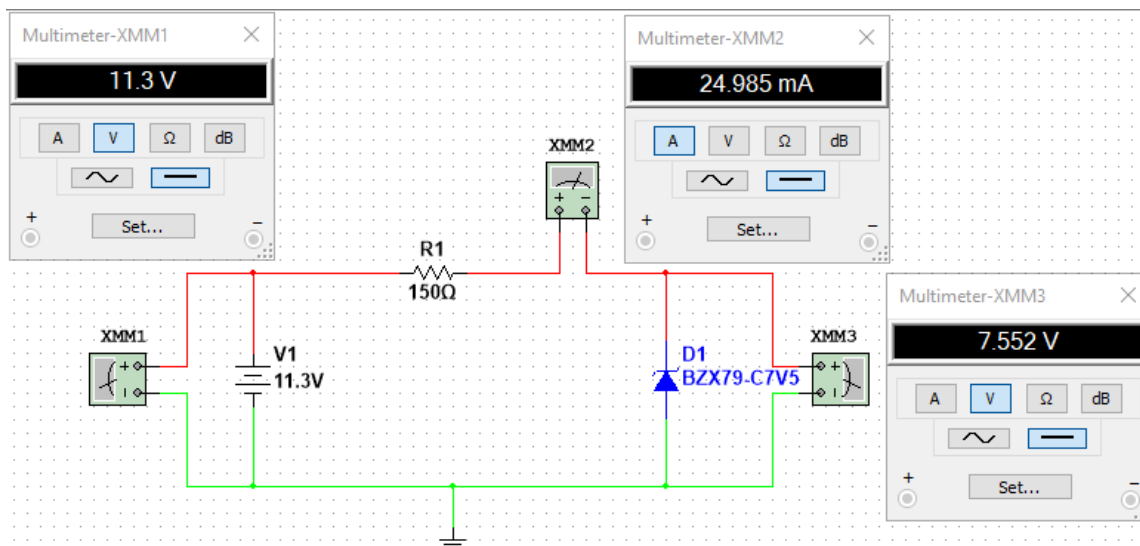


Figure 2.2.12 Reverse biasing of a Zener diode with 25 mA current in "Multisim"

Experimental Solution

Figure 2.2.13 shows the illustration of the Zener diode reverse bias on the "breadboard", where it was carried out in the laboratory.

First, we connect the voltage supply (0-15 Volt) and a digital bench top multimeter set to calculate voltage in Volts, in parallel to the ends of the supply in order to calculate the source voltage. We then connect the 105 Ohm resistor in series with the source, as well as connecting the diode in series with the resistor. The diode will be inversely polarized, which means that the N contact (black line) will be turned towards the source, so that the anode voltage (P contact) will be less than the cathode voltage (N contact). We connect the simple digital multimeter set to calculate current intensity in μA , in series with the resistor and before the diode, in order to calculate the intensity of the current flowing through the diode and therefore the whole circuit. Finally, we connect a digital bench multimeter set to calculate voltage in volts, in parallel to the ends of the diode, in order to calculate the potential difference at the ends of the diode. We also use a wire to connect the earth ground.

ELECTRONICS

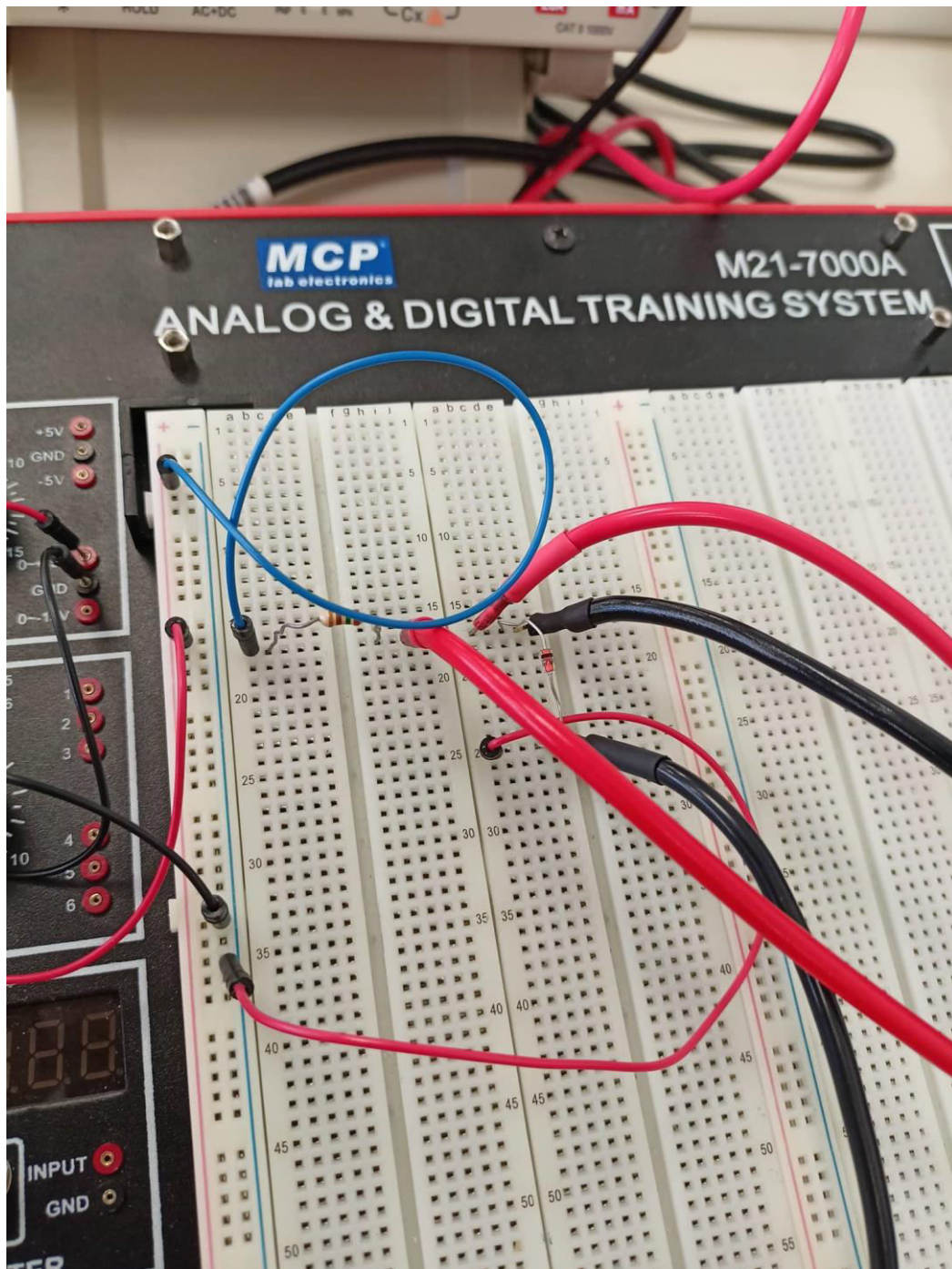


Figure 2.2.13 Experimental illustration of Zener diode reverse bias on the "Breadboard"

In Figures 2.2.14, 2.2.15, 2.2.16, 2.2.17, 2.2.18, 2.2.19, 2.2.20, 2.2.21 and 2.2.22 the recorded measurements of Table 7 are experimentally verified on the "Breadboard" interface board from the laboratory environment.

ELECTRONICS

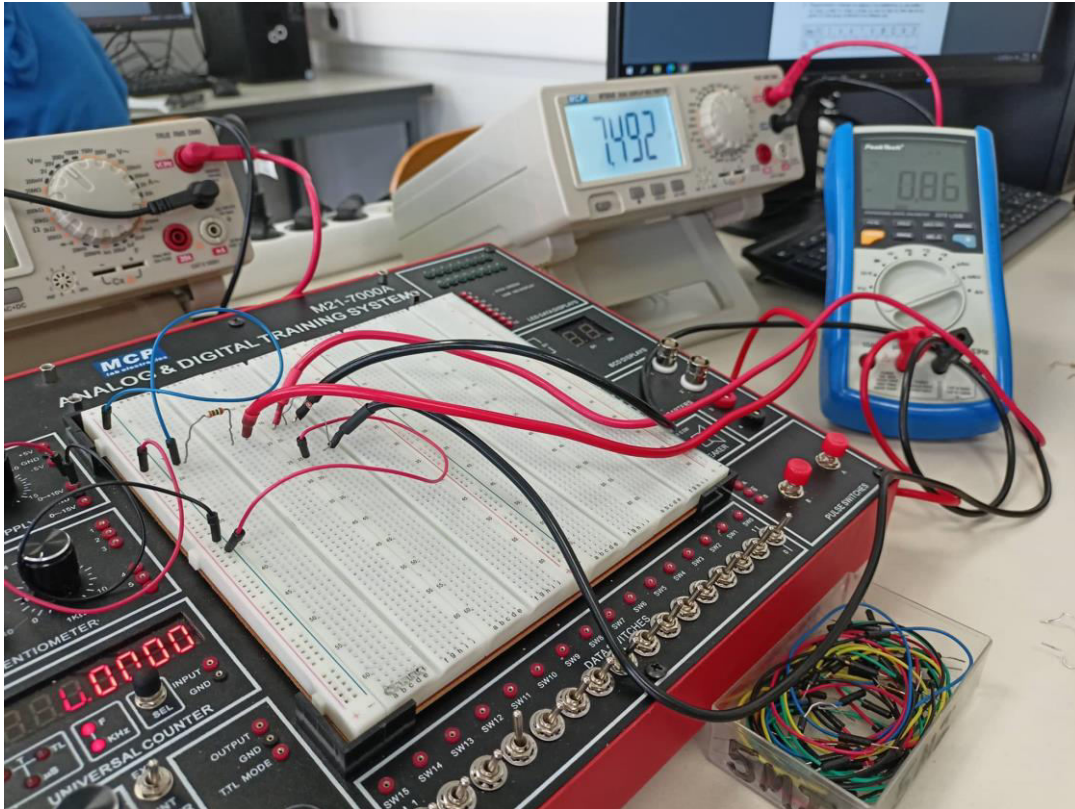


Figure 2.2.14 Inverted polarization of a Zener diode with 1 mA supply voltage on the "Breadboard"

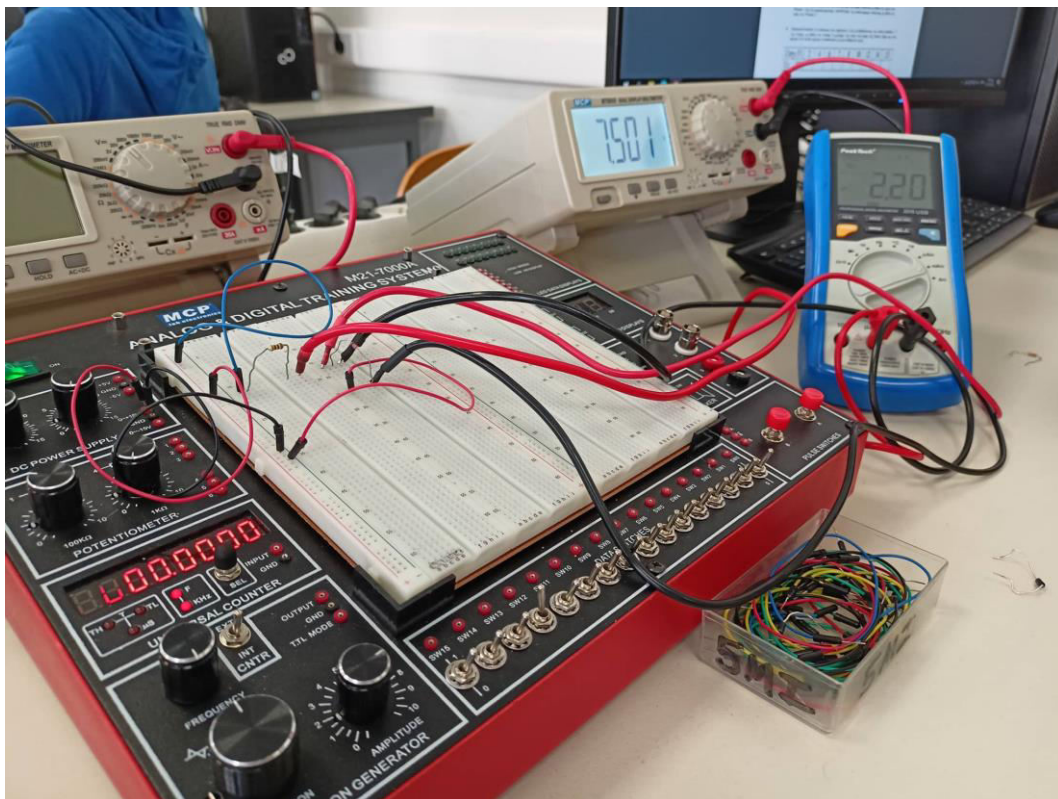


Figure 2.2.15 Inverted polarization of a Zener diode with 2 mA supply voltage on the "Breadboard"

ELECTRONICS

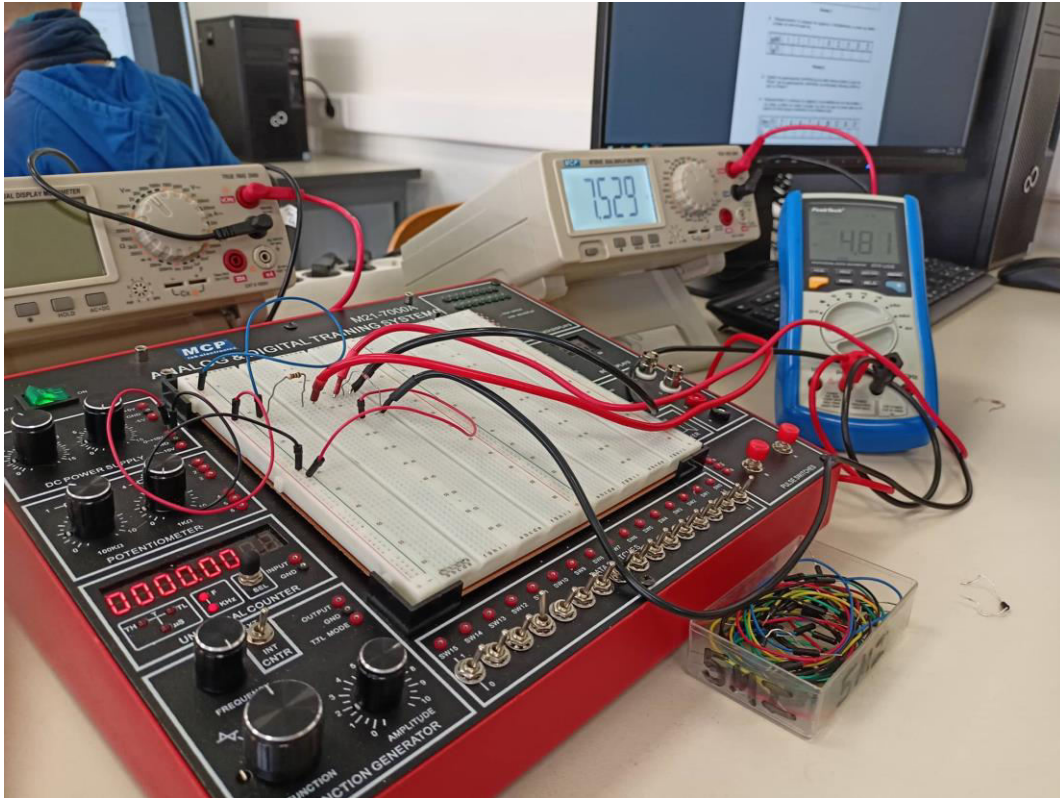


Figure 2.2.16 Inverted polarization of a Zener diode with 5 mA supply voltage on the "Breadboard"

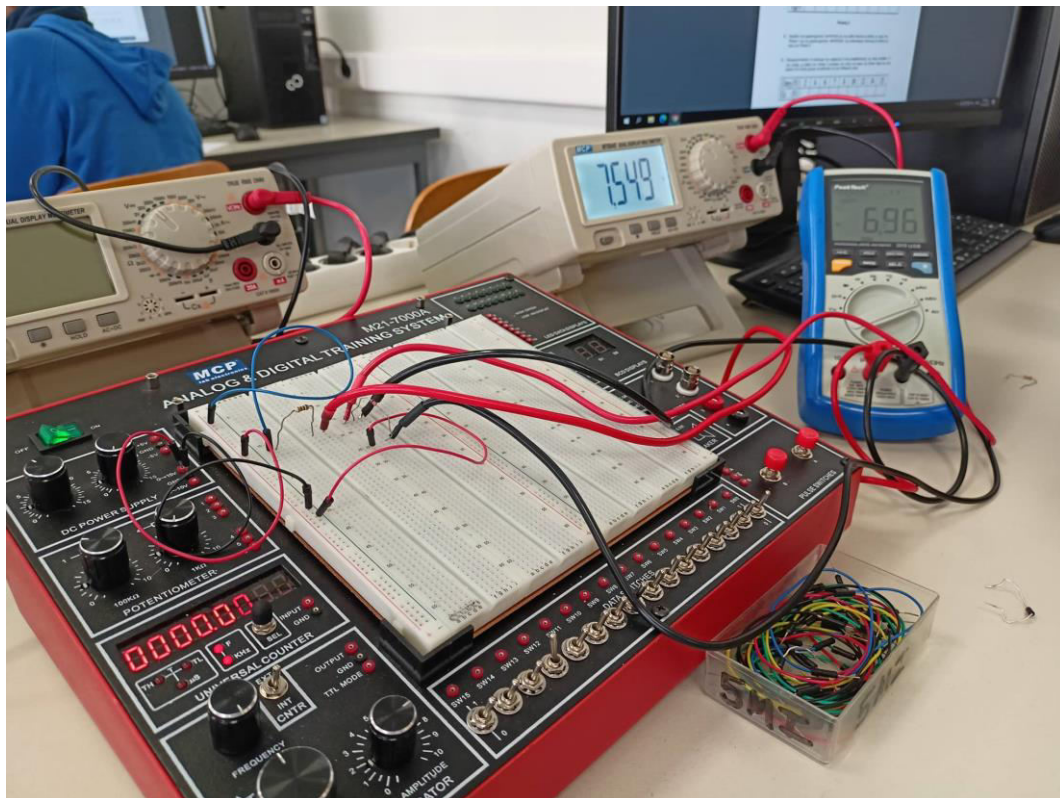


Figure 2.2.17 Inverted polarization of a Zener diode with 7 mA supply voltage on the "Breadboard"

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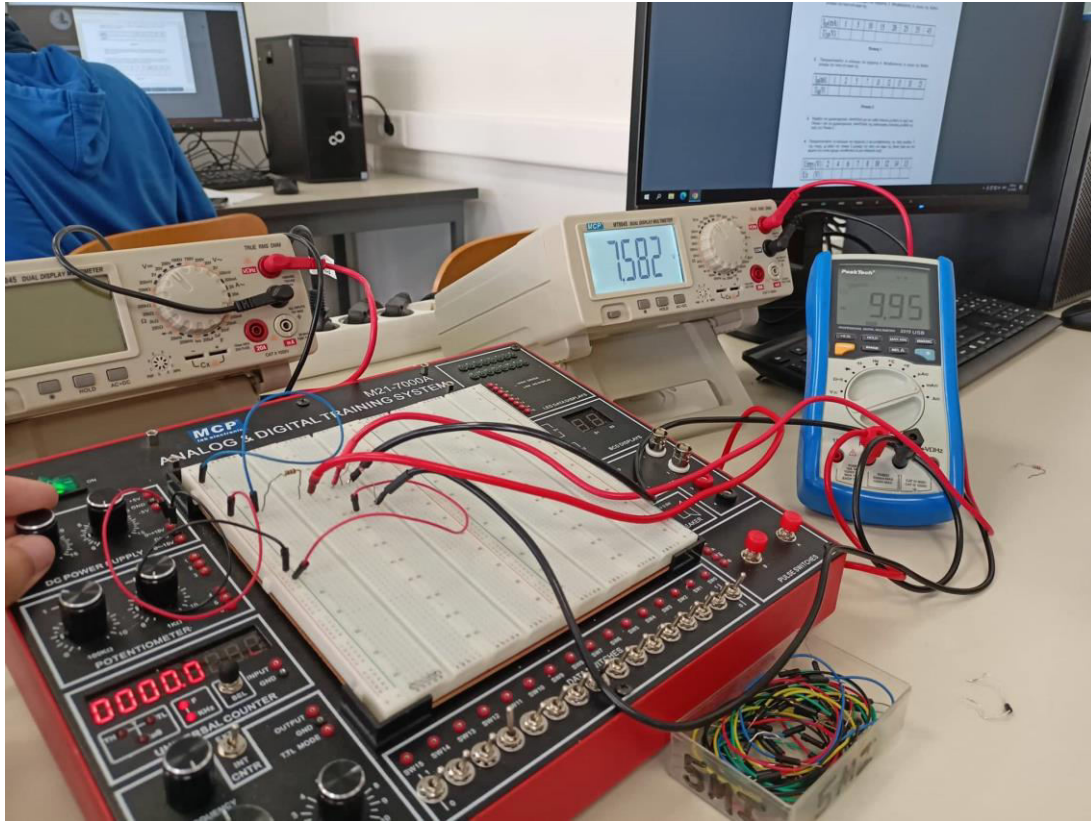


Figure 2.2.18 Inverted polarization of a Zener diode with 10 mA supply voltage on the "Breadboard"

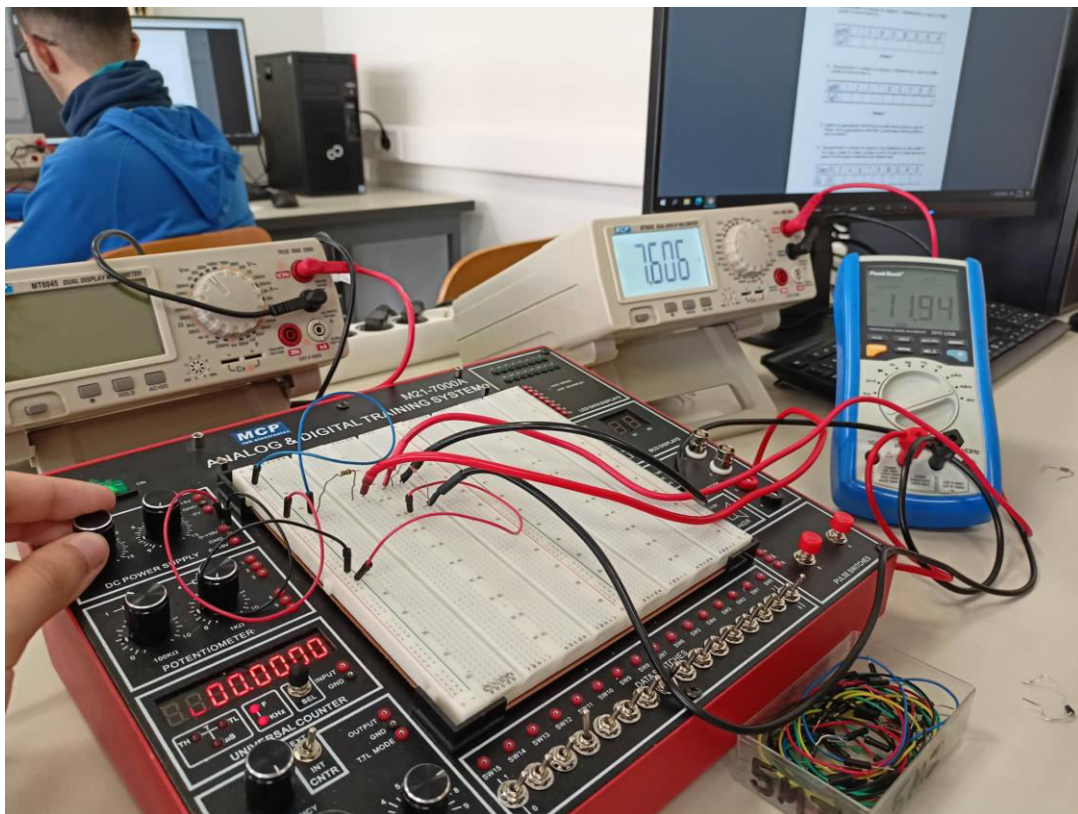


Figure 2.2.19 Inverted polarization of a Zener diode with 12 mA supply voltage on the "Breadboard"

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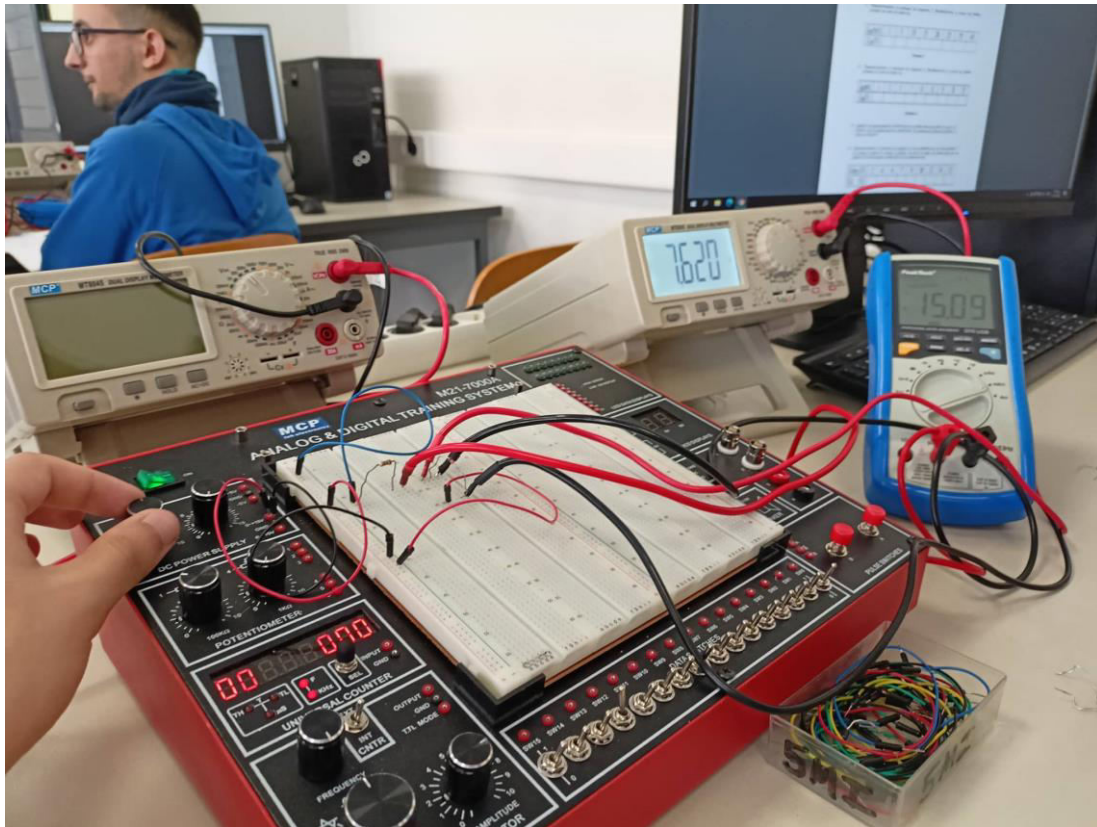


Figure 2.2.20 Inverted polarization of a Zener diode with 15 mA supply voltage on the "Breadboard"

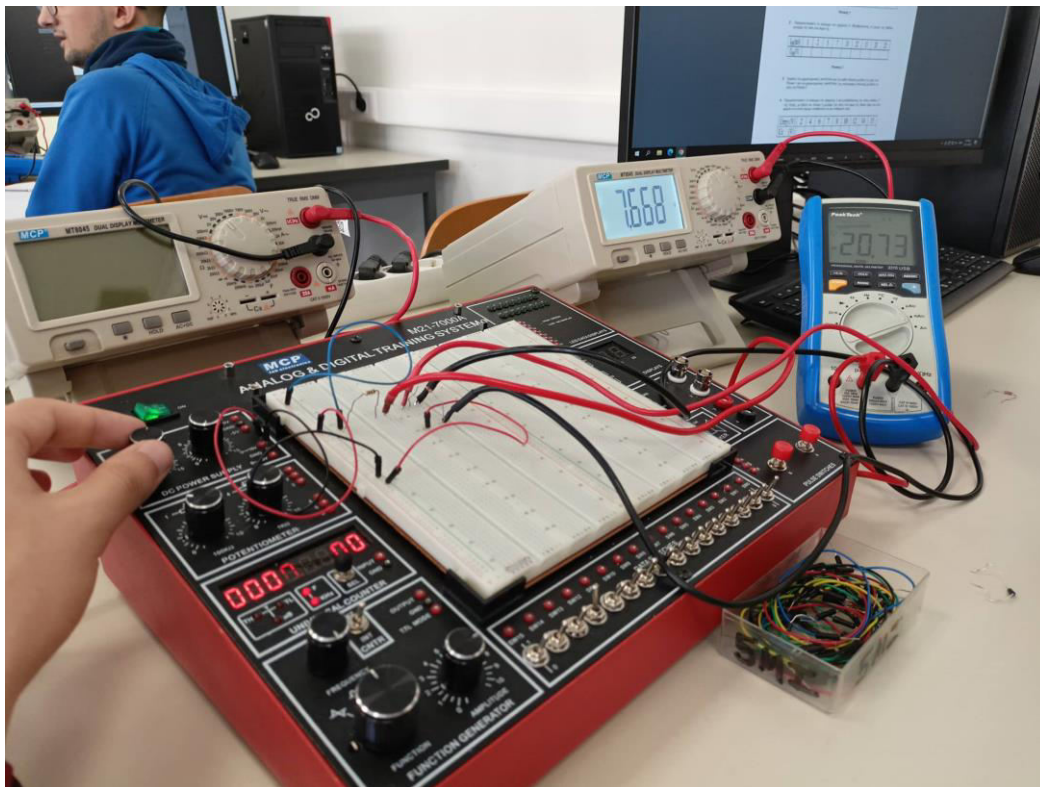


Figure 2.2.21 Inverted polarization of a Zener diode with 20 mA supply voltage on the "Breadboard"

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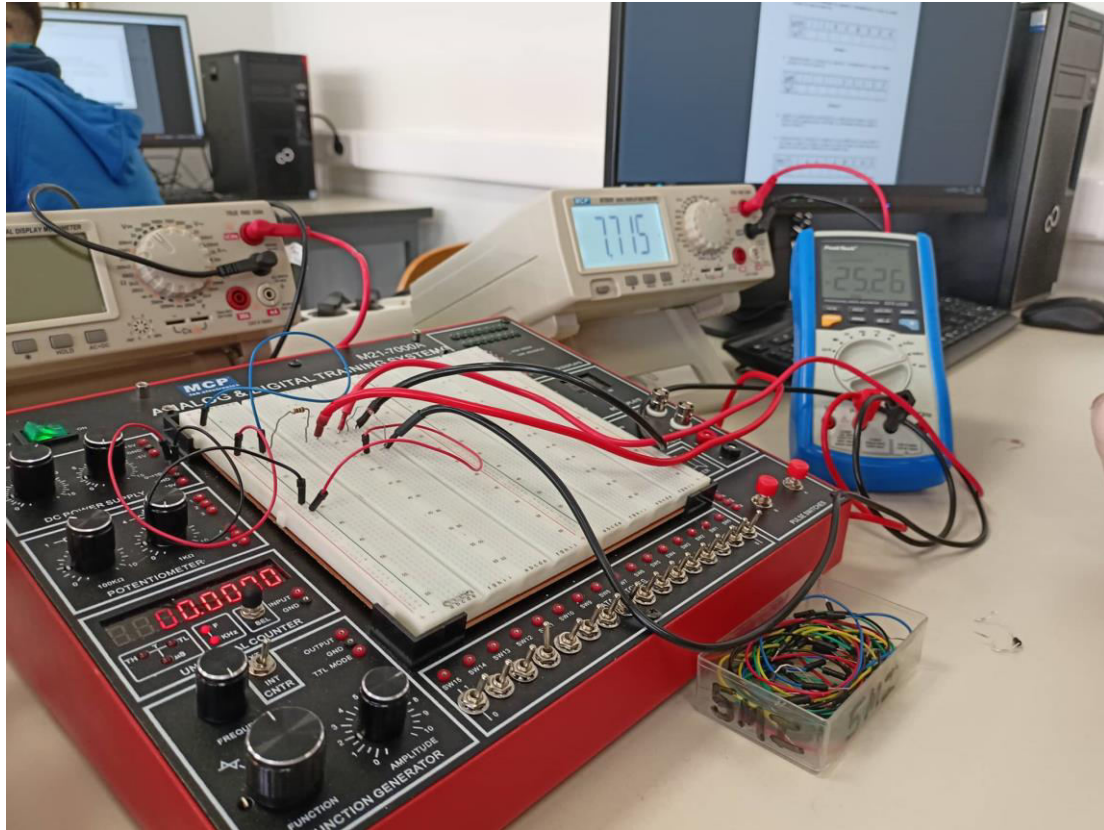


Figure 2.2.22 Inverted polarization of a Zener diode with 25 mA supply voltage on the "Breadboard"

2.3 QUESTIONS

Question 2.3.1

Explain the shape of the characteristic of the Zener diode

The characteristic $I_{DF} - V_{DF}$ of the orthogonally polarized Zener diode is shown in Figure 2.1.2 on page 55. The Zener behaves in orthogonal polarization like the common diode, with gradual increases in current for small supply voltages and a sharp exponential increase when the supply voltage exceeds 0.7 volts. Hence the steep slope in the characteristic from 0.7 Volts onwards. The characteristic $I_{DR} - V_{DR}$ of the inversely polarized silicon diode is illustrated in Figure 2.2.2 on page 65. The current is nearly constant and very small as the stripping region increases and the internal ohmic resistance becomes very large. Consequently, the characteristic is linear. The sharp increase in current for large supplies does not affect the linearity of the characteristic in the inversely polarized Zener and is what distinguishes it from the simple silicon diode.

Question 2.3.3

Where is the Zener diode used? Devise a circuit for each use (except stabilization).

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The Zener diode is used in digital and clipping circuits where the need for voltage stabilization is essential.

Question 2.3.4

In the following circuit we want V_{AB} to be 10 V and the current I_L to have values from 20 mA to 38 mA. Calculate the resistance R of the circuit with $V_{\pi\eta\gamma} = 20$ V and the power consumed by the Zener $P_{z_{max}}$.

$I_{ZMIN} = 2\text{mA}$ and $I_R = I_{ZMIN} + I_{LMAX}$ $I_R = I_{ZMAX} + I_{LMIN}$

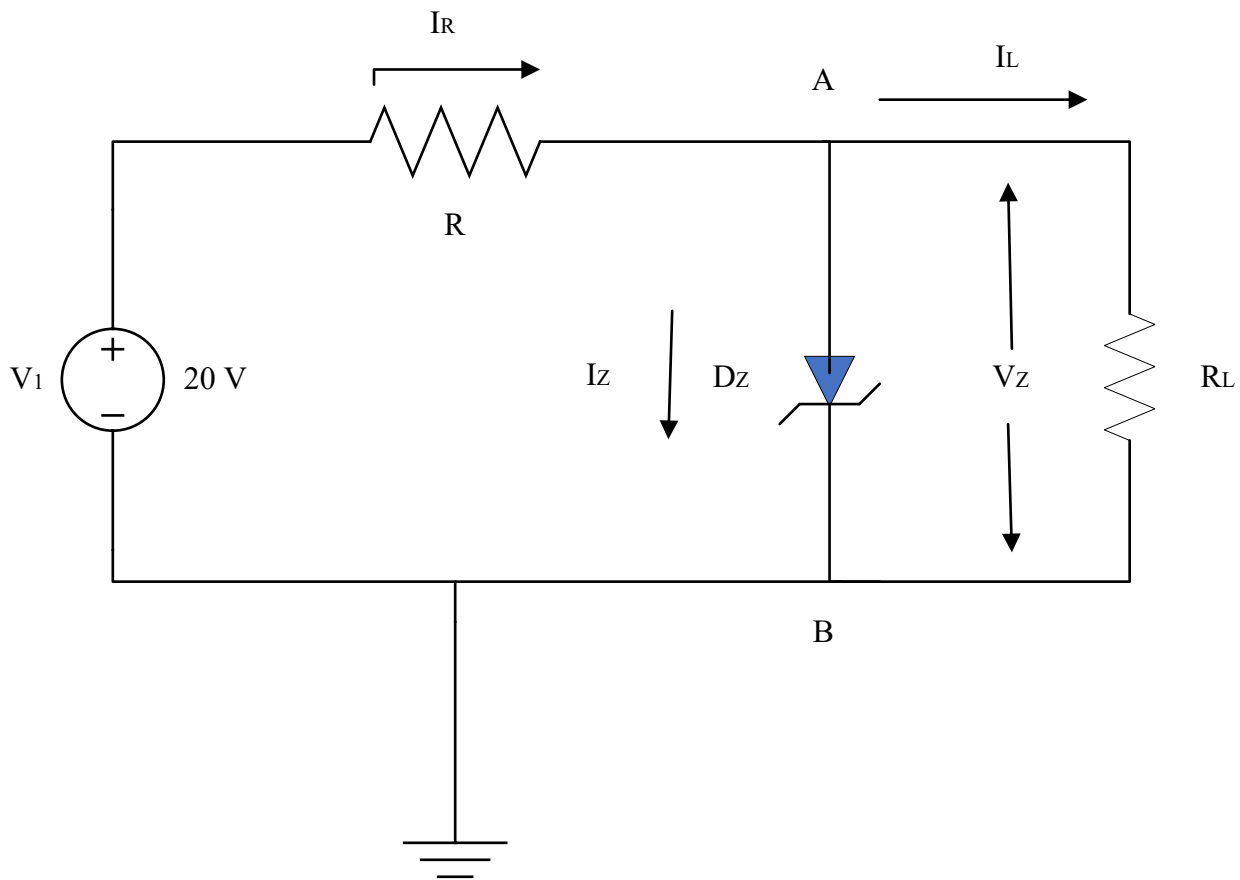


Figure 2.3.1 Schematic representation of the circuit in "MS Visio"

From Kirchhoff's law of voltages it is true that the sum of the potential differences in the individual branches of a loop is equal to zero we have :

$$V1 - V_{AB} = V_R \rightarrow 20 - 10 = I_R \times R \rightarrow (I_{ZMIN} + I_{LMAX}) \times R = 10 \rightarrow \\ (2 \times 10^{-3} + 38 \times 10^{-3}) \times R = 10 \rightarrow R = 250 \text{ Ohm}$$

The power consumed by the Zener is equal to $P_{ZMAX} = V_Z \times I_{ZMAX}$, where $V_Z = I_L \times R_L$. The diode turns on as the voltage at the anode (10 volts) is greater than the voltage at the cathode (0 volts). From the pronunciation we have the following assertions : $I_{ZMIN} = 2 \text{ mA}$, $I_{LMIN} = 20 \text{ mA}$, $I_{LMAX} = 38 \text{ mA}$.

$$I_R = I_{ZMIN} + I_{LMAX} \rightarrow I_R = 2 \text{ mA} + 38 \text{ mA} \rightarrow I_R = 40 \text{ mA}$$

$$I_R = I_{ZMAX} + I_{LMIN} \rightarrow 40 \text{ mA} = I_{ZMAX} + 20 \text{ mA} \rightarrow I_{ZMAX} = 20 \text{ mA}$$

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$$I_L = \frac{I_{LMAX} + I_{LMIN}}{2} \rightarrow I_L = \frac{20 \text{ mA} + 38 \text{ mA}}{2} \rightarrow I_L = 29 \text{ mA}$$

$$P_{ZMAX} = V_Z \times I_{ZMAX} \rightarrow P_{ZMAX} = 0.9 \times 20 \times 10^{-3} \rightarrow P_{ZMAX} = 0.018 \text{ Watt}$$

Question 2.3.5

From the reverse bias characteristic of the Zener you plotted in the paper (Figure 2.2.2 page 65), what is the dynamic resistance of the Zener? (calculate it graphically)

The dynamic resistance of the Zener in reverse bias is the formula :

$$R_{\delta v} = \frac{\Delta V_Z}{\Delta I_Z} \rightarrow R_{\delta v} = \frac{11.3 \text{ Volt} - 7.6 \text{ Volt}}{25 \text{ mA} - 1 \text{ mA}} \rightarrow R_{\delta v} = 0.1541 \text{ Ohm}$$

Where 11.3 Volts is the source supply at 25 mA current (Figure 2.2.12 page 69) and 7.6 Volts is the source supply at 1 mA current (Figure 2.2.4 page 66).

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Thank you for your attention.

